Corrective Overoptimism

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Abstract

Booms in infrastructure investment often coincide with investor exuberance and losses, yet the infrastructure remains long after these episodes. I show that investor overoptimism can correct for a general class of market failure: agents with diagnostic expectations over-extrapolate good news and conduct irreversible investments, pushing what was once a market failure toward the efficient level. This mechanism – corrective overoptimism – is embedded in a dynamic spatial model with endogenous infrastructure and estimated on data from England and Wales, 1700–1881, spanning the canal and railway booms. Corrective overoptimism can be a potent channel for growth.

JEL Classification: D84, E22, E32, N73, O18, O33, R12, R42.

Keywords: Corrective overoptimism, diagnostic expectations, infrastructure investment, spatial development, technology diffusion, Industrial Revolution.

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> Wherever you are in the country, you are just a short distance from the great arterial system. At any moment, you could integrate with this great life-force traversing England; become an essential part of the nation or retreat to the suburban domesticity that you call home.
>
> — Simon Foxell, *Mapping England*

The mass expansion of railways, the electric grid, and the fibre network each coincided with dramatic boom–bust cycles: the Railway Mania of 1846, the Great Depression of 1929, and the Dot-Com Bubble of 2000.[1] Investors suffered enormous losses. Yet from each episode emerged permanent connective infrastructure that reshaped the economy. Janeway (2018) and Morck (2022) argue that this pattern is not coincidental: exuberance-driven technology buildout has been central to the growth process. I formalize this argument and quantify its importance.

The core idea is simple. A monopolist who must sink capital into infrastructure before demand is realized will generically underinvest, because market power drives a wedge between the private return and the social return. If that monopolist over-extrapolates good news – forming beliefs that are too optimistic after positive signals – the resulting over-investment can close the gap. And because the capital is irreversible, the correction survives even after beliefs revert. I call this mechanism *corrective overoptimism*.

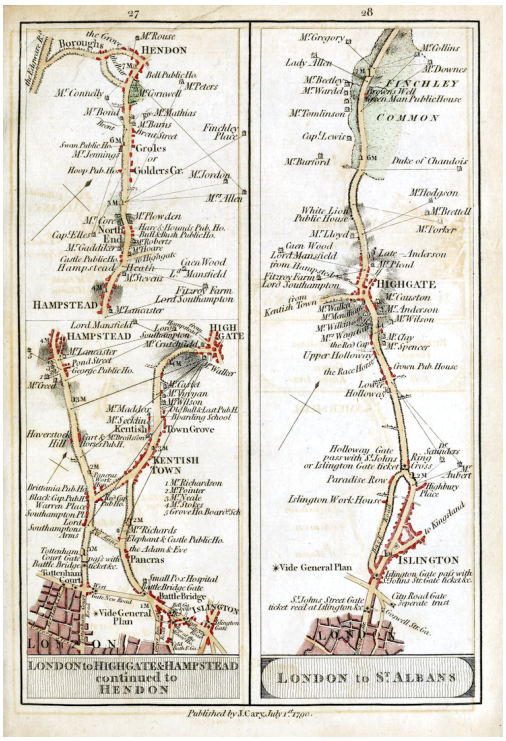
The idea that financial exuberance can be productive has a long history (Keynes, 1936; Kindleberger, 1978; Shiller, 2000). A large literature shows that bubbles can relieve financial constraints, deploying capital more efficiently and crowding in investment for innovation (see, e.g., Martin and Ventura, 2012; Miao and Wang, 2014; Hirano and Yanagawa, 2016; Nanda and Rhodes-Kropf, 2017; Guerron-Quintana et al., 2023). My contribution is to show that behavioral bias can correct not just for financial frictions, but for a broad class of market failures rooted in externalities (Lipsey and Lancaster, 1956). The mechanism I identify operates even in a world without funding constraints: what matters is that an agent invests above their rational choice in a market with positive externalities, and that the resulting investment is sunk.

I model the bias using diagnostic expectations (DE), introduced by Bordalo et al. (2018) as a formalization of the representativeness heuristic of Kahneman and Tversky (1972). Under DE, agents overweight the likelihood of states that have become more likely in light of recent news. A growing literature has shown DE to improve our understanding of credit cycles, asset prices, and business cycle dynamics (Gennaioli et al., 2016; Bordalo et al., 2019, 2020; L'Huillier et al., 2023; Maxted, 2023; Bianchi et al., 2024; Elekdag et al., 2025; Bordalo et al., 2026).[2] I exploit the interaction between the overreactive property of DE and irreversible investment. When a diagnostic infrastructure monopolist receives good news about future demand, it overestimates the return to building and invests beyond the rational level. Once the capital is sunk, the economy benefits permanently.

The simplest form of market failure is a static monopoly. I show corrective overoptimism can close this static gap in Section I. Externalities and spillovers then bloom into many richer forms, particularly through space and time (Lucas, 1988; Romer, 1990; Bresnahan and Trajtenberg, 1995). The rise of the railway coincides with the collapse of the cost of letter delivery and migration;

[1] See, for example, White (1990), Pástor and Veronesi (2009), Odlyzko (2010), Greenstein (2015), and Caoui and Steck (2025).

[2] In complementary work, Bizzarri and d'Arienzo (2024) analyze the social value of information overreaction in abstract coordination settings, while Flynn and Sastry (2025) and Flynn et al. (2026) study contagious narratives and persistence learning over the business cycle.



Great North Road out of London, 1790, John Cary. Courtesy of Duke University via HathiTrust.

telegraphs and fibre networks allow us to exchange ideas at new rates (Mokyr, 2005; Berkes et al., 2026). Each connective revolution increased the ability for each individual to stand on others' shoulders. A single railway company or local turnpike operator cannot capture all the positive benefit they bring to innovation and communication on the other side of the nation. Because these spillovers compound over time, corrective overoptimism can accidentally kickstart a wave of innovations. I contextualize this idea in the structural transformation of England and Wales over 1700–1881 in Section II, showing that regional construction surges systematically reversed below trend and compressed subsequent railway operating yields.

A distinguished literature has studied what transport infrastructure is worth once built. Fogel (1964) and Fishlow (1965) asked what the American economy would have looked like without the railway; Donaldson and Hornbeck (2016) brought this question into a market-access framework; and Hornbeck and Rotemberg (2024) showed that accounting for input distortions substantially magnifies these aggregate productivity gains. A parallel literature characterizes optimal networks in static environments (Fajgelbaum and Schaal, 2020; Allen and Arkolakis, 2022; Allen et al., 2025) and forward-looking spatial dynamics (Allen and Donaldson, 2020; Kleinman et al., 2023).[3] Yet that tradition answers what a network is worth after the tracks are laid. A complementary question is whether the right amount gets built in the first place, and whether the psychological biases of the investors who finance it might, against the usual intuition, improve the outcome. Answering this question requires a model in which the network emerges endogenously from forward-looking investment decisions that are subject to systematic bias.

[3]See also Restuccia and Rogerson (2008), Hsieh and Klenow (2009), and Baqaee and Farhi (2020) on input distortions; Redding and Turner (2015), Ahlfeldt et al. (2015), Donaldson (2018), Hodgson (2018), Heblich et al. (2020), Nagy (2023), and Balboni (2025) on transport networks; and Brancaccio et al. (2020), Bordeu (2025), and Jiang (2026) on decentralized provision.

Section III builds on the spatial growth framework of Desmet and Rossi-Hansberg (2014) and Trew (2020) along four parsimonious margins. First, transport capacity accelerates the spatial diffusion of ideas (Nelson and Phelps, 1966; Cai et al., 2022), capturing dynamic innovation gains beyond static trade-cost reductions. Second, local manufacturing wedges (Desmet et al., 2020; Hornbeck and Rotemberg, 2024) and competitive haulage services (Shaw-Taylor and Wrigley, 2014; Keibek, 2017) capture the allocative gains as cheaper transport shifts labor out of agriculture into industry. Third, reflecting the statutory separation between route construction and carriage in British turnpikes, canals, and early railways (Ward, 1974; Bogart, 2014), forward-looking county infrastructure companies play a dynamic Markov perfect investment game governed by a tractable Tobin marginal-q criterion (Hayashi, 1982). Fourth, within-period worker and firm choices remain invariant to diagnosticity θ: belief overreaction enters the equilibrium exclusively through forward-looking infrastructure investment, where amplified good news permanently expands sunk route capacity while bad-news shortfalls are deferred to the next boom.

Section IV estimates the model by the Simulated Method of Moments on 1710–1881 British data. The estimated diagnosticity parameter is $\widehat{\theta} = 0.739$ (SE $= 0.090$), rejecting rational expectations ($p < 0.0001$) and matching the $\theta \approx 0.74$–0.91 range estimated in macroeconomic and financial data (Bordalo et al., 2018, 2019, 2026). While a re-estimated rational model matches secular output and employment trends, it cannot reproduce the 1830–1850 railway-mania surge, the post-1850 reversal in additions, or the untargeted post-1820 acceleration in real wages (Allen, 2009). Neutralizing diagnosticity ($\theta = 0$) while holding primitives fixed shows that corrective overoptimism expanded the 1881 British transport network by 44 percent, accelerated structural transformation out of agriculture by 13 percentage points, and raised real output per worker by 21 percent and real wages by 13 percent. Nearly three-quarters of that output gain arises from labor reallocation into wedge-distorted manufacturing (30 percent) and faster spatial knowledge diffusion (45 percent) rather than direct transit savings (26 percent).

Moving forward, this paper has three goals. First, to provide a clear illustration of how diagnostic expectations can correct for market failure. Second, to achieve accurate estimates of a model with overoptimism, disciplined by historical context. And third, to assess the empirical relevance of corrective overoptimism. As each goal builds on the last, so do the following sections. Section I provides a simple example. Section II lays out the historical setting. Section III develops the full model, with proofs in Appendix A. Section IV presents the quantitative analysis. Section V concludes.

I A Simple Example

Diagnostic expectations (DE) can overcome market failure. Consider a two-period model of a regional economy that produces wheat and exports it through a monopoly railway.

Demand. Exporters' willingness to pay for shipping is determined by the global commodity price. The inverse demand for railway transport is

$$P_t(Q_t) = Z_t - bQ_t, \qquad b > 0, \tag{1}$$

where Q_t is the volume of wheat shipped, $b > 0$ is the slope of demand, and Z_t is the global price of wheat, which evolves as an AR(1) process $Z_{t+1} = \rho Z_t + \varepsilon_{t+1}$ with $\rho \in (0, 1)$.

Technology. To transport Q_{t+1} units of wheat at date $t+1$, the monopolist must build capacity $K_{t+1} \geq Q_{t+1}$ at date t. Each unit of capacity costs an upfront investment of $c > 0$ to build.

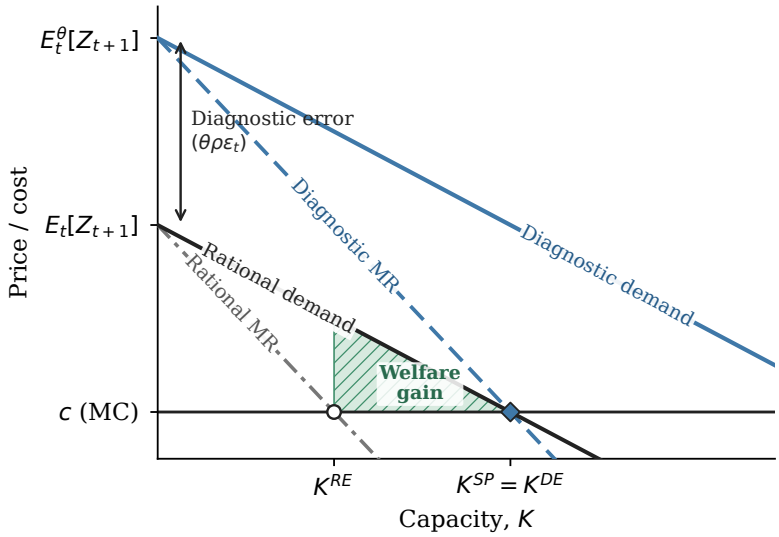


Figure 1: Railways for Wheat Exports

Notes: Linear inverse shipping demand $P = Z - bQ$ and marginal revenue $\text{MR} = Z - 2bQ$. Equating expected marginal revenue rather than marginal social benefit to unit capacity cost c leads rational expectations ($\theta = 0$) to build half the socially efficient capacity ($K^{\text{RE}} = \frac{1}{2}K^{\text{SP}}$), whereas diagnostic overreaction to positive news ($\theta > 0$, $\varepsilon_t > 0$) shifts perceived demand outward and expands sunk capacity toward K^{SP}.

Crucially, this investment is sunk and irreversible. Operating costs are zero.[4]

A The First-Best Baseline

To maximize total welfare, the social planner builds capacity until the expected price exporters are willing to pay for the last unit of transport equals the cost of building it (c):

$$\mathbb{E}_t\left[P_{t+1}(K_{t+1}^{\text{SP}})\right] = c \quad \Longrightarrow \quad K_{t+1}^{\text{SP}} = \frac{\mathbb{E}_t[Z_{t+1}] - c}{b}, \tag{2}$$

by substituting inverse demand $P = Z - bK$.

B The Monopolist's Choice: Rationality vs. Overreaction

Recall that the future wheat price (Z_{t+1}) is the only source of uncertainty. To capture behavioral bias, as in Bordalo et al. (2018), agents' beliefs are subject to diagnostic expectations:

$$\mathbb{E}_t^\theta[Z_{t+1}] = \mathbb{E}_t[Z_{t+1}] + \theta \underbrace{\left(\mathbb{E}_t[Z_{t+1}] - \mathbb{E}_{t-1}[Z_{t+1}]\right)}_{\text{news}}, \tag{3}$$

where $\theta \geq 0$ is the diagnosticity parameter. When $\theta = 0$, we are back in the world of rational expectations. When $\theta > 0$, the railway owner over-updates their belief of the future wheat price based on good and bad news. Under an AR(1) process, $\mathbb{E}_t^\theta[Z_{t+1}] = \mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t$.

[4]Selling installed capacity forward at the market-clearing price $P_{t+1}(K_{t+1})$ (with $\varepsilon_{t+1} \geq -c$) makes $Q_{t+1} = K_{t+1}$ and the first-order conditions below exact. If the owner could instead leave capacity idle ex post, it would ship $\min\{K_{t+1}, Z_{t+1}/(2b)\}$ and the closed forms below would be lower bounds on optimal capacity. Online Appendix S.A analyzes ex-post quantity competition and route entry.

The profit-maximizing monopolist equates expected marginal revenue $\mathrm{MR}_t = Z_t - 2bQ_t$ to cost. Setting $\mathbb{E}_t^\theta[\mathrm{MR}_{t+1}] = c$:

$$K_{t+1}^{\mathrm{DE}} = \frac{\mathbb{E}_t^\theta[Z_{t+1}] - c}{2b} = \frac{\mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t - c}{2b}. \tag{4}$$

Under rational expectations ($\theta = 0$), the monopolist underinvests:

$$K_{t+1}^{\mathrm{RE}} = \frac{1}{2}K_{t+1}^{\mathrm{SP}}. \tag{5}$$

However, diagnostic expectations provide a way out of this failure. Suppose at time t, a positive shock $\varepsilon_t > 0$ arrives: a failed harvest in Europe drives up the global commodity price for wheat. Then

$$K_{t+1}^{\mathrm{DE}} - K_{t+1}^{\mathrm{RE}} = \frac{\mathbb{E}_t^\theta[Z_{t+1}] - \mathbb{E}_t[Z_{t+1}]}{2b} = \frac{\theta\rho\varepsilon_t}{2b} > 0. \tag{6}$$

So the diagnostic investor builds more capital than the rational investor. In particular, if

$$\frac{\theta\rho\varepsilon_t}{2b} = \frac{1}{2}K_{t+1}^{\mathrm{SP}} \quad \Longrightarrow \quad \theta\rho\varepsilon_t = \rho Z_t - c, \tag{7}$$

then we attain the social optimum K_{t+1}^{SP} (and any positive news satisfying $0 < \theta\rho\varepsilon_t < 2(\rho Z_t - c)$ strictly raises expected welfare above K_{t+1}^{RE}).[5] Because capital is irreversible, the level of capacity built during the boom remains in place permanently even after beliefs revert.

II Historical Setting and Motivating Evidence

Britain's Industrial Revolution combines three empirical regularities across England and Wales over 1710–1881 that motivate our dynamic spatial framework. First, aggregate growth reallocated labor out of agriculture from a substantial pre-existing industrial base, sorting manufacturing toward the Midlands and Northern coalfields while expanding distribution and transport employment in every county (Crafts and Harley, 1992; Broadberry et al., 2014; Shaw-Taylor and Wrigley, 2014). Second, turnpikes, canals, and railways were promoted by local committees under Parliamentary acts, separating durable route construction from competitive carriage alongside persistent local manufacturing wedges (Ward, 1974; Bogart, 2014; Trew, 2020; Hornbeck and Rotemberg, 2024). Third, major waves of infrastructure building surged after favorable fundamental news and systematically overshot—producing sharp boom–bust reversals in capacity additions and post-boom compression in operating yields (Odlyzko, 2010; Arnold and McCartney, 2011; Campbell, 2012).

A Spatial Structural Transformation and Distribution Labor

To measure regional structural change on a consistent geography, I place county occupational series on fixed 1831 ancient-county boundaries across twelve benchmark dates from 1710 to 1881, combining Keibek (2017) with the CAMPOP national benchmarks (Shaw-Taylor et al., 2019), London registration-county series (Field and Shaw-Taylor, 2019), and the county population weights of Wrigley (2011) (Online Appendix S.B). Occupations follow the Primary–Secondary–Tertiary (PST) classification of Wrigley (2010): primary covers agriculture, forestry, and fishing; secondary

[5] Expected social surplus relative to rational monopoly is $\mathbb{E}_t[S_{t+1}(K_{t+1}^{\mathrm{DE}}) - S_{t+1}(K_{t+1}^{\mathrm{RE}})] = \frac{\theta\rho\varepsilon_t}{8b}[2(\rho Z_t - c) - \theta\rho\varepsilon_t]$. More generally, for any market failure where marginal social benefit exceeds marginal private benefit at K_{t+1}^{RE} (Lipsey and Lancaster, 1956), the envelope theorem implies that a small diagnostic expansion has zero first-order private loss to the investor while generating a strictly positive first-order social gain.

groups mining with manufacturing and construction because coalfield geography parallels industrial production (Clark and Jacks, 2007; Allen, 2009); and tertiary comprises transport, communications, dealers, and sellers, isolating the distribution and haulage activities required to move goods across space.

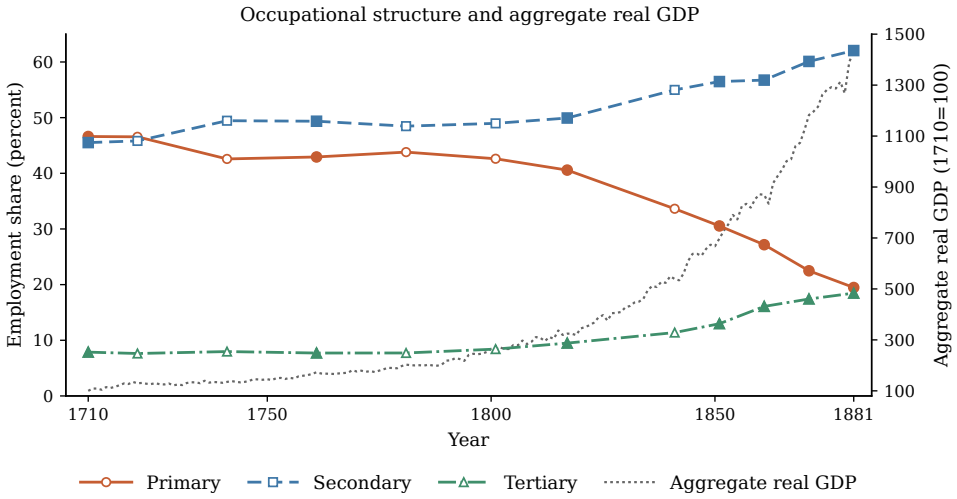


Figure 2: Occupational Structure and Aggregate Real GDP

Notes: Adult male employment shares across primary (excluding mining), secondary (manufacturing, mining, construction), and tertiary (dealers, sellers, transport, communications) sectors: filled markers denote CAMPOP benchmarks (Shaw-Taylor et al., 2019) and open markers denote aligned Keibek (2017) estimates. Right axis: Great Britain real GDP (1710 = 100; Feinstein 1972; Broadberry et al. 2022).

Figures 2 and 3 highlight two features of this transition. Nationally, structural transformation proceeded from a large eighteenth-century industrial base: secondary employment reached half of included employment by 1817, while between 1851 and 1881 the primary share fell from 31 to 19 percent, the secondary share rose from 56 to 62 percent, and the tertiary share expanded from 13 to 18 percent as growing trade volumes required expanding distribution services (Herrendorf et al., 2012; Shaw-Taylor and Wrigley, 2014). Regionally, primary employment remained concentrated in the South and East and secondary industry sorted toward the Midlands and Northern coalfields, whereas distribution and transport employment expanded in every county while remaining thickest around London and the northern industrial hubs (Heblich et al., 2020; Bogart et al., 2022).

B Decentralized Infrastructure Provision, Carriage, and Local Distortions

Unlike the state-planned trunk systems of continental Europe (Milward and Saul, 1973; Smith, 1990), British turnpikes, canals, and railways were promoted decentralizedly by regional committees whose completed links reshaped market access and technology diffusion (Mokyr, 2005; Bogart, 2014; Trew, 2020; Berkes et al., 2026). Across all three eras, statutory organization separated durable route provision from competitive carriage: turnpike trusts, canal companies, and early railway promoters raised capital to build route capacity and collect tolls within parliamentary-authorized corridors, while independent carriers hired operating labor and vehicles to move freight (Ward, 1974; Hadfield, 1981; Turnbull, 1987).

Network expansion also interacted with persistent local distortions in manufacturing. Desmet, Greif and Parente (2020) document how interregional competition eroded local craft resistance to mechanization unevenly across regions, and Hornbeck and Rotemberg (2024) show that transport

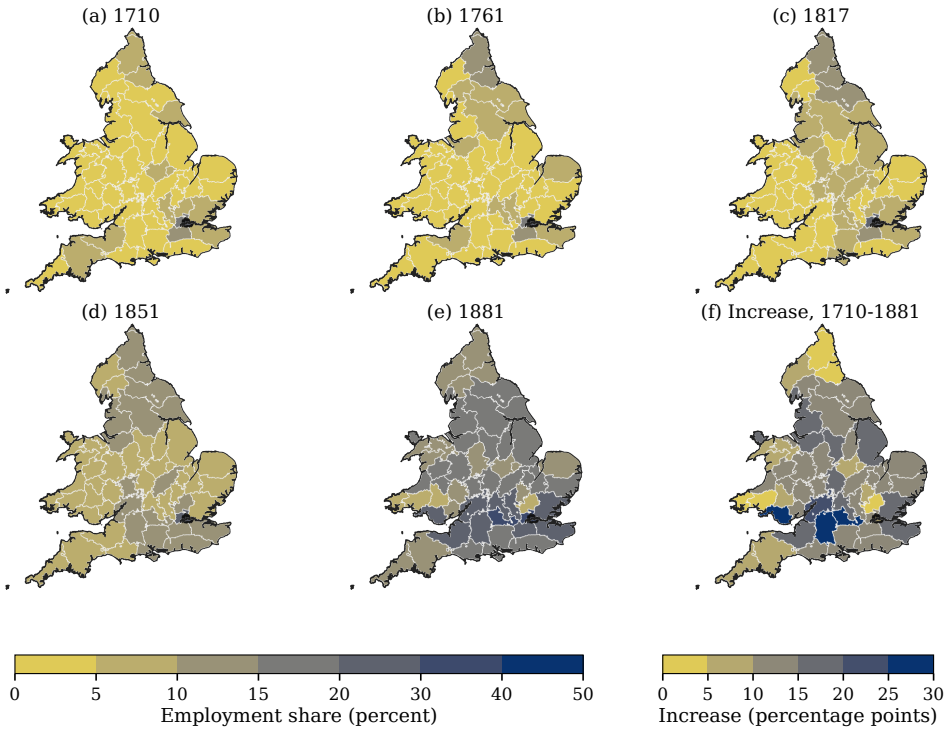


Figure 3: Spatial Expansion of Distribution and Transport Employment, 1710–1881

Notes: Panels (a)–(e) plot county tertiary employment shares at five benchmark dates; Panel (f) plots the 1710–1881 percentage-point increase. County shares through 1851 combine Keibek (2017) and Shaw-Taylor et al. (2019) on 1831 ancient-county boundaries, and 1881 projects the reconstructed latitude-band surface (Online Appendix S.B); the black outline marks London (Field and Shaw-Taylor, 2019).

improvements generate first-order productivity gains when they reallocate inputs toward wedge-distorted manufacturing producers. Figure 4 plots the cumulative multimode transport stock and annual physical additions over 1710–1881, illustrating the clustered construction waves of the 1790s canal mania and the 1830s–1840s railway booms.

C Boom–Bust Reversals and Yield Compression in Infrastructure

Historical accounts of British canal and railway promotion emphasize recurrent cycles of promoter optimism and subsequent disappointment. During the 1830s and the 1845–1848 Railway Mania, prospectuses extrapolated from early trunk successes to advertise 10–15 percent dividends even as realized equity returns averaged under 5 percent and half of Mania-authorized mileage went unbuilt (Odlyzko, 2010). Campbell (2012) shows that weekly railway share prices overreacted to short-run dividend news, while Arnold and McCartney (2011) and Shaen (1847) document systematic cost overruns and post-opening yield shortfalls across early canal and railway companies.

To test whether infrastructure building adjusted smoothly to fundamentals or overshot during booms, let $r^I_{b,t} = (T_{b,t+1} - T_{b,t})/T_{b,t}$ denote the annual growth rate of transport capacity in latitude band b and $c^I_{b,t} = \mathrm{HP}_{6.25}[\log(10^{-3} + r^I_{b,t})]$ its Ravn and Uhlig (2002) cyclical component. Following Jordà (2005) and Bordalo et al. (2018, 2020), I estimate local projections across horizons $h \in \{1, \ldots, 5\}$ for future regional capacity additions $c^I_{b,t+h}$ and aggregate British railway gross operating

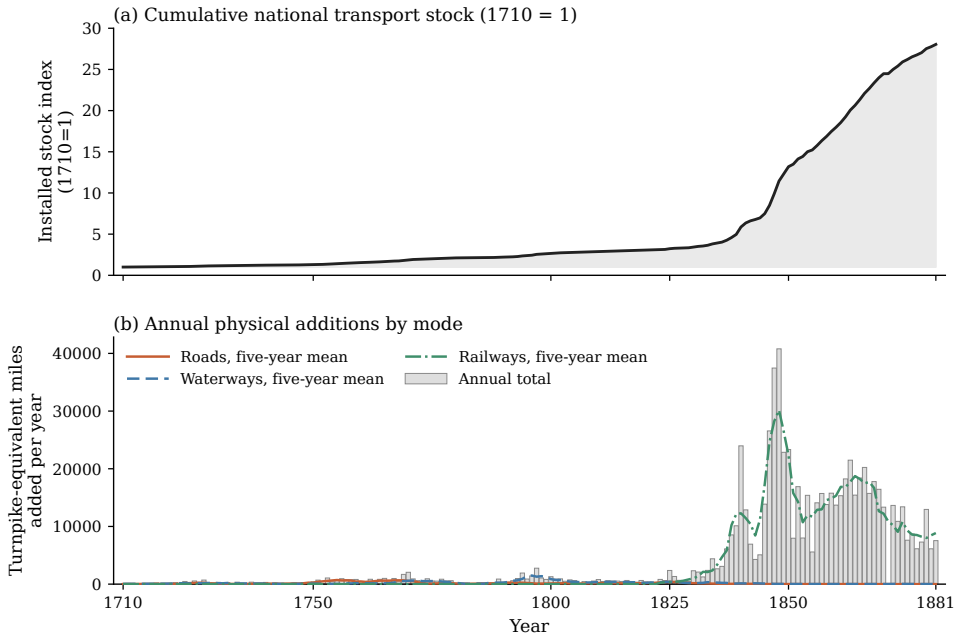


Figure 4: Transport Stock and Installed Additions, 1710–1881

Notes: Turnpikes, waterways, and railways aggregated into turnpike-equivalent miles (road+10×waterway+ 50 × railway). Top panel: cumulative national stock (1710 = 1). Bottom panel: annual physical additions (bars) and five-year moving averages by mode (lines).

yields R_{t+h}/K_{t+h} (working receipts over paid-up capital from Mitchell 1988):

$$c^I_{b,t+h} = \alpha_{bh} + \delta_{th} + \rho_h c^I_{b,t} + u_{b,t+h}, \qquad \frac{R_{t+h}}{K_{t+h}} = \alpha^R_h + \beta^R_h \Delta \ln K_t + u^R_{t+h}. \tag{8}$$

Under rational construction smoothing, a building surge at date t predicts positive future additions above trend ($\rho_h > 0$). Under boom-time overshooting, excess capacity depresses subsequent additions below trend ($\rho_h < 0$) and compresses subsequent operating yields ($\beta^R_h < 0$). Estimating ρ_h directly on the future level $c^I_{b,t+h}$ places no $-c^I_{b,t}$ term on the left-hand side, avoiding mechanical differencing (Jordà, 2005; López-Salido et al., 2017).

Table 1 documents sharp post-boom reversals in both physical construction and realized operating yields. In Panel A, a one-unit cyclical surge in regional capacity additions is followed by below-trend additions over the next three years ($\rho_1 = -0.221$, SE $= 0.019$; $\rho_2 = -0.215$, SE $= 0.020$; $\rho_3 = -0.121$, SE $= 0.020$). Instrumenting regional building with predetermined coalfield and 1710 manufacturing exposure interacted with leave-one-out regional addition cycles (first-stage $F > 172$) yields closely matching reversals ($\rho^{\text{IV}}_1 = -0.273$, SE $= 0.054$; $\rho^{\text{IV}}_2 = -0.202$, SE $= 0.057$), confirming that fundamental-driven booms overshoot and contract.[6] In Panel B, railway paid-up capital surges $\Delta \ln K_t$ predict lower subsequent gross operating yields across all five horizons ($\beta^R_1 = -13.00$ to $\beta^R_5 = -11.15$, $R^2 = 0.40$–0.52), consistent with the 34 percent shortfall between prospectus and realized net traffic yields documented in Shaen (1847)'s parliamentary audit.

[6]Online Appendix S.B (Table S.B.1) shows that these reversals are robust to two-way clustering by latitude band and year, leave-corridor-out ($|k - b| \geq 3$) shift-share instruments, one-sided backward-looking detrending, untransformed level and inverse-hyperbolic-sine addition rates, unfiltered cumulative changes, and separate pre-railway (1710–1829) and railway (1830–1880) subsamples.

Table 1: Local Projections of Infrastructure Investment and Railway Yields

	(1) $h=1$	(2) $h=2$	(3) $h=3$	(4) $h=4$	(5) $h=5$
Panel A. Dependent variable: Regional investment cycle $c^I_{b,t+h}$ (25 latitude bands, 1710–1880)					
Investment cycle $c^I_{b,t}$ (TWFE)	−0.221	−0.215	−0.121	−0.028	−0.011
	(0.019)	(0.020)	(0.020)	(0.020)	(0.019)
Investment cycle $c^I_{b,t}$ (2SLS)	−0.273	−0.202	−0.084	−0.088	−0.004
	(0.054)	(0.057)	(0.054)	(0.056)	(0.028)
Band and year fixed effects	Yes	Yes	Yes	Yes	Yes
First-stage F-statistic (2SLS)	175.6	174.8	174.4	173.2	172.3
Observations	4,250	4,225	4,200	4,175	4,150
Panel B. Dependent variable: Railway gross operating yield R_{t+h}/K_{t+h} (percent, 1843–1881)					
Log paid-up capital growth $\Delta \ln K_t$ (OLS)	−13.00	−13.71	−13.65	−12.49	−11.15
	(2.23)	(2.06)	(2.31)	(2.37)	(2.19)
R^2	0.46	0.52	0.51	0.45	0.40
Observations	38	37	36	35	34

Notes: Local projections (8) at horizons $h = 1, \ldots, 5$ years (Jordà, 2005). Panel A estimates regional regressions across 25 latitude bands (1710–1880) by TWFE and by 2SLS using shift-share instruments $Z_{b,t} = (e^{\text{coal}}_b \bar{c}^{I,\text{coal}}_{-b,t},\ \pi_{M,b,1710}\bar{c}^{I,M}_{-b,t},\ (0.5 + e^{\text{coal}}_b)\bar{c}^{I}_{-b,t})'$ that interact predetermined GIS coalfield exposure e^{coal}_b (Bogart et al., 2022) and 1710 manufacturing shares $\pi_{M,b,1710}$ with leave-one-out regional addition cycles; band-clustered standard errors are in parentheses. Panel B regresses British railway gross working receipts over paid-up capital R_{t+h}/K_{t+h} (percent) on annual log paid-up capital growth $\Delta \ln K_t$ over 1843–1881 (Mitchell, 1988), with Newey–West HAC standard errors in parentheses.

Taken together, medium-horizon construction reversals ($\rho_h < 0$), predictable post-boom yield compression ($\beta^R_h < 0$), and systematic shortfalls relative to prospectus forecasts discriminate against standard rational mechanisms. First, under multi-period time-to-build and convex adjustment costs (Kydland and Prescott, 1982; Hayashi, 1982; Kalouptsidi, 2014), multi-year gestation lags generate positive short-run completion momentum ($\rho_1 = +0.067$ during the railway era and $\rho^{1s}_1 = +0.159$ under one-sided detrending; Online Appendix Table S.B.1), but additions return to trend once projects finish rather than dropping below trend at medium horizons ($\rho_2, \rho_3 < 0$). Nor can known construction lags explain why date-t capital growth $\Delta \ln K_t \in \mathcal{F}_t$ predicts five years of operating-yield compression in Panel B or why realized yields systematically missed promoters' own parliamentary prospectuses (Shaen, 1847). Second, under Bayesian learning and irreversible real options (Arrow, 1968; Jovanovic, 1982; Bernanke, 1983; Dixit and Pindyck, 1994; Pástor and Veronesi, 2009), uncertainty raises the option value of waiting—delaying and smoothing entry rather than bunching it into booms—while martingale belief updating implies gradual continuation ($\rho_h > 0$) and unpredictable ex post return errors conditional on date-t investment. Third, in the aggregate time series, cyclical additions load strongly on current and lagged sectoral output cycles ($b_0 = 2.95$, $b_1 = 1.58$) but have zero predictive power for next-year output ($b_{\text{lead}} = -0.005$, SE $= 0.004$), indicating that promoters extrapolated recent industrial expansions rather than acting on private foresight of future fundamentals (Cochrane, 1991). Finally, year fixed effects δ_{th} in Panel A absorb aggregate interest-rate and risk-premium fluctuations so that the 2SLS reversals reflect cross-regional overreaction to real industrial news within each year, while Mania-era prospectuses advertised 10–15 percent dividends—above, not below, historical norms (Odlyzko, 2010; Campbell, 2012).

III The Model

A The Spatial Economy

Time is discrete, $t \in \{0, 1, 2, \dots\}$. Building on the continuous-space geography of Lucas and Rossi-Hansberg (2002) and Rossi-Hansberg (2005) and its extension to spatial growth and endogenous transport by Desmet and Rossi-Hansberg (2014) and Trew (2020), the economy occupies the unit interval $\ell \in [0, 1]$, partitioned into J county jurisdictions $\mathcal{C}_j = [c_{j-1}, c_j]$, with output, employment, and route infrastructure measured as densities per unit of land. Three sectors operate across space: agriculture (A) and manufacturing (M), which produce the two consumption goods, and transport and distribution services (T). Competitive land leasing assigns each site $\ell \in [0, 1]$ to a single goods sector ($e_A(\ell, t) + e_M(\ell, t) = 1$, $e_i(\ell, t) \in \{0, 1\}$), while competitive transport carriers operate alongside goods producers within each county.

A continuum of worker households of mass $\bar{L}_t$ supplies one unit of labor inelastically and holds equal shares in the national portfolio of land leases, producer net profits, carrier capacity rents, and factor-wedge rebates. Under constant participation, England and Wales population P_t^{EW} determines the household mass (normalized to 100 in 1710):

$$\bar{L}_t = 100 \frac{P_t^{EW}}{P_{1710}^{EW}}, \tag{9}$$

with each household receiving $1/\bar{L}_t$ of aggregate nonlabor income. In each county $j \in \{1, \dots, J\}$, a parliamentary-authorized infrastructure company chooses completion intensity x_{jt} to maximize expected discounted land rents inside $\mathcal{C}_j$ net of manufactured construction expenditure.

As in dynamic spatial general equilibrium frameworks (Kleinman et al., 2023), each period t unfolds in two stages conditional on inherited knowledge $A_{it}(\ell)$, route capacity $T_t(\ell)$, and aggregate productivity gaps $\mathbf{z}_t$, separating forward-looking state transitions from within-period spot market clearing:

1. *Pre-discovery commitments:* County companies commit $\mathbf{x}_t \in \mathcal{X}_t$; goods entrants announce pure sector $e_i(\ell, t)$, research probability $\phi_i(\ell, t)$, innovation bill Ψ_i, and hired production labor $L_i(\ell, t)$ at a fixed local contract wage $w(\ell, t)$; carriers commit operating labor $L_T(\ell, t)$ at $w(\ell, t)$; and households choose residence density $n(\ell, t)$.

2. *Post-discovery spot clearing and transitions:* Local innovation gains $G_{it}(\ell, \omega)$ realize; output utilization $Y_i(\ell, t, \omega)$, county tolls $\tau_j(t, \omega)$, spot prices $p_i(\ell, t, \omega)$, trade flows $H_i(\ell, t, \omega)$, and consumption $c_i(\ell, t, \omega)$ clear markets statewise; and discoveries and completed projects update $A_{i,t+1}(\ell)$ and $T_{t+1}(\ell)$.

Table 2 summarizes the environment; Subsections III.B–III.G derive closed-form rules that reduce the equilibrium to a coupled spatial operator and a dynamic county marginal-q game.

B State and Beliefs

At the start of date t, the persistent spatial state consists of bounded, positive, measurable knowledge profiles $A_{At}(\ell)$ and $A_{Mt}(\ell)$ and installed route capacity $T_t(\ell)$, observed alongside aggregate sectoral productivity gaps $\mathbf{z}_t = (z_{At}, z_{Mt}, z_{Tt})'$ and the predetermined demographic, coalfield energy, manufacturing-wedge, and statutory project schedules in $\mathcal{E}_t \subset \mathcal{F}_t$. Only the active sector at ℓ conducts research there, whereas both sectors adopt discoveries from the spatial frontier; infrastructure built at date t enters operating capacity at $t + 1$ and accelerates knowledge adoption from $t + 1$ onward.

Table 2: Economic Environment

	Households
Lifetime utility	$W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s\geq t}} \mathbb{E}_t^\theta[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s))]$
Period utility	$U(c_A, c_M) = (c_A - \gamma)^\eta c_M^{1-\eta}$
Budget	$p_A c_A + p_M c_M = w + b_t(\omega)$
Nonlabor income	$b_t = \bar{L}_t^{-1} \int_0^1 (R - p_M I + \Pi_F + \Pi_T + \Pi_\psi)\, d\ell$
	Production and Innovation
Bid rent	$R_i = \max_{\substack{0\leq\phi_i<1 \\ \bar{L}_i\geq 0}} \{\mathbb{E}_t^\theta[(p_i - r_i\tau_j)_+ Y_i] - (1+\psi_i) w L_i - \Psi_i(\phi_i; w, \ell)\}$
Production technology	$Y_i = e_i Z_i G_i L_i^{\alpha_i}$
Innovation expenditure	$\Psi_i(\phi_i; w, \ell) = w\left[f_i(\ell) + \frac{\chi_i}{1-\phi_i}\right]$ for $\phi_i > 0$, $\quad \Psi_i(0; w, \ell) = 0$
Research outcome	$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}$, $\quad g \geq 1$
Knowledge after innovation	$A_{it}^+ = A_{it}[1 + e_{it}(G_{it} - 1)]$
	Transport Services
Operating profits	$\max_{L_T\geq 0} \mathbb{E}_t^\theta[\tau_j Y_T - w L_T]$
Transport technology	$Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}$
County capacity and demand	$K_j = \int_{\mathcal{C}_j} Y_T\, d\ell$, $\quad D_j = \int_{\mathcal{C}_j} \sum_i r_i (Y_i + \lvert H_i\rvert)\, d\ell$
	County Infrastructure Companies
Value function	$V_{jt}^\theta = \max_{0\leq x_j\leq \bar{x}_{jt}} \{R_{jt} + \beta \mathbb{E}_t^\theta[V_{j,t+1}^\theta]\}$
Net rents	$R_{jt} = \mathbb{E}_t^\theta \int_{\mathcal{C}_j} [R_t - p_{Mt} I_t(\ell; \mathbf{x}_t)]\, d\ell$
Installed capacity	$T_{t+1} = T_t + \sum_j x_{jt} \iota_{jt}$
	State Transitions
Productivity	$Z_{it} = A_{it} e^{z_{it}} / \mathbb{E}_\pi[e^{z_i}]$, $\quad i \in \{A, M, T\}$
Sectoral shocks	$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma})$
Available frontier	$F_{it}(\ell) = \sup_{r\in[0,1]} e^{-\delta\lvert\ell - r\rvert} A_{it}^+(r)$
Adoption	$A_{i,t+1} = A_{it}^+ + \left[1 - e^{-\lambda_0 (T_t/T_{t_0})^{1-\alpha_T}}\right](F_{it} - A_{it}^+)$
	Market Clearing
Residence	$\mathbb{E}_t^\theta[(w + b - \gamma p_A)/\mathcal{P}] = \bar{u}$
Goods	$\partial_\ell H_i = Y_i - n c_i - e_i \Psi_i / p_i - I_i - \kappa_0 e^{-T_t} \lvert H_i\rvert$
Labor and population	$n = \sum_i e_i L_i + L_T$, $\quad \int_0^1 n\, d\ell = \bar{L}_t$
Services	$D_j \leq K_j$, $\quad \tau_j \geq 0$, $\quad \tau_j(K_j - D_j) = 0$

Notes: Local and date arguments are suppressed; $i \in \{A, M\}$ unless stated otherwise, and j indexes the county. $e_i \in \{0, 1\}$ indicates the highest-bidding goods sector with land rent $R = \max_i R_i$; $f_A(\ell) = f_A$ and $f_M(\ell) = f_M + \xi(\ell)$; construction uses manufactured goods ($I_A = 0$, $I_M = I$). Borders are closed and the numeraire is $p_A(1, t) = 1$. Remaining definitions are in the text.

Aggregate productivity gaps follow a stationary first-order Markov process on a compact symmetric box $\mathcal{B}$:

$$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma}), \tag{10}$$

where $\mathrm{TN}_{\mathcal{B}}$ is a multivariate Gaussian law truncated to $\mathcal{B}$ ($\pm c$ stationary untruncated standard deviations around zero, with $c = 3$ in the benchmark and $c = 4$ for robustness; Online Appendix S.C.1), $\boldsymbol{\rho} = \mathrm{diag}(\rho_A, \rho_M, \rho_T)$ with $\lvert\rho_s\rvert < 1$, and $\boldsymbol{\Sigma}$ is the positive definite innovation covariance matrix with $\sigma_s > 0$. The kernel has a unique stationary distribution π on $\mathcal{B}$, and effective productivity (19) is normalized by $\mathbb{E}_\pi[e^{z_i}]$ so that $e^{z_{it}}/\mathbb{E}_\pi[e^{z_i}]$ has unit stationary physical mean. Write $f(\mathbf{z}' \mid \mathbf{z})$ for this physical kernel and $M(\mathbf{m})$ for the conditional mean on $\mathcal{B}$ of a Gaussian centered at $\mathbf{m}$.

Belief formation follows the diagnostic expectations operator of Bordalo et al. (2018, 2020), which formalizes Kahneman and Tversky (1972)'s representativeness heuristic and underpins macroeconomic models of sentiment-driven cycles (L'Huillier et al., 2023; Maxted, 2023; Bianchi et al., 2024; Bordalo et al., 2026). Relative to the no-news reference forecast $\bar{\mathbf{z}}_t = \mathbb{E}_{t-1}[\mathbf{z}_t] = M(\boldsymbol{\rho}\mathbf{z}_{t-1})$, observed news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$ tilts perceived transition probabilities toward states whose likelihood increased most in light of $\boldsymbol{\varepsilon}_t$, weighted by the representativeness ratio $\Lambda_t(\mathbf{z}') = f(\mathbf{z}' \mid \mathbf{z}_t)/f(\mathbf{z}' \mid \bar{\mathbf{z}}_t)$:

$$\mathbb{E}_t^\theta[X] = \frac{\mathbb{E}_t[X\Lambda_t(\mathbf{z}_{t+1})^\theta]}{\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]}, \qquad \theta \geq 0, \tag{11}$$

where $\theta = 0$ nests rational expectations and $\theta > 0$ indexes belief extrapolation. Because $\mathbf{z}_t$ is observed at the start of date t and local Bernoulli–Pareto discoveries are independent of $\mathbf{z}_{t+1}$, the factor $\Lambda_t(\mathbf{z}_{t+1})^\theta$ cancels from (11) for any within-period payoff conditional on $(\Omega_t, \mathbf{x}_t)$. Exponentiating the Gaussian kernel yields the closed-form diagnostic one-step law and recursive forecast reference (Online Appendix S.C.1):

$$\mathbf{z}_{t+1} \overset{\theta}{\sim} \mathrm{TN}_{\mathcal{B}}(\mathbf{m}_t^\theta, \boldsymbol{\Sigma}), \qquad \mathbf{m}_t^\theta = \boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}(\mathbf{z}_t - \bar{\mathbf{z}}_t), \tag{12}$$

$$\bar{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^\theta), \qquad \boldsymbol{\varepsilon}_{t+1}^{\text{forecast}} = \mathbf{z}_{t+1} - M(\mathbf{m}_t^\theta), \tag{13}$$

whereas in physical time at date $t+1$, the observed reference resets to $\bar{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$. Equation (12) delivers the two empirical regularities documented in Section II: forecasts overreact to current news with location shift $(1+\theta)\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$, and subsequent realizations systematically disappoint after positive news ($\mathbf{m}_t^\theta - \boldsymbol{\rho}\mathbf{z}_t = \theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$). The aggregate state is $\Omega_t = (A_{At}, A_{Mt}, T_t, \mathbf{z}_t, \bar{\mathbf{z}}_t, t)$ (Figure 5).

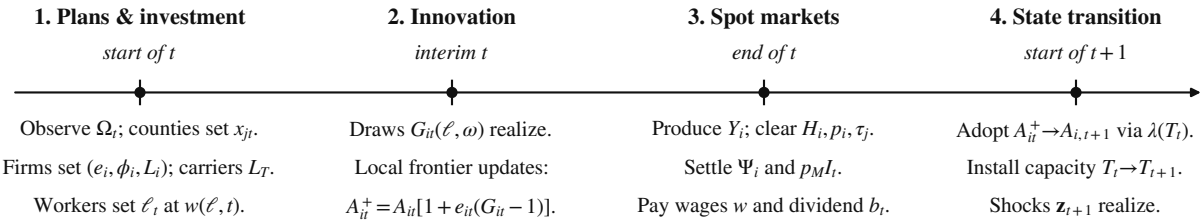


Figure 5: Within-Period Timing

Notes: County investment x_{jt}, producer plans (e_i, ϕ_i, L_i), carrier employment L_T, and worker residence ℓ_t at local contract wage $w(\ell, t)$ are committed before idiosyncratic discoveries $G_{it}(\ell, \omega)$ realize. Spot goods and service markets clear ex post, and completed capacity T_{t+1} and diffused knowledge $A_{i,t+1}$ enter at date $t+1$.

C Workers

Workers live where they work, supply one unit of labor inelastically, and hold equal shares in the national portfolio of land leases, goods producers, transport firms, and factor-wedge rebates, receiving common nonlabor income b_t in addition to the local contract wage $w(\ell, t)$. Goods are nonstorable, and households neither save nor borrow. Each period t, a worker commits residence $\ell_t \in [0, 1]$ before private discoveries at the fixed contract wage $w(\ell, t)$ and chooses spot consumption $(c_A(\ell, t), c_M(\ell, t))$ ex post to solve

$$\begin{aligned} W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s \geq t}} \; & \mathbb{E}_t^\theta\left[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s))\right] \\ \text{s.t. } & p_A(\ell_s, s)c_A(s) + p_M(\ell_s, s)c_M(s) = w(\ell_s, s) + b_s, \quad c_A(s) \geq \gamma, \; c_M(s) \geq 0, \end{aligned} \tag{14}$$

with nonhomothetic Stone–Geary period utility (Kongsamut et al., 2001; Herrendorf et al., 2014; Fajgelbaum and Redding, 2022):

$$U\big(c_A(\ell,t),c_M(\ell,t)\big)=(c_A(\ell,t)-\gamma)^{\eta}c_M(\ell,t)^{1-\eta},\qquad \eta\in(0,1),\quad \gamma>0. \tag{15}$$

For $w+b_t\geq\gamma p_A$, spot optimization yields the Marshallian demands, marginal price index, and realized indirect utility:

$$\begin{aligned} c_A&=\gamma+\frac{\eta(w+b_t-\gamma p_A)}{p_A}, & c_M&=\frac{(1-\eta)(w+b_t-\gamma p_A)}{p_M},\\ \mathcal{P}(p_A,p_M)&=\frac{p_A^{\eta}p_M^{1-\eta}}{\eta^{\eta}(1-\eta)^{1-\eta}}, & u_t(\ell)&=\frac{w(\ell,t)+b_t-\gamma p_A(\ell,t)}{\mathcal{P}(p_A(\ell,t),p_M(\ell,t))}. \end{aligned} \tag{16}$$

The subsistence floor $\gamma>0$ generates Engel's law, shifting expenditure toward manufacturing as real income rises. The demands and $u_t(\ell)$ are defined where $w+b_t\geq\gamma p_A$; statewise feasibility at every site is part of the equilibrium definition (Definition 1) and is delivered by Condition 1 of Section III.G. Because residence is committed before private discoveries, workers evaluate locations by expected current indirect utility $\mathbb{E}_t^{\theta}[u_t(\ell)]$.

PROPOSITION 1: *If subsistence is feasible statewise at every site, expected indirect utility $\mathbb{E}_t^{\theta}[u_t(\ell)]$ is finite for every ℓ, and lifetime utility is finite, then, given the current price, wage and portfolio-income schedules, a worker's optimal dynamic residence choice maximizes expected current-period indirect utility.*[7] [Proof →]

Because annual relocation is costless and portfolio claims do not depend on residence, the continuation value of the worker's problem is invariant to current location, equalizing expected indirect utility across occupied sites:

$$\mathbb{E}_t^{\theta}\big[u_t(\ell)\big]=\bar{u}_t\qquad \text{where } n(\ell,t)>0\quad \big(\mathbb{E}_t^{\theta}[u_t(\ell)]\leq\bar{u}_t \text{ elsewhere}\big), \tag{17}$$

$$w(\ell,t)=\frac{\bar{u}_t-\mathbb{E}_t^{\theta}\big[(b_t-\gamma p_A(\ell,t))/\mathcal{P}(p_A(\ell,t),p_M(\ell,t))\big]}{\mathbb{E}_t^{\theta}\big[1/\mathcal{P}(p_A(\ell,t),p_M(\ell,t))\big]}. \tag{18}$$

Because labor demand is strictly positive at every site for any finite contract wage (carriers hire by (50) whenever $\mathbb{E}_t[\tau_{j(\ell)}]>0$, and otherwise the winning goods sector hires by (30) at a positive expected net price), every site is occupied in equilibrium, (17) holds with equality on all of $[0,1]$, and (18) determines the contract wage at every site.

D Production, Research, and Land

Competitive entrants in sector $i\in\{A,M\}$ bid for one-period pure site leases at $\ell\in[0,1]$.[8] Effective productivity combines permanent local knowledge $A_i(\ell,t)$ (or constant carrier frontier $A_T>0$) with the aggregate productivity factor, and realized output Y_i uses committed labor $L_i(\ell,t)$ under diminishing returns:

$$Z_i(\ell,t)=A_i(\ell,t)e^{z_{it}}/\mathbb{E}_{\pi}[e^{z_i}],\quad i\in\{A,M\},\qquad Z_T(t)=A_Te^{z_{Tt}}/\mathbb{E}_{\pi}[e^{z_T}], \tag{19}$$

$$Y_i(\ell,t,\omega)=e_i(\ell,t)Z_i(\ell,t)G_i(\ell,t,\omega)L_i(\ell,t)^{\alpha_i},\qquad i\in\{A,M\},\quad 0<\alpha_A<\alpha_M<1, \tag{20}$$

where $\alpha_A<\alpha_M$ reflects the higher land intensity of agriculture relative to manufacturing (Valentinyi and Herrendorf, 2008). Entrants announce (e_i,ϕ_i,L_i) taking state-contingent spot

[7] Proofs of the propositions are in Appendix A.

[8] Transport routes occupy narrow corridors and do not crowd out productive land.

prices $p_i(\ell,t,\omega)$, county tolls $\tau_{j(\ell)}(t,\omega)$, and the local contract wage $w(\ell,t)$ as given, earning the net unit price $p_i(\ell,t,\omega) - r_i\tau_{j(\ell)}(t,\omega)$ ex post.

Manufacturing firms face a predetermined local operating-labor wedge,

$$\psi_A(\ell) = 0, \qquad \psi_M(\ell) = \bar{\psi}_M \omega_M(\ell), \qquad \bar{\psi}_M \geq 0, \quad \omega_M(\ell) > 0, \tag{21}$$

paying $(1+\psi_M(\ell))w(\ell,t)$ per hired production worker while agriculture is undistorted ($\psi_A = 0$) and wedge receipts are rebated lump-sum to households. Holding $\omega_M(\ell)$ fixed makes transport improvements a source of first-order allocative efficiency gains as labor reallocates toward distorted industrial producers (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024).

Before private discoveries, an entrant in sector $i \in \{A, M\}$ chooses a research success probability $\phi_i \in [0,1)$. Building on the Pareto idea-flow representation of Kortum (1997) and Eaton and Kortum (2002), a successful draw yields a Pareto productivity multiplier with tail exponent $a_i > 1/(1-\alpha_i)$:

$$\Pr(G_i = 1 \mid \phi_i) = 1 - \phi_i, \tag{22}$$

$$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}, \qquad g \geq 1, \quad a_i > 1. \tag{23}$$

Discoveries are spatially correlated. A success field $X^s_{it}(\ell)$ and a quality field $X^g_{it}(\ell)$ are stationary standard Gaussian Ornstein–Uhlenbeck processes on $[0,1]$, independent of each other, across sectors and dates, and of aggregate productivity gaps, with correlation length $d_0 > 0$; with Φ the standard normal distribution function, an entrant choosing ϕ at ℓ draws

$$\begin{aligned} G_i(\phi;\ell,\omega) &= 1 + \mathbf{1}_{\{\Phi(X^s_{it}(\ell))<\phi\}}\left[\Phi(X^g_{it}(\ell))^{-1/a_i} - 1\right], \\ \mathrm{Cov}(X^h_{it}(\ell), X^h_{it}(r)) &= e^{-|\ell-r|/d_0}, \qquad h \in \{s,g\}. \end{aligned} \tag{24}$$

Because $\Phi(X^h_{it}(\ell))$ is uniform at every site, (24) reproduces (22)–(23) at each ℓ, while neighboring sites draw correlated discoveries. All research intensities share one probability space: $G_i(\phi;\ell,\omega)$ is nondecreasing in ϕ, and $G_i(\phi_n;\ell,\omega) \to G_i(\phi;\ell,\omega)$ almost surely whenever $\phi_n \to \phi$.

Innovation requires a wage-indexed monetary expenditure committed before discovery and settled in the spot market by purchasing own-sector output locally:

$$\Psi_i(\phi_i; w, \ell) = \begin{cases} w\left[f_i(\ell) + \dfrac{\chi_i}{1-\phi_i}\right] & \text{if } \phi_i \in (0,1), \\ 0 & \text{if } \phi_i = 0, \end{cases} \tag{25}$$

$$q^R_i(\omega) = \frac{\Psi_i(\phi_i; w, \ell)}{p_i(\omega)}, \tag{26}$$

where $f_A(\ell) = f_A \geq 0$ and $f_M(\ell) = f_M + \xi(\ell) \geq 0$ combines fixed participation cost $f_i \geq 0$ with predetermined manufacturing energy overhead $\xi(\ell) = \omega_E e(\ell) \geq 0$ ($\omega_E \geq 0$ scaling the standardized coalfield energy exposure schedule $e(\ell)$ of Allen 2009), and $\chi_i > 0$ governs research convexity. Gross marketed output Y_i includes research goods, using local distribution services $r_i Y_i$ once, while only the knowledge transition carries discoveries to future periods.

Firm optimization and innovation threshold.— Risk-neutral firms measure payoffs in the numeraire $p_A(1,t) = 1$ and hold neither future land claims nor patents. Taking state-contingent goods prices $p_i(\omega)$, county tolls $\tau_{j(\ell)}(\omega)$, and the local contract wage w as given, a producer with hired labor L_i and discovery $G_i(\omega)$ chooses spot output utilization to maximize operating surplus:

$$\max_{0 \leq Y \leq Z_i G_i L_i^{\alpha_i}} \left\{ (p_i - r_i\tau_{j(\ell)})Y - (1+\psi_i)wL_i \right\} = (p_i - r_i\tau_{j(\ell)})_+ Z_i G_i L_i^{\alpha_i} - (1+\psi_i)wL_i, \tag{27}$$

where $[x]_+ = \max\{x, 0\}$, so full capacity $Y_i = e_i Z_i G_i L_i^{\alpha_i}$ is operated whenever $p_i(\omega) > r_i \tau_{j(\ell)}(\omega)$. Writing expected net unit revenue per efficiency unit as

$$m_i(\phi_i; \ell) = \mathbb{E}_t^{\theta}\Big[\big(p_i(\ell, \omega) - r_i \tau_{j(\ell)}(\omega)\big)_+ G_i(\phi_i; \ell, \omega)\Big], \tag{28}$$

the entrant maximizes expected land-bid surplus:

$$R_i(\ell) = \max_{\substack{0 \le \phi_i < 1 \\ L_i \ge 0}} \Big\{ Z_i(\ell) m_i(\phi_i; \ell) L_i^{\alpha_i} - (1 + \psi_i(\ell)) w(\ell) L_i - \Psi_i(\phi_i; w(\ell), \ell) \Big\}. \tag{29}$$

First-order optimization in L_i yields the closed-form hiring rule and profiled scalar innovation problem:

$$L_i^*(\phi_i; w, \ell) = \left[\frac{\alpha_i Z_i(\ell) m_i(\phi_i; \ell)}{(1 + \psi_i(\ell)) w(\ell)} \right]^{\frac{1}{1-\alpha_i}}, \tag{30}$$

$$R_i(w, \ell) = \max_{\phi_i \in [0,1)} \Big\{ \Lambda_i(\ell) w(\ell)^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi_i; \ell)^{\frac{1}{1-\alpha_i}} - \Psi_i(\phi_i; w(\ell), \ell) \Big\}, \tag{31}$$

with $L_i^*(\phi_i; w, \ell) = 0$ if $m_i(\phi_i; \ell) = 0$ and $\Lambda_i(\ell) = (1 - \alpha_i) Z_i(\ell) [\alpha_i Z_i(\ell)/(1 + \psi_i(\ell))]^{\alpha_i/(1-\alpha_i)}$. On $\phi_i \in (0, 1)$, Ψ_i is strictly convex with $\partial^2_{\phi\phi} \Psi_i = 2w(\ell)\chi_i/(1 - \phi_i)^3 > 0$ and diverges as $\phi_i \uparrow 1$. By the envelope theorem, an interior research optimum $\tilde{\phi}_i(w, \ell) \in (0, 1)$ satisfies

$$Z_i(\ell) L_i^*(\tilde{\phi}_i; w, \ell)^{\alpha_i} \partial_\phi m_i(\tilde{\phi}_i; \ell) = \frac{w(\ell)\chi_i}{\big(1 - \tilde{\phi}_i(w, \ell)\big)^2}, \tag{32}$$

and for $\chi_i > \bar{\chi}_{it}$ the active objective is strictly concave on $(0, 1)$ with at most one interior maximizer. Dividing (31) by $w(\ell)$ with baseline profitability index $\pi_i^0(w, \ell) \equiv \Lambda_i(\ell)[m_i(0; \ell)/w(\ell)]^{1/(1-\alpha_i)}$ and discovery gain $\Gamma_i(\phi_i; \ell) \equiv [m_i(\phi_i; \ell)/m_i(0; \ell)]^{1/(1-\alpha_i)} - 1$ gives

$$\frac{R_i(w, \ell)}{w(\ell)} = \pi_i^0(w, \ell) + \max\left\{ 0, \max_{\phi_i \in (0,1)} \left[\pi_i^0(w, \ell) \Gamma_i(\phi_i; \ell) - f_i(\ell) - \frac{\chi_i}{1 - \phi_i} \right] \right\}. \tag{33}$$

Comparing $\tilde{\phi}_i(w, \ell)$ with zero research defines a unique reservation threshold $\underline{\pi}_i(\ell) \in (0, \infty]$ (Appendix A.6.1): innovation is active ($\phi_i^*(w, \ell) = \tilde{\phi}_i(w, \ell) > 0$) if and only if $\pi_i^0(w, \ell) > \underline{\pi}_i(\ell)$, and zero otherwise. Transport improvements $T_t(\ell)$ raise net unit prices $p_i(\ell, \omega) - r_i \tau_{j(\ell)}(\omega)$ while coalfield proximity lowers $\xi(\ell)$, pushing $\pi_M^0(w, \ell)$ above $\underline{\pi}_M(\ell)$ in industrializing counties. Since $\pi_i^0(w, \ell) \propto w(\ell)^{-1/(1-\alpha_i)}$, higher contract wages weakly lower $\phi_i^*(w, \ell)$ and strictly lower $L_i^*(\phi_i^*(w, \ell); w, \ell)$ (Appendix A.6.1).

Land competition and spatial sorting.— Competitive bidding assigns each site to the sector with the higher expected bid rent:

$$R(\ell) = \max\{R_A(\ell),\, R_M(\ell)\}, \qquad e_i(\ell) = \begin{cases} 1 & \text{if } R_i(\ell) = R(\ell), \\ 0 & \text{otherwise.} \end{cases} \tag{34}$$

Because $\alpha_A < \alpha_M$ and $\Lambda_i(\ell) \propto Z_i(\ell)^{1/(1-\alpha_i)}(1 + \psi_i(\ell))^{-\alpha_i/(1-\alpha_i)}$, the sign of $R_A(w, \ell) - R_M(w, \ell)$ is governed by the elasticity- and wedge-adjusted comparative-advantage profile $\frac{1}{1-\alpha_M} \log Z_M(\ell) - \frac{1}{1-\alpha_A} \log Z_A(\ell) - \frac{\alpha_M}{1-\alpha_M} \log(1+\psi_M(\ell))$. Under Condition 2 of Section III.G, which requires this profile to rise steeply enough to dominate the spatial variation of wages, net prices, and research activity, $R_A(w, \ell) - R_M(w, \ell)$ changes sign exactly once, at an interior boundary $b_{ct} \in (0, 1)$, assigning $[0, b_{ct}]$

to agriculture and $(b_{ct}, 1]$ to manufacturing; in the frictionless benchmark of Appendix A.6.3, where wages and prices are spatially uniform, an increasing profile is by itself sufficient for at most one crossing. At the winning plan, local employment, worker participation, output, and wedge transfers satisfy

$$n(\ell) = \sum_{i \in \{A,M\}} e_i(\ell) L_i(\ell) + L_T(\ell), \tag{35}$$

$$\bar{u}_t = \mathbb{E}_t^\theta \left[\frac{w(\ell) + b_t - \gamma p_A(\ell)}{\mathcal{P}(p_A(\ell), p_M(\ell))} \right], \tag{36}$$

and realized output, innovation expenditure, and wedge transfers satisfy

$$Y_i = e_i Z_i G_i L_i^{\alpha_i}, \qquad \Psi_i = \Psi_i(\phi_i^*(\ell); w(\ell), \ell), \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i. \tag{37}$$

PROPOSITION 2: *Fix the inherited physical state, dated environment, and a common rule for selecting conditional private equilibria that depends only on the inherited state, the dated environment, the investment profile, and the date-t discoveries. Conditional on an investment profile, current household, firm, and carrier decisions, market prices, and county current dividends are invariant to θ. If two belief specifications select the same profile, the same private discoveries and sectoral shocks generate the same current allocation and next physical state.* [Proof →]

Because $\mathbb{E}_t^\theta$ in (11) reweights only future aggregate shocks $\mathbf{z}_{t+1}$ while leaving the conditional distribution of date-t private discoveries $G_{it}(\ell, \omega)$ unchanged, every within-period expectation conditional on $(\Omega_t, \mathbf{x}_t)$ reduces to rational expectations ($\mathbb{E}_t^\theta[\cdot] = \mathbb{E}_t[\cdot]$ in (18), (28), and (36)); diagnostic expectations affect the allocation solely through county infrastructure investment $\mathbf{x}_t^\theta$.[9]

After early-stage plans are committed, the active producer's realized discovery $G_{it}(\ell) \geq 1$ enters the public knowledge pool, updating local pre-diffusion technology (Cai et al., 2022):

$$A_{it}^+(\ell) = A_{it}(\ell)\big[1 + e_{it}(\ell)(G_{it}(\ell) - 1)\big]. \tag{38}$$

The inactive sector ($e_{it}(\ell) = 0$) retains $A_{it}(\ell)$ at this step, and transitory aggregate gaps $\mathbf{z}_t$ do not enter permanent knowledge. Because spillovers attenuate exponentially with distance (Keller, 2002; Berkes et al., 2025), the technology frontier available at ℓ is the pointwise spatial supremum

$$F_{it}(\ell) = \sup_{r \in [0,1]} e^{-\delta|\ell - r|} A_{it}^+(r), \qquad \delta > 0, \tag{39}$$

which is uniquely defined everywhere under pointwise entrant optimality and land sorting, and finite almost surely because the Gaussian fields generating the discoveries in (24) have continuous sample paths (Online Appendix Lemma S.C.1). Both sectors then close a fraction of their gap to $F_{it}(\ell)$ via the Nelson and Phelps (1966) catch-up law at a speed governed by installed route capacity (Cinnirella et al., 2024; Berkes et al., 2026):

$$A_{i,t+1}(\ell) = A_{it}^+(\ell) + (1 - e^{-\lambda_t(\ell)})[F_{it}(\ell) - A_{it}^+(\ell)], \tag{40}$$

$$\lambda_t(\ell) = \lambda_0 \left[\frac{T_t(\ell)}{T_{t_0}(\ell)} \right]^{1-\alpha_T}, \qquad \lambda_0 > 0, \tag{41}$$

where $T_{t_0}(\ell)$ is the initial reference network (half-life $\log(2)/\lambda_0$ years).

[9] Keeping within-period Bernoulli–Pareto discovery probabilities physical isolates the intertemporal network-investment channel, where construction must be committed a full period before operating returns realize.

PROPOSITION 3: *For bounded positive post-discovery knowledge and finite nonnegative adoption rates, $A_{it}^{+}(\ell) \le A_{i,t+1}(\ell) \le F_{it}(\ell)$. Holding discoveries and reference levels fixed, more inherited capacity weakly raises next-period productivity. If no further discoveries occur and geography is fixed, the available frontier remains unchanged and*

$$F_{it}(\ell) - A_{i,t+h}(\ell) = [F_{it}(\ell) - A_{it}^{+}(\ell)] \exp\left[-\sum_{k=t}^{t+h-1} \lambda_k(\ell)\right], \qquad h \ge 1. \tag{42}$$

[Proof →]

Localized research in industrial hubs pushes out $F_{it}(\ell)$, while county transport investments raise $\lambda_t(\ell)$ and propagate discoveries across space. Because neither private entrants nor county promoters internalize how $T_t(\ell)$ accelerates future technology absorption, (40)–(41) generate a dynamic spatial adoption externality. Online Appendix S.C.2 proves that the frontier is finite with finite moments of every order below a_i and gives the exact conditional mean of next-period knowledge.

E Transport and Distribution

Moving goods over land requires both durable route infrastructure (Redding and Turner, 2015; Donaldson, 2018) and operating carriage services priced in local transport markets (Brancaccio et al., 2020). Transport firms combine installed route capacity $T(\ell, t)$ with hired labor $L_T(\ell, t)$ committed at the local contract wage $w(\ell, t)$ before private discoveries, selling operating services ex post at the county toll τ_j:

$$Y_T(\ell, t) = Z_T(t) L_T(\ell, t)^{\alpha_T} T(\ell, t)^{1-\alpha_T}, \qquad Z_T(t) > 0, \quad 0 < \alpha_T < 1. \tag{43}$$

Each county $\mathcal{C}_j$ operates a competitive operating-service market at a common toll $\tau_{j(\ell)}$: services produced anywhere in $\mathcal{C}_j$ meet distribution or transit demand anywhere within $\mathcal{C}_j$ without additional cost, whereas services cannot be dispatched across county boundaries.

Shipping good i across space incurs both physical transit degradation at rate $\kappa(\ell, t)$ and r_i units of county operating services per unit of distance at toll $\tau_{j(\ell)}(t)$, yielding the effective ad valorem transport-cost rate $\tilde{\kappa}_i(\ell, t)$ and spatial no-arbitrage law for signed trade flow $H_i(\ell, t)$ (positive rightward):

$$\tilde{\kappa}_i(\ell, t) = \kappa(\ell, t) + \frac{r_i \tau_{j(\ell)}(t)}{p_i(\ell, t)}, \tag{44}$$

$$|\partial_\ell p_i| \le \kappa p_i + r_i \tau_{j(\ell)}, \qquad \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)}) \operatorname{sign}(H_i) \quad \text{if } H_i \ne 0. \tag{45}$$

Because (45) bounds the price gradient by the bounded quantity $\kappa p_i + r_i \tau_{j(\ell)}$, spot prices $p_i(\cdot, t)$ are Lipschitz, hence absolutely continuous, across space: a step in $\tau_{j(\ell)}$ across a county border changes the slope bound and kinks $\partial_\ell p_i$ without creating a price jump. Installed infrastructure lowers physical transit losses and bilateral iceberg delivery factors $d_t(\ell, r)$ exponentially:

$$\kappa(\ell, t) = \kappa_0 e^{-T(\ell, t)}, \qquad \kappa_0 > 0, \quad T(\ell, t) > 0, \tag{46}$$

$$d_t(\ell, r) = \exp\left\{\kappa_0 \left| \int_r^\ell e^{-T_t(u)}\, du \right|\right\}. \tag{47}$$

Each unit of marketed output $Y_i(\ell, t)$ requires r_i units of county service for local origin distribution and r_i units per unit of distance shipped ($r_M = 1$, $r_A > 0$), so county-wide service demand

and operating capacity are

$$D_j(\omega) = \int_{\mathcal{C}_j} \sum_{i\in\{A,M\}} r_i(Y_i(\ell,\omega) + |H_i(\ell,\omega)|)\, d\ell, \qquad K_j = \int_{\mathcal{C}_j} Y_T(\ell,t)\, d\ell. \tag{48}$$

Carrier optimization and county service clearing.— Before private discoveries, a competitive carrier at $\ell \in \mathcal{C}_j$ chooses $L_T(\ell)$ taking $T(\ell)$, $w(\ell)$, and the distribution of τ_j as given:

$$\max_{L_T\geq 0} \mathbb{E}_t^\theta\left[\tau_j\, Y_T - wL_T\right], \qquad Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}, \tag{49}$$

$$w = \alpha_T Z_T L_T^{\alpha_T-1} T^{1-\alpha_T}\, \mathbb{E}_t^\theta[\tau_j] \quad\Longleftrightarrow\quad L_T^*(w,\ell) = \left[\frac{\alpha_T Z_T\, \mathbb{E}_t^\theta[\tau_{j(\ell)}]}{w(\ell)}\right]^{\frac{1}{1-\alpha_T}} T(\ell), \tag{50}$$

where $\mathbb{E}_t^\theta[\tau_j] = \mathbb{E}_t[\tau_j]$ conditional on $(\Omega_t, \mathbf{x}_t)$ by Proposition 2. After private discoveries realize, predetermined county capacity K_j clears against county service demand D_j via complementary slackness, and carrier operating surplus accrues to the household portfolio:

$$D_j \leq K_j, \qquad \tau_j \geq 0, \qquad \tau_j(K_j - D_j) = 0, \tag{51}$$

$$\Pi_T(\ell,t) = \tau_{j(\ell)}(t) Y_T(\ell,t) - w(\ell,t) L_T(\ell,t). \tag{52}$$

F County Infrastructure Companies

Unlike centralized national network planners (Fajgelbaum and Schaal, 2020; Allen and Arkolakis, 2022), historical British turnpikes, canals, and railways were promoted by local county companies through parliamentary acts (Bogart, 2014). Each county $j \in \{1,\ldots,J\}$ has an infrastructure company holding a long-lived claim on land-lease rents inside $\mathcal{C}_j$. Roads, waterways, and railways accumulate into a composite route stock $T(\ell,t)$ under one-period time-to-build (Kydland and Prescott, 1982): before private discoveries at date t, company j commits completion intensity $x_{jt} \in [0, \bar{x}_{jt}]$ within the resource-feasible statutory box $\mathcal{X}_t = \prod_j [0,\bar{x}_{jt}] \subseteq \prod_j [0, o_{jt}]$ ($o_{jt} \in \{0,1\}$), and completed increments enter operating route capacity at date $t+1$:

$$x_{jt} \in [0,\bar{x}_{jt}], \qquad \operatorname{supp} \iota_{jt} \subseteq \mathcal{C}_j, \tag{53}$$

$$T_{t+1}(\ell) = T_t(\ell) + \sum_{j=1}^{J} x_{jt}\iota_{jt}(\ell), \qquad T_0(\ell) \geq \underline{T} > 0. \tag{54}$$

Construction uses manufactured goods with quadratic adjustment costs on the county engineering profile (e_{jt}, k_{jt}) ($\operatorname{supp} k_{jt} \subseteq \mathcal{C}_j$, $e_{jt} > 0$):

$$I_t(\ell;\mathbf{x}_t) = \sum_{j=1}^{J} m_I e_{jt} k_{jt}(\ell)\left(x_{jt} + \frac{\chi_I}{2} x_{jt}^2\right), \qquad m_I > 0, \quad \chi_I > 0, \tag{55}$$

purchased at the realized spot price $p_M(\ell,t,\omega)$. Full-package physical increments grow at two percent per year, $\iota_{jt} = 1.02^{t-t_0}\iota_{j,t_0}$; after the last observed schedule year $t = 1881$, each county repeats its 1881 opportunity $o_{j,1881}$ and engineering profile $(e_j k_j)_{1881}$ at the same (m_I, χ_I) while physical increments continue to grow as $\iota_{j,1881+h} = 1.02^h \iota_{j,1881}$.

Promoter optimization, marginal-q, and stage Nash equilibrium.— Taking rival counties' intensities $\mathbf{x}_{-j,t}$ as given, company j's current payoff is expected county land rent net of spot manufacturing construction expenditure:

$$R_{jt}(\mathbf{x};\Omega_t) = \mathbb{E}_t^\theta\left[\int_{\mathcal{C}_j} \{R_t(\ell;\mathbf{x},\Omega_t) - p_{Mt}(\ell,\omega;\mathbf{x},\Omega_t) I_t(\ell;\mathbf{x})\}\, d\ell\right]. \tag{56}$$

By Proposition 2, current returns conditional on $(\Omega_t, \mathbf{x})$ depend only on physical private discoveries ($\mathbb{E}_t^\theta = \mathbb{E}_t$ in (56)), whereas future aggregate shocks enter the continuation value under $\mathbb{E}_t^\theta$ with perceived next-period reference state $\bar{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^\theta)$ from (13) (while the realized physical state at date $t+1$ resets to $\bar{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$):

$$\Pi_{jt}^\theta(\mathbf{x}; \Omega_t) = R_{jt}(\mathbf{x}; \Omega_t) + \beta \mathbb{E}_t^\theta\left[V_{j,t+1}^\theta(\Omega_{t+1})\right], \qquad V_{jt}^\theta(\Omega_t) = \Pi_{jt}^\theta\left(\mathbf{x}_t^\theta(\Omega_t); \Omega_t\right), \tag{57}$$

$$x_{jt}^\theta(\Omega_t) \in \arg\max_{0 \le u \le \bar{x}_{jt}} \Pi_{jt}^\theta(u, \mathbf{x}_{-j,t}^\theta(\Omega_t); \Omega_t), \qquad j = 1, \ldots, J. \tag{58}$$

If the pure-strategy Nash set $\mathcal{N}_t^\theta$ contains multiple profiles, parliamentary coordination selects the profile maximizing $\sum_{j=1}^J \Pi_{jt}^\theta$ (with deterministic tie-breaking). Because the aggregate spatial knowledge frontier and future knowledge trajectories $(A_{As}, A_{Ms})_{s \ge t+1}$ reflect atomistic firm discoveries and frontier adoption across the national continuum, each county company takes the aggregate knowledge path as external to its own corridor completion choice x_{jt} when evaluating continuation values. Where differentiable, county j's perceived marginal continuation value q_{jt}^θ (differentiating with respect to own route capacity T_{t+1} at the given knowledge path) and marginal construction cost c_{jt} satisfy (Hayashi, 1982):

$$q_{jt}^\theta(\mathbf{x}) = \beta \partial_{x_j} \mathbb{E}_t^\theta[V_{j,t+1}^\theta(\Omega_{t+1})], \qquad c_{jt}(\mathbf{x}) = -\partial_{x_j} R_{jt}(\mathbf{x}; \Omega_t), \tag{59}$$

$$q_{jt}^\theta(\mathbf{x}_t^\theta) = c_{jt}(\mathbf{x}_t^\theta) \tag{60}$$

at an interior best response $x_{jt}^\theta \in (0, \bar{x}_{jt})$, with complementary slackness at $\{0, \bar{x}_{jt}\}$. Because every engineering profile lies inside its own county ($\operatorname{supp} k_{kt} \subseteq \mathcal{C}_k$), county j pays only for its own construction, and (56) splits into gross county land rent and a construction bill, $R_{jt}(\mathbf{x}; \Omega_t) = \widetilde{R}_{jt}(\mathbf{x}) - m_I(x_{jt} + \frac{\chi_I}{2} x_{jt}^2) D_{jt}^I(\mathbf{x})$, with

$$\widetilde{R}_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} R_t(\ell; \mathbf{x}, \Omega_t)\, d\ell, \qquad D_{jt}^I(\mathbf{x}) = \int_{\mathcal{C}_j} \mathbb{E}_t[p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t)]\, e_{jt} k_{jt}(\ell)\, d\ell > 0, \tag{61}$$

where D_{jt}^I is the county construction-price index. Differentiating gives the marginal construction cost c_{jt} and, collecting the marginal benefits of capacity, the gross marginal value $\widetilde{q}_{jt}^\theta$:

$$c_{jt} = m_I(1 + \chi_I x_{jt}) D_{jt}^I - \partial_{x_j} \widetilde{R}_{jt} + m_I\Big(x_{jt} + \frac{\chi_I}{2} x_{jt}^2\Big) \partial_{x_j} D_{jt}^I,$$

$$\widetilde{q}_{jt}^\theta = q_{jt}^\theta + \partial_{x_j} \widetilde{R}_{jt} - m_I\Big(x_{jt} + \frac{\chi_I}{2} x_{jt}^2\Big) \partial_{x_j} D_{jt}^I.$$

Condition (60) therefore reads $\widetilde{q}_{jt}^\theta = m_I(1 + \chi_I x_{jt}) D_{jt}^I$, and with complementary slackness it becomes the clipped investment rule

$$x_{jt}^\theta = \min\left\{\bar{x}_{jt},\ \max\left\{0,\ \frac{1}{\chi_I}\left[\frac{\widetilde{q}_{jt}^\theta(\mathbf{x}_t^\theta)}{m_I D_{jt}^I(\mathbf{x}_t^\theta)} - 1\right]\right\}\right\}. \tag{62}$$

A county builds only when the marginal value of capacity per unit of construction outlay exceeds one, and lower adjustment convexity χ_I makes buildout more responsive to that ratio. Diagonal dominance of own adjustment-cost curvature over cross-county general-equilibrium spillovers yields a unique stage equilibrium (Vives, 1999):

PROPOSITION 4: *Fix $(\Omega_t, \mathcal{E}_t)$ and $\theta \ge 0$ on a nonempty compact box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ where every profile has a unique admissible private allocation and continuous county payoffs Π_{jt}^θ. If each Π_{jt}^θ is*

concave in x_{jt}, $\mathcal{N}_t^\theta$ is nonempty and compact. If, in addition, Π_{jt}^θ is C^2 and there exist $m_j > 0$, $L_{jk} \geq 0$, and $v_j > 0$ on $\mathcal{X}_t$ such that

$$-\partial_{jj}\Pi_j^\theta \geq m_j > 0, \qquad |\partial_{jk}\Pi_j^\theta| \leq L_{jk} \quad (k \neq j), \tag{63}$$

$$\gamma_I = \max_j \frac{\sum_{k\neq j} L_{jk} v_k}{m_j v_j} < 1, \tag{64}$$

then the county Nash equilibrium is unique and simultaneous best responses converge to it from every initial profile in $\mathcal{X}_t$. If instead Π_{jt}^θ is C^1 and the pseudogradient $\mathbf{g}_t^\theta = (\partial_{x_j}\Pi_{jt}^\theta)_j$ is strongly monotone on $\mathcal{X}_t$, $\sum_j (x_j - y_j)[g_{jt}^\theta(\mathbf{y}) - g_{jt}^\theta(\mathbf{x})] \geq \mu\|\mathbf{x}-\mathbf{y}\|^2$ for some $\mu > 0$, the Nash equilibrium is likewise unique. In either case, if Π_{jt}^θ is jointly measurable in Ω_t, the selection $\Omega_t \mapsto \mathbf{x}_t^\theta(\Omega_t)$ is measurable. [Proof →]

By Proposition 2, θ enters the dated allocation solely through $q_{jt}^\theta(\mathbf{x})$ in (60). Differentiating the diagnostic kernel (11)–(12) holding $V_{j,t+1}^\theta$ fixed shows that current news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$ shifts perceived marginal-q directly and—together with $dq_{jt}^\theta/d\theta = \partial_\theta q_{jt}^\theta\big|_{V_{t+1}} + \beta\mathbb{E}_t^\theta[\partial_{x_j}\partial_\theta V_{j,t+1}^\theta]$—shifts interior buildout ($\mathcal{I}_t = \{j : x_{jt}^\theta \in (0, \bar{x}_{jt})\}$) according to

$$\begin{aligned} \partial_\theta q_{jt}^\theta(\mathbf{x})\Big|_{V_{t+1}} &= \beta\,\mathrm{Cov}_t^\theta\Big(\partial_{x_j} V_{j,t+1}^\theta(\Omega_{t+1}),\, \mathbf{z}_{t+1}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\rho}\boldsymbol{\varepsilon}_t\Big), \\ \frac{d\mathbf{x}_{t,\mathcal{I}_t}^\theta}{d\theta} &= -\big[D_{\mathbf{x}_{\mathcal{I}_t}}(\mathbf{q}_{t,\mathcal{I}_t}^\theta - \mathbf{c}_{t,\mathcal{I}_t})\big]^{-1}\frac{d\mathbf{q}_{t,\mathcal{I}_t}^\theta(\mathbf{x}_t^\theta)}{d\theta}. \end{aligned} \tag{65}$$

Because each county company captures only local land rents inside $\mathcal{C}_j$ and takes aggregate knowledge evolution as external, its private marginal-q omits three positive social returns to route capacity: (i) cross-county market-access and transit savings $\sum_{k\neq j}\partial_{x_j}\Pi_{kt}^0 > 0$ (Fogel, 1964; Krugman, 1991; Donaldson and Hornbeck, 2016; Fajgelbaum and Schaal, 2020; Allen and Arkolakis, 2022); (ii) the allocative-efficiency gain $\int_0^1 \psi_M(\ell)w(\ell)\partial_{x_j}L_{Mt}(\ell)\,d\ell > 0$ from reallocating labor into wedge-distorted manufacturing (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024); and (iii) faster spatial knowledge diffusion (Nelson and Phelps, 1966; Lucas, 1988; Romer, 1990; Aghion and Howitt, 1992; Mokyr, 2005; Desmet and Rossi-Hansberg, 2014), since higher $T_s(\ell)$ at $s \geq t+1$ raises next-period knowledge at given discoveries by

$$\frac{\partial A_{i,s+1}(\ell)}{\partial T_s(\ell)} = (1-\alpha_T)\frac{\lambda_s(\ell)}{T_s(\ell)}e^{-\lambda_s(\ell)}\big[F_{is}(\ell) - A_{is}^+(\ell)\big] \geq 0, \tag{66}$$

which is strictly positive wherever the available frontier exceeds local technology. Rational promoters ($\theta = 0$) therefore omit three positive social returns and, to that extent, underinvest relative to a planner who internalizes them, whereas favorable manufacturing and transport news that raises the perceived continuation value, $q_{jt}^\theta > q_{jt}^0$ in (65), expands irreversible network buildout under diagnostic expectations ($\theta > 0$).

G Equilibrium and State Transitions

Fix $(\Omega_t, \mathcal{E}_t)$ and a feasible county investment profile $\mathbf{x}_t \in \mathcal{X}_t$. Before private discoveries, pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w)$ clear local and national labor markets, while ex-post goods

flows $H_i(\ell, t)$ satisfy spatial conservation with closed borders:

$$n = \sum_{i \in \{A,M\}} e_i L_i + L_T, \tag{67}$$

$$\int_0^1 n \, d\ell = \bar{L}_t, \tag{68}$$

$$\partial_\ell H_i = Y_i - n c_i - e_i \frac{\Psi_i}{p_i} - I_i - \kappa |H_i|, \qquad H_i(0,t) = H_i(1,t) = 0, \quad i \in \{A, M\}. \tag{69}$$

Realized producer net profit Π_F, contractual land rent R, origin carrier surplus Π_T, manufacturing wedge rebates Π_ψ, net land rent R^{net}, and the common portfolio dividend b_t are

$$\Pi_F = \sum_i e_i \Big\{ (p_i - r_i \tau_{j(\ell)}) Y_i - (1 + \psi_i) w L_i - \Psi_i(\phi_i; w, \ell) \Big\} - R, \tag{70}$$

$$R = \sum_i e_i R_i, \qquad \Pi_T = \tau_{j(\ell)} Y_T - w L_T, \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i, \tag{71}$$

$$R^{\text{net}} = R - p_M I, \tag{72}$$

$$b_t = \frac{1}{\bar{L}_t} \int_0^1 \{ R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi \} \, d\ell. \tag{73}$$

Free entry implies $\mathbb{E}_t[\Pi_F] = 0$, while ex-post windfalls $\Pi_F(\ell, \omega)$ accrue to households. Carrier tolls and wedge rebates are pecuniary transfers, whereas research goods Ψ_i/p_i, construction I, and iceberg degradation $\kappa|H_i|$ absorb physical output.

DEFINITION 1 (Conditional dated equilibrium): Given $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$, a conditional dated equilibrium consists of pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w, R, I)$ and a measurable post-discovery family $(Y_i, c_i, p_i, \tau_j, H_i, b_t)$ such that: (i) entrants solve (29), land is assigned by (34), and utilization solves (27); (ii) households choose residence by (17) and consumption by (16); (iii) carriers solve (49) and county services clear by (51); and (iv) for almost every realization, spatial no-arbitrage, goods and labor clearing, and (73) hold. At absolutely continuous versions of $p_i(\cdot, \omega)$ and $H_i(\cdot, \omega)$ (Lipschitz, by (45)) and continuous $w(\cdot)$ on $[0, 1]$ (with $p_A(1, t) = 1$), entrant optimality (29) and pure-site sorting (34) hold at every $\ell \in [0, 1]$ (breaking isolated ties to the left sector), differential flow and no-arbitrage relations hold almost everywhere, subsistence is feasible statewise, and expectations in agent objectives are finite.

In a competitive continuum economy (Aumann, 1966), individual entrants and carriers evaluate alternative plans at the quoted price and contract-wage functions (w, p, τ) without recomputing local market clearing, while equilibrium announcements jointly clear labor, goods, and county service markets.

PROPOSITION 5: *For every realization with absolutely continuous prices and flows satisfying local labor clearing, spatial goods balance, spatial no-arbitrage, county service clearing* (51)*, and closed national boundaries, the ledgers* (70)*–*(73) *imply the aggregate household budget identity. By Walras' Law, there is no additional independent national resource constraint.* [Proof →]

Building on classical general equilibrium analysis (Arrow and Hahn, 1971) and the nonlinear spatial operator representation of Allen and Arkolakis (2014) and Allen et al. (2024), substituting the closed-form household demand and mobility rules (16)–(18), producer hiring and research rules (30)–(32), land sorting (34), and carrier hiring (50) into market clearing (67)–(73) reduces the conditional dated equilibrium to a fixed point $(w, p, \tau) = \mathcal{T}_t(w, p, \tau)$ on the quote domain $\mathcal{W}_t \times$

$\mathcal{P}_t \times \mathcal{M}_t$ of positive continuous wages $w \in C([0,1];[\underline{w}_t, \overline{W}_t])$, statewise continuous spot prices $p = (p_A, p_M)$ ($p_A(1,\omega) = 1$), and nonnegative county tolls $\tau = (\tau_1, \ldots, \tau_J)$ with positive expected tolls $\mathbb{E}_t[\tau_j] \in [\underline{\tau}_{jt}, \overline{\tau}_{jt}]$ ($\underline{\tau}_{jt} > 0$):

$$\begin{aligned} \widetilde{w}(\ell) = (\mathcal{T}_t^w(w,p,\tau))(\ell) &= \frac{\bar{u}_t(w,p,\tau) - \mathbb{E}_t\big[(b_t(\omega;w,p,\tau) - \gamma p_A(\ell,\omega))/\mathcal{P}(\ell,\omega)\big]}{\mathbb{E}_t\big[1/\mathcal{P}(\ell,\omega)\big]}, \\ (\widetilde{p}(\cdot,\omega), \widetilde{\tau}(\omega)) &= \mathcal{T}_t^p(w,p,\tau;\omega), \\ \bar{L}_t &= \int_0^1 \Big[\sum_{i \in \{A,M\}} e_i(\ell) L_i^*(\phi_i^*(\widetilde{w}(\ell),\ell);\widetilde{w}(\ell),\ell) + L_T^*(\widetilde{w}(\ell),\ell) \Big] d\ell, \end{aligned} \tag{74}$$

where national labor clearing at $\widetilde{w} = \mathcal{T}_t^w(w,p,\tau)$ pins down $\bar{u}_t(w,p,\tau)$ and $\mathcal{T}_t^p(w,p,\tau;\omega)$ maps predetermined capacities and construction demand $I_t(\cdot;\mathbf{x}_t)$ into the unique spot prices and county tolls supporting the strictly concave Negishi-weighted log-surplus spatial dispatch problem under (45), (51), and (69)–(73).

Existence and uniqueness of the conditional dated equilibrium rest on three primitive conditions:

1. *(Resource feasibility and finite innovation tails.)* On the contract-wage collar $[\underline{w}_t, \overline{W}_t]$ where zero-research labor demand crosses $\bar{L}_t$, net deliverable agricultural capacity exceeds subsistence ($\underline{Y}_{At} > \gamma \bar{L}_t$), net deliverable manufacturing capacity exceeds construction demand ($\underline{Y}_{Mt} > \int_0^1 I_t(\ell;\mathbf{x}_t)\, d\ell$), county carrier capacity covers distribution demand uniformly on $\mathcal{X}_t$, and $a_i > 1/(1-\alpha_i)$ for $i \in \{A, M\}$.

2. *(Innovation-cost convexity and Ricardian single-crossing.)* $\chi_i > \bar{\chi}_{it}$ (Appendix A.6.1) so active research (31) is strictly concave on $(0,1)$, $\pi_i^0(w(\ell),\ell) - \underline{\pi}_i(\ell)$ crosses zero transversally at most finitely many times when $f_i > 0$, and $\partial_\ell[\frac{1}{1-\alpha_M} \log Z_{Mt}(\ell) - \frac{1}{1-\alpha_A} \log Z_{At}(\ell) - \frac{\alpha_M}{1-\alpha_M} \log(1+\psi_M(\ell))] > 0$ dominates spatial factor-price and toll gradients on $\mathcal{U}_t$ so the bid-rent difference $R_A(\ell) - R_M(\ell)$ satisfies $R_A(0) > R_M(0)$, $R_A(1) < R_M(1)$, and $\partial_\ell[R_A(b_{ct}) - R_M(b_{ct})] < 0$ at a unique cutoff $b_{ct} \in (0,1)$.

3. *(Norm contraction.)* In a weighted Banach norm $\|\cdot\|_{\mathcal{U}_t}$, the Jacobian $\mathcal{D}_t(u)$ of $\mathcal{T}_t$ satisfies $\sup_{u \in \mathcal{U}_t} \|\mathcal{D}_t(u)\|_{\mathcal{U}_t} \le \bar{\rho}_t < 1$, so direct local labor-demand elasticities $1/(1-\alpha_i)$ dominate indirect spatial trade, toll, and dividend feedbacks.

PROPOSITION 6: *Consider the economy of Definition 1 with $0 < \alpha_A < \alpha_M < 1$, $\alpha_T \in (0,1)$, bounded positive piecewise-continuous (A_{At}, A_{Mt}), $\eta \in (0,1)$, $\gamma > 0$, and $\phi_{it}(\ell) \in [0,1)$.*

1. (Existence, active-research threshold, and pure-site sorting.) *Under Conditions 1 and 2, every $\mathbf{x}_t \in \mathcal{X}_t$ admits a regular conditional dated equilibrium in $\mathcal{U}_t$. At the continuous price and wage versions, optimal research $\phi_{it}^*(\ell) \in [0,1)$ is uniquely determined at every $\ell \in [0,1]$ by the threshold rule (32)–(33), and land partitions at a unique cutoff $b_{ct} \in (0,1)$ with agriculture on $[0, b_{ct}]$ and manufacturing on $(b_{ct}, 1]$, uniquely determining $A_{it}^+(\ell,\omega)$ and the almost surely finite frontier $F_{it}(\ell,\omega)$ (Online Appendix Lemma S.C.1) everywhere on $[0,1]$.*

2. (Global uniqueness.) *Under Condition 3, the regular conditional dated equilibrium is globally unique for each $\mathbf{x}_t \in \mathcal{X}_t$.*

3. (Constructive verification from primitives.) *Around the closed-form frictionless spatial benchmark, backward county food-anchor induction and regular perturbation verify a stock-uniform wage collar $[\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}]$ and Conditions 1–3 directly from primitives across all admissible county infrastructure paths $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$ (Appendix A.6.3).* [Proof →]

DEFINITION 2 (Dynamic equilibrium): Given Ω_0, diagnosticity $\theta \geq 0$, $\{\mathcal{E}_t\}_{t\geq 0}$, and the physical shock laws, a dynamic equilibrium is an adapted process of states, investments, conditional dated equilibria, and county values such that: (i) each dated allocation satisfies Definition 1; (ii) county policies satisfy (58) under values (57) and perceived forecast-reference update (13), taking the aggregate knowledge trajectory (A_{At}, A_{Mt}) as given; and (iii) realized states follow (10), (38)–(40), and (54) with observed reference $\bar{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$.

Substituting the unique conditional dated equilibrium rent and manufacturing price schedules from Proposition 6 into (56) yields county j's reduced stage payoff

$$R_{jt}(\mathbf{x}_t; \Omega_t) = \widetilde{R}_{jt}(\mathbf{x}_t) - m_I \left(x_{jt} + \frac{\chi_I}{2} x_{jt}^2 \right) D_{jt}^I(\mathbf{x}_t),$$

with gross county land rent $\widetilde{R}_{jt}$ and construction-price index D_{jt}^I from (61), county mean route stock $\overline{T}_{j,t+1} = \overline{T}_{jt} + d_{jt} x_{jt}$ ($\overline{T}_{jt} = |\mathcal{C}_j|^{-1} \int_{\mathcal{C}_j} T_t(\ell)\, d\ell$ and $d_{jt} = |\mathcal{C}_j|^{-1} \int_{\mathcal{C}_j} \iota_{jt}(\ell)\, d\ell$ by (54)) and knowledge transition (38)–(40). After 1881, household mass remains at $\bar{L}_{1881+h} = \bar{L}_{1881}$ while the 1881 project schedule repeats with physical package growth $g = 1.02$. Write $\bar{Z}_T \equiv A_T / \mathbb{E}_\pi[e^{z_T}]$, $\varepsilon = e^{-z_{Tt}/\alpha_T} / (\bar{Z}_T)^{1/\alpha_T}$ for the carrier-friction scale, and $D_{\min} > 0$ for the lower bound on the construction-price index D_{jt}^I.

PROPOSITION 7: *Suppose the conditions of Proposition 6 hold along feasible state histories.*

1. (Finite- and infinite-horizon Markov perfect equilibrium.) *Whenever own adjustment-cost convexity $m_I \chi_I D_{\min}$ dominates cross-county price and continuation spillovers, the J-county stage pseudogradient is strongly monotone on $\mathcal{X}_t$ with modulus $\mu = \frac{1}{8} m_I \chi_I D_{\min} > 0$, attenuating cross-county policy sensitivity to inherited county stock $\overline{T}_{it}$:*

$$\left\| \frac{\partial \mathbf{x}_t^{(H),\theta}}{\partial \overline{T}_{it}} \right\| \leq \frac{C_u \varepsilon}{\mu \min_r \overline{T}_{rt}^{(1+\alpha_T)/\alpha_T}}. \tag{75}$$

   For every finite horizon $H \geq 1$ and $\theta \in [0, \bar{\theta}]$, backward induction yields a unique pure-strategy Markov perfect equilibrium $(\mathbf{x}_t^{(H),\theta}, \mathbf{V}_t^{(H),\theta})_{t=0}^H$, which converges geometrically as $H \to \infty$ to a unique infinite-horizon dynamic equilibrium $(\mathbf{x}_t^\theta, \mathbf{V}_t^\theta)_{t\geq 0}$ whenever the state-weight drift κ and policy sensitivity satisfy $\beta\kappa + BL_a < 1$ (Theorem A.1).

2. (Constructive verification on the rescaled corridor.) *Under the terminal schedule with $\beta g^{1-\alpha_T} < 1$, regular perturbation around the frictionless benchmark verifies stage-game strong monotonicity, horizon-invariant continuation concavity, and geometric convergence to a unique interior infinite-horizon equilibrium on a forward-invariant rescaled stock corridor (Appendix A.7).* [Proof →]

IV Quantitative Analysis

This section estimates the model on historical evidence from England and Wales over 1710–1881 and uses the estimated model to quantify the contribution of diagnostic expectations to transport accumulation, structural transformation, real wages, prices, and frontier knowledge during the industrial takeoff (Crafts and Harley, 1992; Broadberry et al., 2014). Spatial allocations are computed on $N = 500$ locations partitioned into $J = 10$ county jurisdictions, solving at each annual date for the conditional dated equilibrium and the Markov perfect county investment profile.

A From Historical Districts to Model Space

Because England and Wales is predominantly north–south oriented, latitude provides a natural one-dimensional ordering of regional economic activity on $\ell \in [0,1]$ (Trew, 2020; Bogart et al., 2022). For each of twenty-five equal-latitude bands, historical employment and route quantities are summed across the band's east–west extent and divided by land width, with $\ell = 0$ at the southern coast and $\ell = 1$ at the northern border.

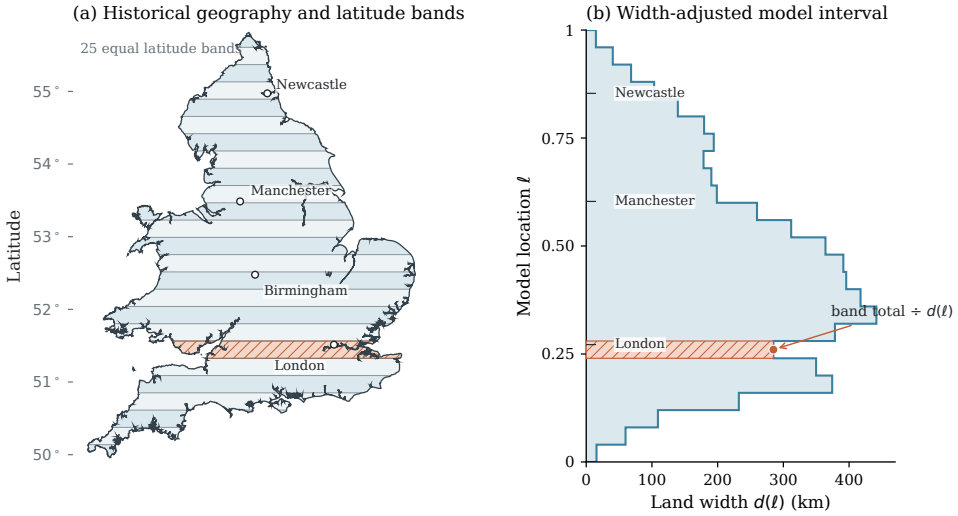


Figure 6: From Historical Geography to the Model Interval

Notes: Panel (a) partitions England and Wales into 25 equal-latitude bands. Panel (b) divides band totals by east–west land width $d(\ell)$ to form densities on $\ell \in [0,1]$.

This projection captures the relative positions of metropolitan London in the South, the West Midlands, and the northern coalfield belt, together with the reallocation of labor across them, while abstracting from east–west variation within a latitude band (Online Appendix S.B).

B Data and External Calibration

Aggregate household mass $\bar{L}_t$ follows the annual England and Wales population series of Thomas and Dimsdale (2017) over 1710–1881, normalized to $\bar{L}_{1710} = 100$ and held constant after 1881 ($\bar{L}_{1881+h} = \bar{L}_{1881}$). Installed transport capacity is reconstructed annually across the twenty-five latitude bands from turnpike, waterway, and railway GIS records. Letting M^r_{hy}, M^w_{hy}, and M^g_{hy} denote permanent route miles of turnpike road (Rosevear et al., 2026), navigable waterway (Cambridge Group for the History of Population and Social Structure, 2016a), and railway (Mitchell, 1988; Cambridge Group for the History of Population and Social Structure, 2016b; Coll Ardanuy et al., 2021) in band h and year y, and d_h the band's mean east–west land width, the measured transport stock is

$$\mathcal{I}^T_{hy} = \frac{(M^r_{hy} + 10M^w_{hy} + 50M^g_{hy})/d_h}{25^{-1}\sum_{k=1}^{25}(M^r_{k,1710} + 10M^w_{k,1710} + 50M^g_{k,1710})/d_k}, \tag{76}$$

where the service-capacity weights $(1, 10, 50)$ follow Trew (2020) and the 1710 cross-band mean is normalized to unity. The 1710 cross-section initializes $T_0(\ell)$, while subsequent stocks are generated endogenously by county investment (53)–(54).

Following Hornbeck and Rotemberg (2024), the manufacturing operating wedge $\psi_M(\ell) = \bar{\psi}_M \omega_M(\ell)$ combines a scale parameter $\bar{\psi}_M$ with a predetermined spatial exposure profile $\omega_M(\ell)$ from geocoded fire-insurance registers (Study 1838), normalized to unit 1710 manufacturing-labor-weighted mean ($\frac{\int_0^1 \omega_M(\ell)\omega_{M0}(\ell)\,d\ell}{\int_0^1 \omega_{M0}(\ell)\,d\ell} = 1$ with $\omega_{M0}(\ell) \equiv e_{M,1710}(\ell)L_{M,1710}(\ell)$). By the conditional hiring rule (30), an active manufacturing site with payroll $w(\ell)L_M(\ell)$ and expected gross operating surplus $\mathbb{E}_t[F_{M\ell}] = \mathbb{E}_t\big[[p_M(\ell) - r_M\tau_{j(\ell)}]_+ Y_M(\ell)\big] - w(\ell)L_M(\ell)$ satisfies

$$1 + \psi_M(\ell) = \alpha_M\left(1 + \frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)}\right), \qquad \text{or} \qquad \frac{\alpha_M}{1-\alpha_M}\frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)} = 1 + \frac{\psi_M(\ell)}{1-\alpha_M}, \tag{77}$$

directly calibrating $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from the Clark (2002) nonagricultural labor share, with digitized firm operating accounts (Clough records) providing an external check on $\mathbb{E}_t[F_{M\ell}]/[w(\ell)L_M(\ell)]$.

C Estimation and Identification

The estimation strategy is the Simulated Method of Moments (Hansen, 1982; Bordalo et al., 2026), which jointly estimates $p = 26$ free structural parameters under diagnostic expectations ($\theta \geq 0$ plus the 16 preference, innovation, spatial diffusion, transport, and construction-cost parameters and the 9 sectoral shock parameters in Table 5), while calibrating the manufacturing wedge scale $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from (77) and concentrating out the eight shape parameters of the five-knot initial 1710 productivity profiles $A_A(\ell, 1710)$ and $A_M(\ell, 1710)$ ($\int_0^1 A_i(\ell, 1710)\,d\ell = 1$) on the 1710 occupational cross-section. To evaluate belief extrapolation, the model is also estimated under the rational expectations restriction ($\theta = 0$, $p = 25$ free SMM parameters). Both specifications share the fixed parameters in Table 3 and minimize the weighted distance between the $M = 30$ historical moments and simulated moments averaged across B common primitive shock histories $\{u_b\}_{b=1}^B$:

$$Q(\Theta) = \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big]' W \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big], \qquad \overline{\mathbf{m}}(\Theta) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{m}(\Theta; u_b), \tag{78}$$

where $\mathbf{m}^{\text{data}} \in \mathbb{R}^{30}$ stacks the aggregate, transport-network, spatial-covariance, sectoral-cycle, and investment-dynamics moments in Panels A–E of Table 6, and $W = \text{diag}(1/\widehat{\text{SE}}_1^2, \ldots, 1/\widehat{\text{SE}}_{30}^2)$ weights each residual by its inverse empirical variance so that $Q(\Theta) = \sum_{k=1}^{30}\big[(\overline{m}_k(\Theta) - m_k^{\text{data}})/\widehat{\text{SE}}_k\big]^2$ is measured in studentized units (Online Appendix S.D and Table S.D.2 report the formula, standard error $\widehat{\text{SE}}_k$, and weight W_{kk} for each moment).

Table 4 summarizes how the moments discipline each parameter block. By Proposition 2, the allocation at each date depends on θ only through county construction, so within-period margins discipline the static parameters given the network, while network dynamics discipline beliefs and construction costs. Sectoral labor shares map into the fixed production elasticities (α_A, α_M), and the nonagricultural labor share, given fire-insurance exposure, calibrates $\bar{\psi}_M$ (77). The 1710 occupational shares, the relative price path, and dated spatial employment covariances discipline (η, γ), $(\alpha_T, \kappa_0, r_A)$, and $A_i(\ell, 1710)$. Aggregate output growth and the shift from agricultural to industrial output identify the innovation and diffusion parameters $(a_i, f_i, \chi_i, \omega_E, \delta, \lambda_0)$, while HP-filtered sectoral output variances, autocovariances, and cross-covariances pin down $(\boldsymbol{\rho}, \boldsymbol{\Sigma})$. Finally, the linear and quadratic trends and subperiod growth rates of log transport capacity (Panel B of Table 6) and the investment-cycle regressions (Panel E) separate θ from $(m_I, \chi_I, \boldsymbol{\rho})$. Under $\theta = 0$, lowering χ_I or

Table 3: Externally Fixed Parameters and Normalizations

Parameter	Value	Description
β	1/1.05	Annual discount factor (real interest rate of 5%)
$\alpha_A,\ \alpha_M$	0.390, 0.710	Production labor elasticities (agriculture, manufacturing)
r_M	1.000	Manufacturing distribution requirement (normalization)
c	3	Truncation bound for aggregate productivity innovations (in s.d.)
d_0	0.100	Spatial Ornstein–Uhlenbeck correlation length for discovery draws
g_m	log(1.02)	Post-1881 growth rate of physical project increments
$\bar{L}_{1710}$	100	Initial aggregate labor supply (normalization)
$N,\ J$	500, 10	Number of spatial grid cells and county jurisdictions

Notes: Numeraire $p_A(1,t) = 1$; initial productivity profiles satisfy $\int_0^1 A_{i,1710}(\ell)\,d\ell = 1$ $(i \in \{A, M\})$; and the 1710 transport stock has cross-band mean one. Production labor elasticities (α_A, α_M) follow sectoral value-added shares (Herrendorf et al., 2015) (Proposition S.E.2), and demographic mass $\bar{L}_t$ follows (9).

raising $\boldsymbol{\rho}$ increases investment–output comovement only by raising the first-order autocorrelation ρ_1 of cyclical investment, whereas $\theta > 0$ amplifies boom-time marginal q, raises 1830–1850 transport growth, and lowers ρ_1 as transitory news subsides. These sign restrictions are verified numerically at the estimates.[10]

Table 4: Parameter Blocks and Identifying Moments

Parameter block	Parameters	Primary identifying moments
Preferences	$\eta,\ \gamma$	Linear trends in sectoral employment shares and log relative price $\ln(p_M/p_A)$
Operating wedge	$\bar{\psi}_M$	Nonagricultural labor share and fire-insurance exposure (77)
Innovation	$a_i,\ f_i,\ \chi_i,\ \omega_E$	Linear trends in log real GDP and $\ln(Y_M/Y_A)$; manufacturing share–energy covariance $C_{e,y}$
Spatial diffusion	$\delta,\ \lambda_0$	Manufacturing share–latitude covariance $C_{l,y}$ and spatial autocovariance $C_{N,y}$
Transport	$\alpha_T,\ \kappa_0,\ r_A$	Tertiary employment share trend, cross-band variance $V^{\text{occ}}_{T,y}$, and $\ln(p_M/p_A)$ trend
Construction costs	$m_I,\ \chi_I$	Linear and quadratic trends of log transport stock $\ln \overline{T}_t$; cross-band variance $V_{T,t}$
Diagnostic beliefs	θ	Subperiod transport growth (1830–1850), post-1850 slope change, investment–output slopes (b_0, b_1), autocorrelation (ρ_1), and county lag slopes (g_1, g_2)
Initial productivity	$A_i(\ell, 1710)$	1710 occupational shares and cross-sectional employment covariances
Productivity process	$\boldsymbol{\rho},\ \boldsymbol{\Sigma}$	Variances, first-order autocovariances, and cross-covariances of HP-filtered log sectoral output

Notes: Primary identifying moments for each parameter block (Table 6 and Online Appendix S.D); all blocks are estimated jointly via (78) under DE $(\theta \geq 0)$ and RE $(\theta = 0)$. Online Appendix S.E states the local rank and sign restrictions.

Following Bordalo et al. (2026), Table 5 reports the parameter estimates and standard errors under diagnostic (DE) and rational (RE, $\theta = 0$) expectations, and Table 6 compares the empirical and simulated moments across all five blocks. Diagnosticity is estimated at $\widehat{\theta} = 0.739$ (SE =

[10] Online Appendix S.E states the local rank conditions (Rothenberg, 1971; Andrews et al., 2017) and sign restrictions; Replication Appendix R.A details the sequence-space Jacobian evaluation of (78) (Auclert et al., 2021).

0.090), rejecting $\theta = 0$ ($t = 8.21$, $p < 0.0001$). This estimate lies within the benchmark range across financial and macroeconomic settings (Bordalo et al., 2026): slightly below the $\theta \approx 0.8$–0.9 estimated from earnings expectations and credit spreads (Bordalo et al., 2018, 2019), above the $\theta \approx 0.5$–0.6 from macroeconomic surveys (Bordalo et al., 2020), and aligned with the $\theta \approx 0.74$–1.08 in structural models of investment and credit cycles (Maxted, 2023; Bianchi et al., 2024; Bordalo et al., 2026). When the remaining $p = 25$ parameters are re-estimated under $\theta = 0$, the restricted RE estimator lowers adjustment convexity ($\widehat{\chi}_I = 1.579$ versus 1.920) and construction scale ($\widehat{m}_I = 0.020$ versus 0.023), raises manufacturing persistence ($\widehat{\rho}_M = 0.561$ versus 0.533) and volatility ($\widehat{\sigma}_M = 0.042$ versus 0.039), and adjusts spatial catch-up and transport costs ($\widehat{\lambda}_0 = 0.175$ versus 0.160, $\widehat{\kappa}_0 = 0.073$ versus 0.080, $\widehat{\delta} = 1.245$ versus 1.350) to compensate for the absence of extrapolation.

Table 5: SMM Parameter Estimates: Diagnostic versus Rational Expectations

Parameter	Description	DE	RE ($\theta = 0$)
Panel A. Preferences, production, and trade costs			
η	Stone–Geary food expenditure share	0.520 (0.028)	0.540 (0.031)
γ	Subsistence food requirement	0.042 (0.003)	0.039 (0.003)
α_A	Agricultural labor elasticity	0.390 [fixed]	0.390 [fixed]
α_M	Manufacturing labor elasticity	0.710 [fixed]	0.710 [fixed]
$\bar{\psi}_M$	Manufacturing labor wedge scale	0.251 (0.032) [calibrated]	0.251 (0.032) [calibrated]
α_T	Transport labor elasticity	0.450 (0.029)	0.473 (0.033)
κ_0	Iceberg transport cost scale	0.080 (0.007)	0.073 (0.007)
r_A	Agricultural distribution requirement ($r_M = 1$)	0.200 (0.015)	0.216 (0.018)
Panel B. Innovation and spatial diffusion			
a_A	Agricultural innovation Pareto shape	2.200 (0.150)	2.122 (0.156)
a_M	Manufacturing innovation Pareto shape	1.800 (0.112)	1.887 (0.126)
f_A	Agricultural fixed innovation cost	0.040 (0.004)	0.043 (0.005)
f_M	Manufacturing fixed innovation cost	0.030 (0.003)	0.028 (0.003)
χ_A	Agricultural variable innovation cost	0.080 (0.008)	0.085 (0.009)
χ_M	Manufacturing variable innovation cost	0.060 (0.006)	0.055 (0.006)
ω_E	Coalfield innovation cost loading	0.0040 (0.0006)	0.0045 (0.0007)
λ_0	Baseline technological catch-up rate	0.160 (0.014)	0.175 (0.016)
δ	Spatial knowledge decay rate	1.350 (0.097)	1.245 (0.097)
Panel C. Infrastructure investment and expectations			
θ	Diagnostic expectations parameter	0.739 (0.090)	0 [fixed]
m_I	Infrastructure construction cost scale	0.023 (0.006)	0.020 (0.005)
χ_I	Infrastructure adjustment cost convexity	1.920 (0.233)	1.579 (0.213)
Panel D. Aggregate productivity process			
ρ_A	AR(1) persistence: Agriculture	0.168 (0.042)	0.174 (0.043)
ρ_M	AR(1) persistence: Manufacturing	0.533 (0.124)	0.561 (0.108)
ρ_T	AR(1) persistence: Transport	0.303 (0.076)	0.318 (0.079)
σ_A	Innovation s.d.: Agriculture	0.067 (0.004)	0.070 (0.004)
σ_M	Innovation s.d.: Manufacturing	0.039 (0.002)	0.042 (0.002)
σ_T	Innovation s.d.: Transport	0.023 (0.002)	0.025 (0.002)
ϱ_{AM}	Innovation correlation: (A, M)	−0.058 (0.015)	−0.082 (0.020)
ϱ_{AT}	Innovation correlation: (A, T)	0.094 (0.024)	0.077 (0.019)
$\varrho_{MT\|A}$	Partial innovation correlation: $(M, T \mid A)$	0.337 (0.084)	0.373 (0.093)

Notes: SMM estimates minimizing (78) across $M = 30$ moments (Table 6) under diagnostic (DE, $\theta \geq 0$, $p = 26$) and rational (RE, $\theta = 0$, $p = 25$) expectations across $B = 1{,}000$ common shock histories, with misspecification-robust sandwich standard errors (S.D.11) in parentheses. The wedge scale $\bar{\psi}_M = 0.251$ (SE = 0.032) is calibrated via (77), the eight 1710 productivity spline coefficients are concentrated out on the 1710 occupational cross-section, and fixed parameters are in Table 3.

Because those parameter shifts distort other margins, the fit improvement under DE spans Table 6. Both models match long-run output and employment trends (Panel A) and dated spatial sorting covariances (Panel C). The restricted RE model, however, underpredicts the linear and

Table 6: Empirical Moments and Model Fit

Empirical moment	Window	Data	DE	RE
Panel A. Aggregate growth and structural transformation				
Log real GDP trend (%/yr)	1710–1881	1.473 (0.210)	1.425	1.364
Log relative output $\ln(Y_M/Y_A)$ trend (%/yr)	1710–1881	1.085 (0.572)	0.970	0.799
Log relative price $\ln(p_M/p_A)$ trend (%/yr)	1710–1870	−0.757 (0.616)	−0.632	−0.496
Agricultural employment share trend (pp/yr)	1710–1881	−0.146 (0.023)	−0.139	−0.116
Tertiary employment share trend (pp/yr)	1710–1881	0.060 (0.007)	0.058	0.055
Panel B. Transport network growth and spatial dispersion				
Log transport stock: linear trend (%/yr)	1710–1881	1.948 (0.103)	1.943	1.758
Log transport stock: quadratic trend (×100)	1710–1881	2.923 (0.217)	2.952	2.144
Log transport stock: 1830–1850 growth (%/yr)	1830–1850	7.470 (0.759)	7.410	4.801
Log transport stock: post-1850 slope change (%/yr)	1830–1881	−5.101 (0.881)	−5.060	−2.461
Cross-band variance of log transport stock	1710–1881	0.469 (0.148)	0.437	0.410
Panel C. Cross-sectional employment covariances				
Manufacturing share–latitude covariance	1710–1881	0.077 (0.026)	0.070	0.065
Manufacturing share–energy covariance	1710–1881	−0.070 (0.024)	−0.064	−0.059
Spatial autocovariance of manufacturing share	1710–1881	0.030 (0.006)	0.028	0.026
Cross-band variance of tertiary share	1710–1881	0.0029 (0.0011)	0.0027	0.0025
Panel D. HP-filtered log sectoral output moments ($\times 10^4$)				
Variance: Agriculture (A)	1710–1881	34.79 (4.60)	35.81	39.80
Variance: Manufacturing (M)	1710–1881	9.64 (1.40)	9.96	12.12
Variance: Transport (T)	1710–1881	5.66 (0.93)	5.86	7.37
First-order autocovariance: Agriculture (A)	1710–1881	−7.27 (2.41)	−6.86	−6.75
First-order autocovariance: Manufacturing (M)	1710–1881	−0.49 (0.68)	−0.35	−0.08
First-order autocovariance: Transport (T)	1710–1881	−0.63 (0.39)	−0.55	−0.39
Cross-sector covariance: (A, M)	1710–1881	−2.59 (1.43)	−2.89	−3.89
Cross-sector covariance: (A, T)	1710–1881	−0.55 (0.83)	−0.71	−1.20
Cross-sector covariance: (M, T)	1710–1881	4.74 (0.78)	4.95	6.34
Panel E. Infrastructure investment dynamics				
Standard deviation of cyclical investment, $\sigma(c_t^I)$	1710–1880	0.431 (0.043)	0.427	0.345
Investment on current output cycle (b_0)	1711–1880	2.954 (1.431)	3.091	2.063
Investment on lagged output cycle (b_1)	1711–1880	1.585 (1.257)	1.662	3.142
First-order autocorrelation of investment (ρ_1)	1711–1880	−0.325 (0.074)	−0.344	−0.183
Lead output cycle on investment (b_{lead})	1711–1880	−0.0051 (0.0039)	−0.0049	−0.0035
County investment on population growth, lag 1 (g_1)	1791–1851	−0.162 (0.050)	−0.190	−0.082
County investment on population growth, lag 2 (g_2)	1791–1851	−0.496 (0.050)	−0.484	−0.144

Notes: Empirical moments (with source-bootstrap or Newey–West HAC standard errors in parentheses; Online Appendix S.D.8 and Table S.D.2) and simulated means across $B = 1{,}000$ common shock histories at the DE and RE estimates in Table 5. Sources: Broadberry et al. (2022) and Feinstein (1972) (Panels A, D), Shaw-Taylor et al. (2019) and Keibek (2017) (Panels A, C), and the measured transport stock (76) (Panels B, E).

quadratic trends of the log transport stock (1.758 versus 1.948 percent per year and 2.144 versus 2.923 in Panel B), the 1830–1850 railway boom (4.801 versus 7.410 percent per year under DE and 7.470 in the data), and the post-1850 slowdown (−2.461 versus −5.060 and −5.101). Raising $\widehat{\rho}_M$ and $\widehat{\sigma}_M$ under RE also overstates manufacturing and transport cycle variances ($v_M = 12.12$ versus 9.96 under DE and 9.64 in the data; $v_T = 7.37$ versus 5.86 and 5.66 in Panel D), shifts investment comovement from current to lagged output ($b_0 = 2.063$ and $b_1 = 3.142$ under RE, versus 3.091 and 1.662 under DE and 2.954 and 1.585 in the data), and mutes both the negative serial correlation of investment ($\rho_1 = -0.183$ versus -0.344 under DE and -0.325 in the data) and county investment reversals after population growth ($g_1 = -0.082$ and $g_2 = -0.144$ under RE, versus -0.190 and -0.484 under DE and -0.162 and -0.496 in the data; Panel E). Online Appendix Table S.E.1 confirms that this fit gain does not stem from an extra parameter or moment weighting: the Rivers and Vuong (2002) equal-parameter test ($p = 25$ free parameters in every column) favors $\theta = 0.9$ ($Q = 5.54$) over $\theta = 0$ ($Q = 117.86$) with $z = 4.35$ ($p < 0.0001$). The Andrews et al. (2017)

sensitivity measures show that Panels E and B supply 85.7 percent (49.8 and 35.9 percent) of the identifying variation for $\widehat{\theta}$, while leave-one-block-out estimations yield $\widehat{\theta} \in [0.666, 0.747]$ and equal-block studentized weights yield $\widehat{\theta}_{\text{eq}} = 0.748$ (SE $= 0.104$).

D Aggregate Takeoff and Spatial Reallocation

Figures 7 and 8 plot the historical transition alongside the mean simulated trajectories and 5th–95th percentile bands across 1,000 Monte Carlo draws under DE and RE. Total real GDP rose 14.33-fold between 1710 and 1881, averaging 1.56 percent per year (Figure 7, Panel a), as annual growth accelerated from 1.02 percent in 1710–1760 and 1.16 percent in 1760–1800 to 1.60 percent in 1800–1830 and 2.37 percent in 1830–1881. Because no moment of the real wage series enters the SMM objective in Table 6, Panel (b) of Figure 7 provides an untargeted check on the model's price and wage predictions. Historically, real wages $w/\mathcal{P}$ rose modestly in the early eighteenth century, plateaued at 1.35–1.55 times their 1710 level during the late eighteenth and early nineteenth centuries—the "Engels' pause" of Allen (2009)—and then accelerated after 1820 to 3.28 times their 1710 level by 1881 as transport integration and industrial productivity gains lowered delivered prices. The industrial-to-agricultural output ratio Y_M/Y_A (Panel c) follows the same two-phase timing, drifting to 0.60 times its 1710 level in 1781 as population growth raised food demand before accelerating at 2.61 percent per year after 1813 to 5.77 times its 1710 level by 1881.[11] The estimated DE model reproduces this two-phase acceleration in aggregate output, Y_M/Y_A, and the untargeted real wage path $w/\mathcal{P}$ (1.41 in 1760, 1.75 in 1820, and 3.11 in 1881), whereas the restricted RE trajectory matches the eighteenth-century plateau (1.35 in 1760 and 1.47 in 1820) but stalls at 1.72 by 1881 alongside a muted rise in Y_M/Y_A.

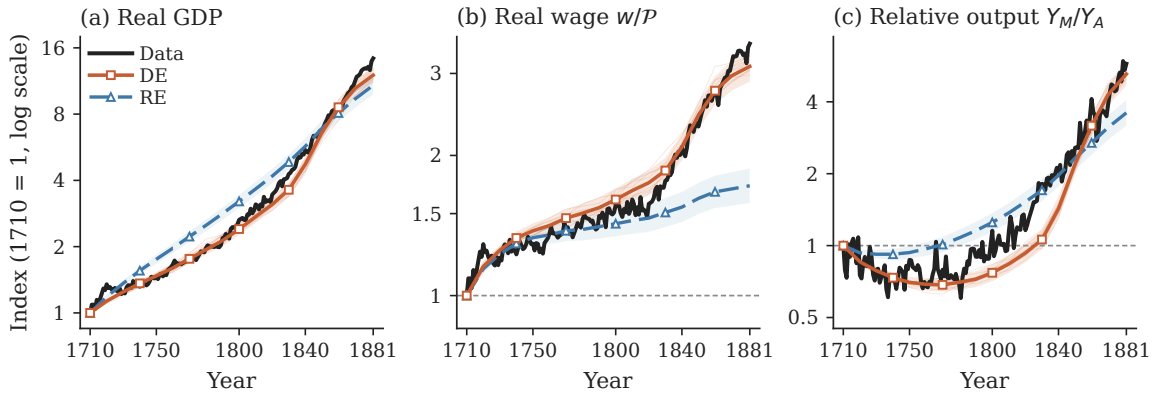


Figure 7: Aggregate Transition, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under diagnostic (DE, $\widehat{\theta} = 0.739$) and restricted rational (RE, $\theta = 0$) expectations (1710 = 1, log scale). Panels (a) and (c) plot real GDP and relative industrial output Y_M/Y_A (Broadberry et al., 2022; Feinstein, 1972); Panel (b) plots the untargeted aggregate real wage $w/\mathcal{P}$ (Clark, 2002; Broadberry et al., 2022).

Transport capacity expanded nonlinearly over the sample. The measured stock (76) rose 27.70-fold between 1710 and 1881 (Figure 8, Panel c). Before 1830 it grew at 0.99 percent per year to 3.28 times its 1710 level, with turnpike-era bursts of 1.79 percent per year in 1750–1770 and 1.68 percent in the 1790s. Railway construction raised annual log growth to 5.90 percent in the 1830s and 8.43

[11] The relative price p_M/p_A (Online Appendix Figure S.D.1, Panel c) mirrors this shift inversely, remaining above its 1710 level through much of the eighteenth century (peaking at 1.47 in 1733) before falling below unity after 1799 to 0.71 in 1800, 0.59 in 1830, 0.39 in 1855, and 0.49 in 1870.

percent in the 1840s (7.47 percent per year over 1830–1850), lifting the stock to 5.91 by 1840 and 13.73 by 1850, before annual growth slowed by 5.10 percentage points to 2.37 percent over 1850–1881 (24.36 in 1870). Disciplined by Panel B of Table 6, the DE path tracks both gradual eighteenth-century accumulation and the 1830–1850 surge and post-1850 moderation, whereas the RE path grows too smoothly and misses the mid-century acceleration. The 1881 band profile (Panel d) is multi-peaked: the West Riding and Lancashire band carried 53.8 times the 1710 national mean, followed by London (46.0) and the Midlands–Northern industrial bands (35.1–40.1), while Cornwall (3.5) and Northumberland (8.2) remained least served. Cross-band log-stock variance $V_{T,y}$ fell from 0.713 in 1710 to 0.295 in 1881 in the data (0.658 to 0.240 under DE; Figure S.D.1, Panel f), with a sample mean of $\overline{V}_T = 0.469$ (0.437 under DE and 0.410 under RE; Panel B). Under DE, counties build more aggressively during the canal and railway booms across the Midlands and Northern corridor (bands 12–22) than under RE, matching the 1881 cross-sectional peaks.

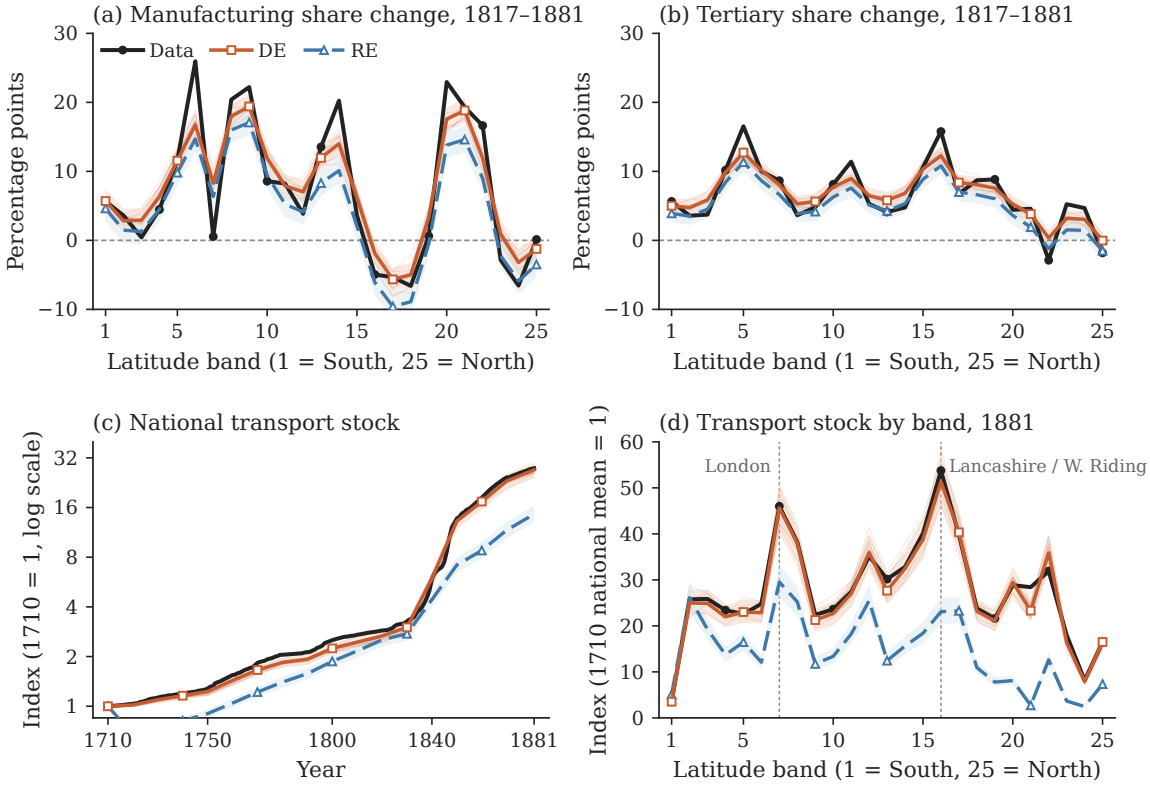


Figure 8: Spatial Reallocation and Infrastructure, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under DE ($\widehat{\theta} = 0.739$) and RE ($\theta = 0$). Panels (a)–(b) plot 1817–1881 percentage-point changes in manufacturing and tertiary employment shares across the 25 latitude bands (1 = South, 25 = North; Online Appendix S.B); Panels (c)–(d) plot the national mean transport stock (76) (1710 = 1, log scale) and the 1881 band profile (1710 national mean = 1).

Panels (a)–(b) of Figure 8 compare the 1817–1881 change in manufacturing and tertiary employment shares across the twenty-five latitude bands with the DE and RE predictions (Online Appendix S.B). As the national manufacturing share rose from 49.9 to 62.1 percent, industrial employment diffused across space: the manufacturing share rose in 20 of 25 bands and its change correlates negatively with the 1817 share (−0.81), with 20–26 percentage point gains in initially peripheral bands (bands 6, 8, and 20–22) and 5–7 point declines in the Lancashire–Yorkshire core (bands 16–18, which remained above 79 percent manufacturing in 1881). Tertiary employment

expanded in 23 of 25 bands as the national transport and distribution share nearly doubled from 9.5 to 18.5 percent, led by the Southern, London, and Lancashire–West Riding corridors. Although the estimation targets only aggregate occupational series and dated spatial covariances in Panels A and C, the DE model captures both manufacturing diffusion and tertiary expansion more closely than RE.[12]

E Counterfactual Analysis: Diagnostic versus Rational Expectations

Re-estimating the model under $\theta = 0$ alters technology and construction-cost parameters alongside beliefs. To isolate diagnostic expectations, the belief-neutralization counterfactual (R | DE) holds all other parameters at their diagnostic estimates, together with the 1710 state, statutory project opportunities, and primitive shock draws $\{u_b\}_{b=1}^{B}$, sets $\theta = 0$, and recomputes county investment policies $\mathbf{x}_t^0$ and the general equilibrium path from 1710 to 1881.

By Proposition 2, household and firm decisions and prices at each date are invariant to θ given county investment, so every difference between the DE and R | DE paths originates in county construction and propagates through the installed network. Construction absorbs manufactures when built; completed capacity lowers route losses $\kappa(T)$ and expands carrier service capacity at later dates; and, by Proposition 3, higher capacity accelerates spatial knowledge catch-up.

Figure 9 and Table 7 report the counterfactual.[13] By 1881, diagnostic expectations expand the aggregate transport stock $\overline{T}$ by +43.79 percent (3.001 versus 2.087; Panel e), with regional gains of +29.53 percent in the Southern belt (Counties 1–5), +54.47 percent in the Midlands–North corridor (Counties 6–9), and +61.33 percent in the Northern border (County 10). This network expansion raises 1881 real output Y (and output per worker y) by +20.96 percent (4.774 versus 3.947; Panel a) and lifts the population-weighted real wage $w/\mathcal{P}$ by +12.97 percent nationally (1.158 versus 1.025; +12.78 to +13.59 percent across regions; Panel b). Larger route capacity accelerates structural transformation (Trew, 2020), lowering the 1881 agricultural employment share by −12.9 percentage points (13.0 versus 25.9 percent) and shifting +11.4 percentage points into manufacturing (64.0 versus 52.6 percent) and +1.5 percentage points into tertiary services (23.0 versus 21.5 percent; Panel c). Faster spatial diffusion lifts manufacturing frontier productivity F_M by +10.90 percent (2.657 versus 2.396; Panel d), while cheaper distribution and higher manufacturing productivity lower p_M/p_A by −13.79 percent (0.492 versus 0.571) and the Stone–Geary price index $\mathcal{P}$ by −5.97 percent (0.720 versus 0.765; Panel f).

Two features of Table 7 highlight the model's spatial mechanisms. First, in Panel B, the counterfactual percentage gain in the available frontier $F_{Mt}(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell-r|} A_{Mt}^{+}(r)$ is spatially uniform at +10.90 percent across all three regions because the spatial supremum scales proportionally with the northern industrial peak. Local applied manufacturing productivity $A_{Mt}(\ell)$, governed by the transport-dependent catch-up rate $\lambda_t(\ell) = \lambda_0[T_t(\ell)/T_{t_0}(\ell)]^{1-\alpha_T}$ in (40)–(41), expands most where route buildout is largest (+7.84 percent in the Southern belt, +13.42 percent in the Midlands–North corridor, and +14.18 percent in the Northern border, or +11.65 percent nationally). Second, decomposing the +20.96 percent 1881 output gain sequentially across the three uninternalized social return channels of Section III.F shows why static "social savings" calculations (Fogel, 1964) capture only a fraction of the long-run return to connective infrastructure: direct carrier and iceberg transit savings holding spatial sorting and technology at their R | DE paths (Fogel, 1964; Donaldson and Hornbeck, 2016) account for +5.42 percentage points (25.9 percent of the gain);

[12] Online Appendix S.D (Figure S.D.1) plots the complete historical and simulated series for every moment block.

[13] Level indices in Figure 9 and Table 7 are in balanced-growth-detrended units (1710 = 1), and employment shares are model-domain shares $\mathcal{L}_o(t)/\bar{L}_t$ (S.D.2). Because exogenous package-scale and frontier drift factors match across DE and R | DE, percentage differences $100\,(\text{DE/R} \mid \text{DE} - 1)$ are identical in un-detrended units.

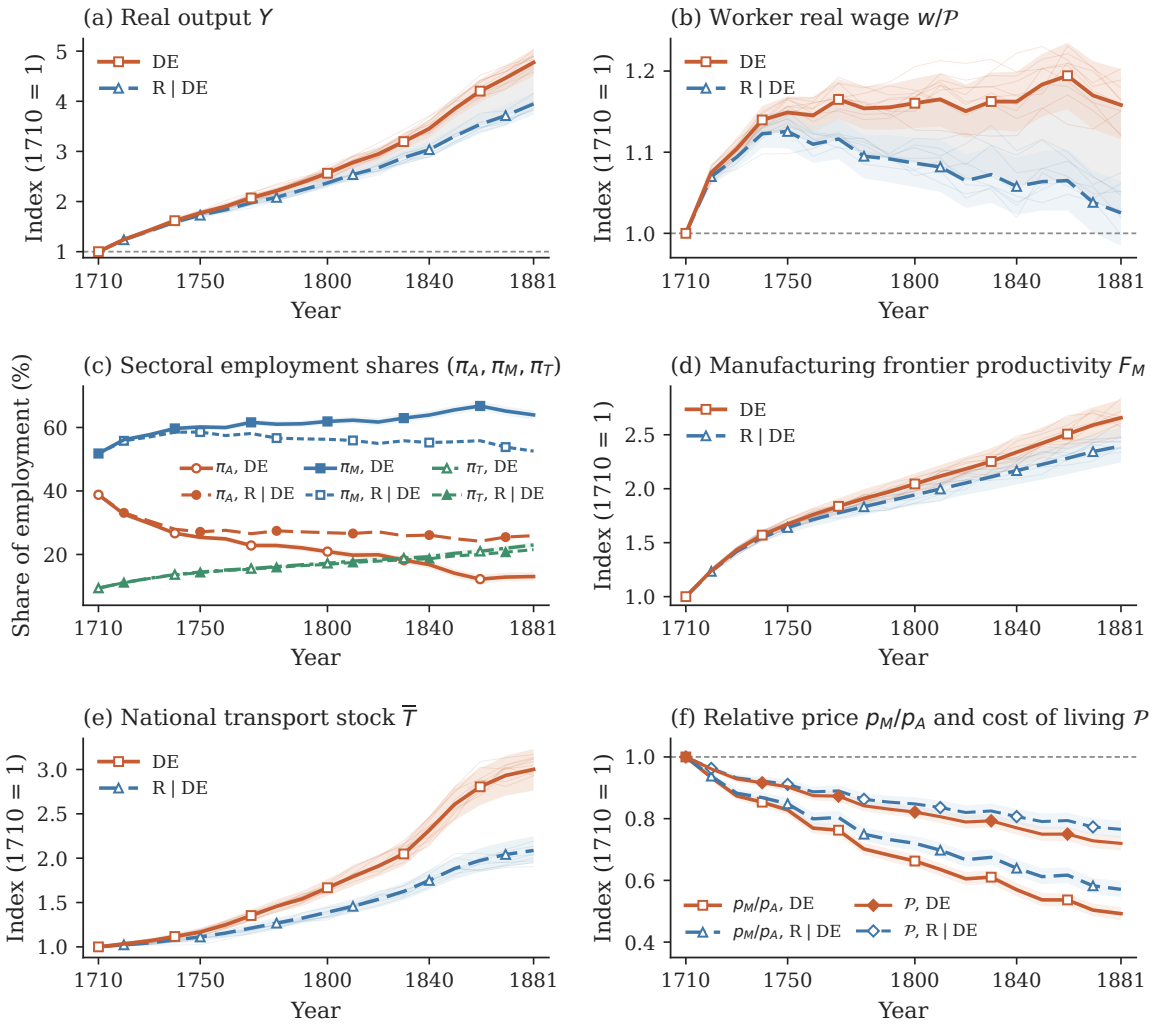


Figure 9: Counterfactual Effects of Diagnostic Expectations, 1710–1881

Notes: Simulated means and 5th–95th percentile bands (1,000 draws) under the baseline diagnostic economy (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE: $\theta = 0$ holding all other parameters and shock histories fixed). Panels (a), (b), (d), (e), and (f) report balanced-growth-detrended indices (1710 = 1) for real output Y, real wage $w/\mathcal{P}$, manufacturing frontier F_M, transport stock $\overline{T}$, and prices ($p_M/p_A, \mathcal{P}$); Panel (c) reports sectoral employment shares π_A, π_M, π_T (percent).

spatial labor reallocation into wedge-distorted manufacturing ($\psi_M(\ell) > 0$; Hsieh and Klenow 2009; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024) accounts for +6.18 percentage points (29.5 percent); and transport-mediated spatial knowledge diffusion and induced innovation (66) (Nelson and Phelps, 1966; Romer, 1990; Mokyr, 2005; Desmet and Rossi-Hansberg, 2014) account for +9.36 percentage points (44.7 percent).

Table 7: Counterfactual Effects of Diagnostic Expectations in 1881

1881 outcome (1710 = 1, except shares)	DE ($\widehat{\theta} = 0.739$)	R | DE ($\theta = 0$)	Difference
Panel A. Aggregate outcomes			
Real output (Y)	4.774	3.947	+20.96%
Population-weighted real wage ($w/\mathcal{P}$)	1.158	1.025	+12.97%
Agricultural employment share (π_A, %)	13.0	25.9	−12.9 pp
Manufacturing / tertiary shares (π_M/π_T, %)	64.0 / 23.0	52.6 / 21.5	+11.4 / +1.5 pp
Manufacturing frontier productivity (F_M)	2.657	2.396	+10.90%
Aggregate transport stock ($\overline{T}$)	3.001	2.087	+43.79%
Relative price of manufactures (p_M/p_A)	0.492	0.571	−13.79%
Stone–Geary price index ($\mathcal{P}$)	0.720	0.765	−5.97%
Panel B. Regional outcomes ($w/\mathcal{P}$ / F_M / $\overline{T}$)			
Southern belt (Counties 1–5)	1.188 / 1.888 / 2.470	1.053 / 1.703 / 1.907	+12.78% / +10.90% / +29.53%
Midlands–North corridor (Counties 6–9)	1.156 / 3.299 / 3.549	1.023 / 2.975 / 2.298	+13.00% / +10.90% / +54.47%
Northern border (County 10)	1.116 / 3.935 / 3.463	0.982 / 3.548 / 2.146	+13.59% / +10.90% / +61.33%

Notes: 1881 outcomes averaged across $B = 1{,}000$ common shock histories under baseline diagnostic expectations (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE, $\theta = 0$ holding all other parameters and shock histories fixed). Level indices are in balanced-growth-detrended units (1710 = 1; percentage differences $100\,(\text{DE}/\text{R} \mid \text{DE} - 1)$ are identical in historical units), and employment shares are in percent (differences in percentage points).

V Conclusion

Economic historians and financial economists have long viewed the recurrent investment manias of the British Industrial Revolution—and subsequent general-purpose network buildouts from electrification to fiber-optic telecommunications (Bresnahan and Trajtenberg, 1995)—through two seemingly incompatible lenses. Financial histories emphasize promoter over-extrapolation, post-boom construction reversals, and disappointing realized yields (Shaen, 1847; Keynes, 1936; Kindleberger, 1978; Shiller, 2000; Odlyzko, 2010; Arnold and McCartney, 2011; Campbell, 2012); growth and spatial histories emphasize that the physical networks left behind permanently integrated national markets and accelerated industrialization (Fogel, 1964; Mokyr, 2005; Bogart, 2014; Donaldson and Hornbeck, 2016; Trew, 2020). Embedding a Markov perfect county infrastructure game under diagnostic expectations into a continuous-space endogenous growth model of England and Wales over 1710–1881 shows that both regularities are two sides of a single second-best mechanism: *corrective overoptimism* (Lipsey and Lancaster, 1956). Because decentralized promoters appropriate only local land rents inside their jurisdictions, rational expectations chronically underinvest relative to the social returns from inter-county transit savings, labor reallocation into wedge-distorted manufacturing, and spatial technology diffusion. Diagnostic overreaction to favorable productivity news induces promoters to build aggressively during booms, and physical irreversibility (Arrow, 1968; Dixit and Pindyck, 1994) locks in the resulting route capacity long after expectations revert.

Estimating the model via the Simulated Method of Moments yields a diagnosticity parameter of $\widehat{\theta} = 0.739$ (SE = 0.090), matching the $\theta \approx 0.74$–0.91 range documented in financial and macroeconomic expectations (Bordalo et al., 2018, 2019, 2026) and identified by the coexistence of strong boom-time investment comovement with sharp post-boom reversals. In the belief-neutralization counterfactual (R | DE), diagnostic expectations expand the 1881 aggregate transport stock by +43.79 percent (+54.47 percent in the Midlands–North corridor), accelerate structural transformation out of agriculture by −12.9 percentage points, lift manufacturing frontier productivity by +10.90 percent, and raise 1881 real output by +20.96 percent and worker real wages by +12.97 percent.

These findings carry three broader implications. First, they recast the "social savings" debate initiated by Fogel (1964): static freight-cost savings capture only about one-quarter (25.9 percent, or +5.42 percentage points) of the long-run output gain from belief-driven network expansion, while nearly three-quarters arises from reallocating labor into wedge-distorted manufacturing (29.5 percent, or +6.18 percentage points; Restuccia and Rogerson 2008; Hsieh and Klenow 2009; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024) and accelerating the spatial diffusion of frontier discoveries (44.7 percent, or +9.36 percentage points; Nelson and Phelps 1966; Lucas 1988; Romer 1990; Krugman 1991; Aghion and Howitt 1992; Mokyr 2005). Second, they establish a primitive boundary condition separating *corrective* from *destructive* belief overreaction (Kahneman and Tversky, 1972; Bordalo et al., 2018, 2020, 2026): when extrapolative beliefs fund non-durable consumption or zero-sum speculation without technological externalities, the bust leaves balance-sheet distress with little productive legacy; when overoptimism funds irreversible connective capital whose social returns exceed private appropriation, private financial disappointment coexists with permanent productivity gains. Finally, corrective overoptimism highlights a comparative institutional trade-off (North, 1990): continental European states addressed infrastructure underprovision through centralized planning and sovereign dividend guarantees (Milward and Saul, 1973; Bogart, 2014), whereas Britain and the United States relied on decentralized private promotion amplified by speculative manias (Janeway, 2018; Morck, 2022). Policies that lean uniformly against investor exuberance—or insulate incumbent network providers from entry—risk suppressing the decentralized investment waves that overcome chronic underprovision.

subsection

A Proofs of Main Results

A.1 Proof of Proposition 1

Proof. Because $\mathbb{E}_t^\theta$ is the expectation under the tilted probability in (11), it is linear, so the objective in (14) equals $\mathbb{E}_t^\theta[U(c_A(t), c_M(t))] + \mathbb{E}_t^\theta[\sum_{s\geq t+1} \beta^{s-t} U(c_A(s), c_M(s))]$. Goods are nonstorable, households neither save nor borrow, relocation is costless, portfolio income b_s is residence independent, and households are atomistic, so the set of feasible continuation plans $\{\ell_s, c_A(s), c_M(s)\}_{s\geq t+1}$ does not depend on $(\ell_t, c_A(t), c_M(t))$ and the supremum C_t of the second term is the same for every current residence. Ex-post consumption optimization gives $U(c_A(t), c_M(t)) = u_t(\ell_t, \omega)$ statewise by (16), so $W_t = \sup_{\ell_t \in [0,1]} \mathbb{E}_t^\theta[u_t(\ell_t)] + C_t$, where finiteness of W_t and of every $\mathbb{E}_t^\theta[u_t(\ell)]$ rules out an $\infty - \infty$ ambiguity. Hence occupied locations maximize $\mathbb{E}_t^\theta[u_t(\ell)]$ at the common value $\bar{u}_t$ and unoccupied locations satisfy $\mathbb{E}_t^\theta[u_t(\ell)] \leq \bar{u}_t$. No Bellman recursion is used: diagnostic expectations are not dynamically consistent, so $\mathbb{E}_t^\theta$ applied to the date-$(t+1)$ value formed under $\mathbb{E}_{t+1}^\theta$ need not equal the date-t perceived continuation. [← Back to Proposition 1] □

A.2 Proof of Proposition 2

Proof. Let $\mathcal{F}_t^+$ be the σ-field generated by $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$ and the date-t discovery fields $(X_{it}^s, X_{it}^g)_i$. By the selection rule in the statement, the conditional dated equilibrium of Definition 1 is $\mathcal{F}_t^+$-measurable, and $\mathbf{z}_{t+1}$ is independent of $\mathcal{F}_t^+$ conditional on Ω_t because the discovery fields are independent across dates and of the aggregate gaps. In (11) the likelihood ratio $\Lambda_t(\mathbf{z}_{t+1})^\theta$ is a function of $\mathbf{z}_{t+1}$ alone, so for every integrable $\mathcal{F}_t^+$-measurable X_t, $\mathbb{E}_t[X_t \Lambda_t(\mathbf{z}_{t+1})^\theta] = \mathbb{E}_t[X_t]\,\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]$ and $\mathbb{E}_t^\theta[X_t] = \mathbb{E}_t[X_t]$. Every within-period objective in Section III—the worker's expected indirect utility (36), the producer's expected net revenue (28), the carrier's expected toll (50), and the county current dividend (56)—is of this form, so the conditional dated equilibrium is invariant to θ once $\mathbf{x}_t$ is fixed, and the next physical state follows from the same discoveries and shocks through (38)–(40) and (54). [← Back to Proposition 2] □

A.3 Proof of Proposition 3

Proof. Let $(\mathcal{D}A^+)(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell - r|} A^+(r)$ and $F = \mathcal{D}A^+$. Taking $r = \ell$ gives $A^+ \leq F$, and the triangle inequality $|\ell - s| \leq |\ell - r| + |r - s|$ implies $\mathcal{D}F = F$. Because $A' = e^{-\lambda}A^+ + (1 - e^{-\lambda})F$ is a convex combination of A^+ and F for $\lambda \geq 0$, we have $A^+ \leq A' \leq F$ and hence $F = \mathcal{D}A^+ \leq \mathcal{D}A' \leq \mathcal{D}F = F$. Without further discoveries, F is unchanged and iterating $F - A' = e^{-\lambda}(F - A^+)$ gives (42); since $\lambda_t(\ell)$ increases in $T_t(\ell)$ and $F - A^+ \geq 0$, higher $T_t(\ell)$ weakly raises $A_{i,t+1}(\ell)$. [← Back to Proposition 3] □

A.4 Proof of Proposition 4

Proof. Since $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ is nonempty, compact, and convex and each Π_{jt}^θ is continuous in $\mathbf{x}$ and concave in x_{jt}, Kakutani's fixed-point theorem yields a nonempty compact Nash equilibrium set $\mathcal{N}_t^\theta$. When Π_{jt}^θ is differentiable, concavity makes the variational inequality

$$g_{jt}^\theta(\mathbf{x}^*)(u - x_j^*) \leq 0 \quad \forall\, u \in [0, \bar{x}_{jt}], \quad j = 1, \ldots, J, \qquad g_{jt}^\theta = \partial_j \Pi_{jt}^\theta, \tag{A.1}$$

necessary and sufficient. Under (63), each best response $b_j(\mathbf{y}_{-j})$ is unique, and comparing $u = b_j(\mathbf{y}_{-j})$ with $v = b_j(\mathbf{z}_{-j})$ in (A.1) gives $(u - v)\{g_j(u, \mathbf{y}_{-j}) - g_j(v, \mathbf{z}_{-j})\} \geq 0$, hence $m_j|u - v| \leq$

$\sum_{k\neq j} L_{jk}|y_k - z_k|$. In the weighted maximum norm $\|\mathbf{q}\|_v = \max_j |q_j|/v_j$, we obtain $\|\mathbf{b}(\mathbf{y}) - \mathbf{b}(\mathbf{z})\|_v \leq \gamma_I \|\mathbf{y} - \mathbf{z}\|_v$ with $\gamma_I < 1$ by (64), so Banach's fixed-point theorem yields a unique Nash equilibrium and geometric convergence of best responses. Under strong monotonicity instead, summing the J inequalities (A.1) at two equilibria $\mathbf{x}$ and $\mathbf{y}$ (with $u = y_j$ and $u = x_j$, respectively) gives $\sum_j (x_j - y_j)[g_j(\mathbf{y}) - g_j(\mathbf{x})] \leq 0$, so $\mu\|\mathbf{x} - \mathbf{y}\|^2 \leq 0$ and $\mathbf{x} = \mathbf{y}$ (Vives, 1999). Measurability of the selection in Ω_t follows in the first case because the equilibrium is the pointwise limit of the measurable best-response iterates (each a measurable maximizer of a Carathéodory payoff), and in the second because it is the unique solution of the variational inequality with a Carathéodory pseudogradient. [← Back to Proposition 4] □

A.5 Proof of Proposition 5

Proof. Fix a realization ω with absolutely continuous $p_i(\cdot, \omega)$ and $H_i(\cdot, \omega)$, so that $p_i H_i$ is absolutely continuous and the fundamental theorem of calculus applies to it. Writing local carrier demand as $D = \sum_i r_i(Y_i + |H_i|)$ and using local labor clearing $n = \sum_i e_i L_i + L_T$, summing the local ledgers (70)–(72) gives

$$R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi = \sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| + \tau_{j(\ell)}(Y_T - D) - wn - \sum_i e_i \Psi_i - p_M I. \quad \text{(A.2)}$$

Because τ_j is constant within each county $\mathcal{C}_j$, county service clearing (51) implies

$$\int_{\mathcal{C}_j} \tau_j Y_T \, d\ell = \tau_j K_j = \tau_j D_j = \int_{\mathcal{C}_j} \tau_j D \, d\ell, \quad \text{(A.3)}$$

so integrating (A.2) over $[0, 1]$ yields

$$b_t \bar{L}_t = \int_0^1 (R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi) d\ell = \int_0^1 \Big[\sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| - wn - \sum_i e_i \Psi_i - p_M I \Big] d\ell. \quad \text{(A.4)}$$

Meanwhile, combining goods conservation (69) with $H_i \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)})|H_i|$ from spatial no-arbitrage (45) cancels the physical loss terms $\pm \kappa p_i |H_i|$ in $\partial_\ell(p_i H_i) = p_i \partial_\ell H_i + H_i \partial_\ell p_i$, giving $\partial_\ell(p_i H_i) = p_i(Y_i - nc_i - e_i \Psi_i / p_i - I_i) + r_i \tau_{j(\ell)} |H_i|$ ($I_A = 0$, $I_M = I$). Integrating over $[0, 1]$ under $H_i(0) = H_i(1) = 0$, summing over $i \in \{A, M\}$, and substituting (A.4) gives $\int_0^1 \sum_i p_i n c_i \, d\ell = \int_0^1 wn \, d\ell + b_t \bar{L}_t$. [← Back to Proposition 5] □

A.6 Proof of Proposition 6

A.6.1 Part 1: Firm attainment, active-research threshold, and pure-site sorting

Fix $\ell \in [0, 1]$ and sector $i \in \{A, M\}$ at bounded quotes $0 \leq p_i(\omega) \leq \bar{p} < \infty$, $\tau_{j(\ell)}(\omega) \geq 0$, and $w > 0$, and write $m_i(\phi) = \mathbb{E}_t[(p_i - r_i \tau_{j(\ell)})_+ G_i(\phi)]$.

LEMMA A.1: *Under (22)–(23) with $a_i > 1/(1 - \alpha_i) > 1$, $m_i(\phi)$ is continuous on $[0, 1]$ and satisfies*

$$\bar{m}_i := \sup_{0 \leq \phi \leq 1} m_i(\phi) \leq \bar{p} \sup_{0 \leq \phi \leq 1} \mathbb{E}[G_i(\phi)] = \bar{p} \, \frac{a_i}{a_i - 1} < \infty. \quad \text{(A.5)}$$

For each $\phi \in [0, 1)$, optimal production labor is unique and given by

$$L_i^*(\phi) = \left[\frac{\alpha_i Z_i m_i(\phi)}{(1 + \psi_i) w} \right]^{1/(1 - \alpha_i)}, \quad \text{(A.6)}$$

and the profiled bid rent

$$R_i(\phi; w) = \Lambda_i w^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi)^{1/(1-\alpha_i)} - \Psi_i(\phi; w), \qquad \Lambda_i = (1-\alpha_i) Z_i \left[\frac{\alpha_i Z_i}{1+\psi_i}\right]^{\alpha_i/(1-\alpha_i)}, \tag{A.7}$$

is finite and attained on $[0, 1)$.

Proof. Since $G_i(\phi; \ell, \omega) = 1 + \mathbf{1}_{\{U_i < \phi\}}(V_i - 1)$ with $U_i = \Phi(X_{it}^s(\ell))$ independent of $V_i = \Phi(X_{it}^g(\ell))^{-1/a_i}$ by construction and, a single site being atomistic, independent of the quoted (p_i, τ_j) at ℓ (a maintained restriction on the quote fields), $m_i(\phi) = m_i(0) + \phi\, \mathbb{E}_t[(p_i - r_i \tau_{j(\ell)})_+ (V_i - 1)] = m_i(0)[1 + \kappa_i(\ell)\phi]$ is affine on $[0, 1]$ with $\mathbb{E}[G_i(\phi)] = 1 + \phi/(a_i - 1) \le a_i/(a_i - 1) < \infty$, giving (A.5). Strict concavity in L^{α_i} yields (A.6)–(A.7), and $\Psi_i(\phi; w) \ge w\chi_i/(1-\phi) \to \infty$ as $\phi \uparrow 1$ ensures attainment on $[0, 1)$. □

At an interior optimum $\tilde{\phi}_i \in (0, 1)$, the envelope theorem gives

$$Z_i L_i^*(\tilde{\phi}_i)^{\alpha_i}\, \partial_\phi \mathbb{E}_t\Big[(p_i - r_i \tau_{j(\ell)})_+ G_i(\tilde{\phi}_i)\Big] = \partial_\phi \Psi_i(\tilde{\phi}_i; w) = \frac{w\chi_i}{(1-\tilde{\phi}_i)^2}, \tag{A.8}$$

matching (32). For $w_2 > w_1$ with optimal plans (ϕ_2, L_2) and (ϕ_1, L_1), writing $R_i(\phi; w, \ell)/w = \pi_i^0(w, \ell)[1 + \Gamma_i(\phi; \ell)] - C_i(\phi)$ with $C_i(\phi) = \Psi_i(\phi; w, \ell)/w$ and summing the revealed-preference inequalities gives

$$[\pi_i^0(w_2, \ell) - \pi_i^0(w_1, \ell)]\,[\Gamma_i(\phi_2; \ell) - \Gamma_i(\phi_1; \ell)] \ge 0. \tag{A.9}$$

Since $\pi_i^0(w_2, \ell) < \pi_i^0(w_1, \ell)$ and $\Gamma_i(\cdot; \ell)$ is strictly increasing, $\phi_2 \le \phi_1$, $m_i(\phi_2; \ell) \le m_i(\phi_1; \ell)$, and $L_2 = L_i^*(\phi_2; w_2, \ell) < L_i^*(\phi_1; w_1, \ell) = L_1$: thus $\phi_i^*(w, \ell)$ is weakly decreasing and $L_i^*(\phi_i^*(w, \ell); w, \ell)$ is strictly decreasing in w. Since $\partial^2_{\phi\phi}[m_i(\phi)^{1/(1-\alpha_i)}] = \frac{\alpha_i}{(1-\alpha_i)^2} m_i(\phi)^{(2\alpha_i - 1)/(1-\alpha_i)} [\partial_\phi m_i]^2$ with $\partial_\phi m_i = m_i(0; \ell)\kappa_i(\ell)$, strict concavity $\partial^2_{\phi\phi} R_i(\phi; w, \ell) < 0$ on $(0, 1)$ holds for $\chi_i > \bar{\chi}_{it} \equiv \frac{\alpha_i}{2(1-\alpha_i)^2} \sup_{\ell,\, w \in [\underline{w}_t, \overline{W}_t],\, \phi \in (0,1)} (1-\phi)^3 [m_i(0; \ell)\kappa_i(\ell)]^2 \Lambda_i(\ell) w^{-1/(1-\alpha_i)} m_i(\phi; \ell)^{(2\alpha_i - 1)/(1-\alpha_i)}$, so at most one interior stationary point $\tilde{\phi}_i(w, \ell)$ exists, and (33) uniquely selects $\phi_i^*(w, \ell) \in \{0, \tilde{\phi}_i(w, \ell)\}$. The reservation threshold is well defined: $S_i(\pi) \equiv \sup_{\phi \in (0,1)} [\pi \Gamma_i(\phi; \ell) - f_i(\ell) - \chi_i/(1-\phi)]$ is convex and nondecreasing in π as a supremum of affine functions with nonnegative slopes, $S_i(0) = -f_i(\ell) - \chi_i < 0$, and $S_i(\pi) \to \infty$ when $\kappa_i(\ell) > 0$, so S_i has a unique root $\underline{\pi}_i(\ell) \in (0, \infty)$ and research is active exactly when $\pi_i^0(w, \ell) > \underline{\pi}_i(\ell)$; at equality both 0 and $\tilde{\phi}_i$ are optimal and the convention $\phi_i^* = 0$ applies. If $\kappa_i(\ell) = 0$, research is never active ($\underline{\pi}_i(\ell) = \infty$), and if $m_i(0; \ell) = 0$ the sector is dormant at ℓ with $R_i = L_i^* = \phi_i^* = 0$. Under Condition 2, $R_A(\ell) - R_M(\ell)$ crosses zero transversally at a unique $b_{ct} \in (0, 1)$, assigning $[0, b_{ct}]$ to agriculture and $(b_{ct}, 1]$ to manufacturing. With e_{it} and ϕ_{it}^* fixed at every ℓ, (38) determines $A_{it}^+(\ell, \omega)$ pointwise, and Lemma S.C.1 (Online Appendix) ensures $F_{it}(\ell, \omega)$ in (39) is finite a.s. with finite mean everywhere on $[0, 1]$.

A.6.2 Part 2: Existence and norm-contraction global uniqueness

Proof. On $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ (Condition 1), $G_i(\phi_i^*; \ell, \omega) \ge 1$ implies $L_i^*(\phi_i^*; w, \ell) \ge L_i^*(0; w, \ell) \propto w^{-1/(1-\alpha_i)}$ ($L_T^* \propto w^{-1/(1-\alpha_T)}$), whereas $\Psi_i(\phi_i^*; w, \ell) = O(w)$ vanishes as $w \downarrow 0$; hence aggregate labor demand exceeds $\bar{L}_t$ at $\underline{w}_t > 0$ and falls below $\bar{L}_t$ at $\overline{W}_t < \infty$. Given predetermined capacities and Condition 1's subsistence, construction, and service slack, maximizing the strictly concave Negishi-weighted Stone–Geary log-surplus $\int_0^1 \vartheta(\ell, \omega) n(\ell) [\eta \log(c_A(\ell, \omega) - \gamma) + (1-\eta) \log c_M(\ell, \omega)]\, d\ell$ at weights $\vartheta(\ell, \omega) = w(\ell) + b_t(\omega) - \gamma p_A(\ell, \omega)$ subject to (69) and $D_j(\omega) \le K_j$ (51) yields unique KKT spot prices $\widetilde{p}_i(\cdot, \omega) \in W^{1,\infty}([0, 1])$ ($\widetilde{p}_A(1, \omega) = 1$) and county tolls $\widetilde{\tau}_j(\omega) \ge 0$. Since $|\partial_\ell \widetilde{p}_i| \le \kappa(T_t)\widetilde{p}_i + r_i \widetilde{\tau}_{j(\ell)}$ a.e., substituting $(\widetilde{p}(w), \widetilde{\tau}(w))$ into $\mathcal{T}_t^w$ in (74) defines a reduced wage operator $\widetilde{\mathcal{T}}_t^w : \mathcal{W}_t \to \mathcal{W}_t$

on $C([0,1])$ with equicontinuous image. By Arzelà–Ascoli and Schauder, $\mathcal{T}_t$ admits a fixed point $q^* = (w^*, p^*, \tau^*)$, which is unique under Condition 3 since $\|q_1 - q_0\| = \|\mathcal{T}_t(q_1) - \mathcal{T}_t(q_0)\| \le \bar{\rho}_t \|q_1 - q_0\|$ with $\bar{\rho}_t < 1$. □

A.6.3 Part 3: Constructive verification from primitives

Proof. Step 1 (Frictionless benchmark). When $\kappa_0 = r_A = r_M = m_I = 0$ and $f_i + \chi_i > \bar{f}_i \equiv \sup_{\ell,\phi\in(0,1)}(1-\phi)\pi_i^0(w_t^0, \ell)\Gamma_i(\phi;\ell)$, prices and wages equalize across space ($p_{At} \equiv 1$, $p_{Mt} \equiv p_{Mt}^0 > 0$, $\tau_{jt} \equiv 0$, $\phi_{it}^* \equiv 0$, $w_t \equiv w_t^0$). Opposite-signed log-knowledge slopes ($\partial_\ell \log A_{At} < 0 < \partial_\ell \log A_{Mt}$) make $\log(R_{At}(\ell)/R_{Mt}(\ell))$ strictly decreasing in ℓ, pinning down a unique cutoff $b_{ct}^0 \in (0,1)$ with $\partial_\ell[R_{At}(b_{ct}^0) - R_{Mt}(b_{ct}^0)] < 0$, while labor clearing and Stone–Geary expenditure shares determine $(w_t^0, p_{Mt}^0, b_{ct}^0)$ in closed form.

Step 2 (Uniform price and wage bounds). For any accumulated stock $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$, higher $s_{j(\ell),t}$ expands carrier capacity K_j and reduces route loss $\kappa_0 e^{-T_t}$, so county tolls are bounded by their initial-stock ceilings $\overline{\tau}_j(0)$ and bounded below in expectation by $\underline{\tau}_{jt} > 0$. Integrating (45) from $p_{At}(1,\omega) = 1$ across counties $j = J, \dots, 1$ bounds $p_{it}(\ell,\omega) \in [\underline{p}_i, \overline{p}_i]$ and $w(\ell) \in [\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}]$ uniformly across all stock paths $\{s_{jt}\}_{j=1}^J$, verifying Conditions 1 and 2.

Step 3 (Contraction under small frictions). Since b_{ct} and any active-research thresholds cross zero transversally, $\mathcal{T}_t$ is C^1 on $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ and contracts at the frictionless benchmark with modulus $\rho_0 < 1$ governed by diminishing returns to labor ($-1/(1-\alpha_i) < 0$) and Stone–Geary demand monotonicity. By continuity of $\mathcal{D}_t$, $\sup_q \|\mathcal{D}_t(q)\| \le \frac{1}{2}(1+\rho_0) < 1$ persists for small $(\kappa_0 e^{-T_0}, r_A, r_M, m_I)$ uniformly along every admissible infrastructure path, verifying Condition 3. [← Back to Proposition 6] □

A.7 Proof of Proposition 7

A.7.1 Part 1: Recursive county contraction and state transitions

Let $K_{\mathbf{x}}^\theta$ be the one-period forecasting kernel on Ω under (11)–(13), with stage value $Q_j(\Omega, \mathbf{x}; V) = R_j(\mathbf{x}; \Omega) + \beta(K_{\mathbf{x}}^\theta V_j)(\Omega)$. The weight $W(\Omega_t) = 1 + \sum_{i\in\{A,M\}} \|A_{it}\|_\infty + \sum_{j=1}^J \overline{T}_{jt}$ has finite one-step conditional mean $(K_{\mathbf{x}}^\theta W)(\Omega_t) \le \kappa W(\Omega_t)$ because $\sup_{\ell\in[0,1]} A_{i,t+1}(\ell) \le \sup_{\ell\in[0,1]} F_{it}(\ell)$ has finite conditional mean by Lemma S.C.1 (Online Appendix).

THEOREM A.1: *On a forecast-invariant state domain where each $\mathbf{x} \in \mathcal{X}_t$ has a unique measurable private equilibrium, let $W \ge 1$ be a measurable weight and $\mathcal{C}$ a closed class of county-value vectors equipped with the weighted C^1 route-stock norm $\|V\|_{W,1} = \max_j \sup_\Omega W(\Omega)^{-1}(|V_j(\Omega)| + \sum_{k=1}^J \overline{T}_k |\partial_{\overline{T}_k} V_j(\Omega)|)$. Suppose $|R_j(\mathbf{x};\Omega)| \le C_R W(\Omega)$ and $(K_{\mathbf{x}}^\theta W)(\Omega) \le \kappa W(\Omega)$. If for every $U, V \in \mathcal{C}$ the stage game has a unique measurable Nash profile $a_V(\Omega)$ satisfying $\|a_V(\Omega) - a_U(\Omega)\|_v \le L_a \|V - U\|_{W,1}$, the maximized payoff and its stock derivatives satisfy the Lipschitz bound with modulus $BW(\Omega)$ in $\|\cdot\|_v$, and the Nash Bellman operator $(\mathcal{B}^\theta V)_j(\Omega) = \max_{x_j} Q_j(\Omega, x_j, a_{V,-j}(\Omega); V)$ maps $\mathcal{C}$ into itself with $\beta\kappa + BL_a < 1$, then $\mathcal{B}^\theta$ has a unique fixed point $V^\theta \in \mathcal{C}$ and a unique Markov perfect policy a_{V^θ}.*

Proof. Holding actions fixed, $Q_j(\Omega, \mathbf{x}; V) - Q_j(\Omega, \mathbf{x}; U)$ and its stock derivatives are bounded by $\beta\kappa W(\Omega)\|V - U\|_{W,1}$, while shifting rival actions from $a_{V,-j}$ to $a_{U,-j}$ (and applying the envelope theorem for $a_{V,j}$) adds at most $BW(\Omega)L_a\|V - U\|_{W,1}$. Thus $\|\mathcal{B}^\theta V - \mathcal{B}^\theta U\|_{W,1} \le (\beta\kappa + BL_a)\|V - U\|_{W,1}$, yielding a unique fixed point $V^\theta \in \mathcal{C}$. □

COROLLARY A.1: *Under Theorem A.1, every realization of the physical shock and discovery laws from Ω_0 generates a unique path of allocations, knowledge frontiers, productivity gaps, and route stocks.*

Proof. Composing the unique conditional dated equilibrium map with the unique county policy a_{V^θ} fixes the allocation and investment profile at each state, and iterating (10), (38)–(40), and (54) determines the realized path by induction. □

A.7.2 Part 2: Constructive verification on the rescaled stock corridor

Proof. Fix the $J = 10$ counties $\mathcal{C}_j = [(j-1)/10, j/10]$ and the primitive regime of Proposition 6. At date t with $n \geq 1$ periods remaining in horizon H, county j maximizes $\Pi_{jt}^{(n),\theta}(\mathbf{x}_t; \Omega_t) = R_{jt}(\mathbf{x}_t; \Omega_t) + \beta \mathbb{E}_t^\theta[V_{j,t+1}^{(n-1),\theta}(\Omega_{t+1})]$ ($V_H^{(0),\theta} \equiv 0$), where $R_{jt} = \widetilde{R}_{jt} - m_I(x_{jt} + \frac{1}{2}\chi_I x_{jt}^2) D_{jt}^I$ with $D_{jt}^I \geq D_{\min} > 0$ from (61).

Step 1 (Stage-game diagonal dominance and attenuated cross-county sensitivity). Since $\mathbf{x}_t$ enters the current dated equilibrium only through $I_t(\ell; \mathbf{x}_t) = \sum_{r=1}^J e_{rt} k_{rt}(\ell) X_{rt}$ with $X_{rt} = m_I(x_{rt} + \frac{1}{2}\chi_I x_{rt}^2)$, the chain rule gives $\partial^2_{x_{jt}x_{jt}} \widetilde{R}_{jt} = (\partial_{X_{jt}} \widetilde{R}_{jt}) m_I \chi_I + (\partial^2_{X_{jt}X_{jt}} \widetilde{R}_{jt}) m_I^2 (1 + \chi_I x_{jt})^2$. When construction absorbs a small fraction of manufacturing capacity ($m_I = O(\varepsilon)$ with χ_I large so $|\partial_{X_{jt}} \widetilde{R}_{jt}| \leq \frac{1}{2} D_{jt}^I$), differentiating R_{jt} twice yields $\partial^2_{x_{jt}x_{jt}} R_{jt} \leq -\frac{1}{2} m_I \chi_I D_{jt}^I + O(m_I^2 (1 + \chi_I)^2)$ and $\partial^2_{x_{it}x_{jt}} R_{jt} = O(m_I^2(1 + \chi_I)^2)$ ($i \neq j$). When $H_{jt}^{(n-1)}(\overline{\mathbf{T}}_{t+1}; \mathbf{z}_t, \overline{\mathbf{z}}_t) = \mathbb{E}_t^\theta[V_{j,t+1}^{(n-1),\theta}]$ is concave in $\overline{T}_{j,t+1}$ with $|\partial^2_{\overline{T}_i \overline{T}_j} H_{jt}^{(n-1)}| \leq C_2 \varepsilon^2 / (\overline{T}_{i,t+1} \overline{T}_{j,t+1})^{(1+\alpha_T)/\alpha_T}$, the stage-$n$ pseudogradient $g_{jt}^{(n),\theta} = \partial_{x_j} \Pi_{jt}^{(n),\theta}$ is strongly monotone with modulus $\mu = \frac{1}{8} m_I \chi_I D_{\min} > 0$. By Proposition 4, $\mathbf{x}_t^{(n),\theta}$ is unique, and the implicit function theorem on each active interior set $\mathcal{A} = \{r : 0 < x_{rt}^{(n),\theta} < \bar{x}_{rt}\}$ yields the attenuated stock-sensitivity bound (75): $\|\partial_{\overline{T}_{it}} \mathbf{x}_t^{(n),\theta}\| \leq C_u \varepsilon / (\mu \min_r \overline{T}_{rt}^{(1+\alpha_T)/\alpha_T})$.

Step 2 (Backward induction on county value concavity). Differentiating $V_{jt}^{(n),\theta}(\Omega_t) = \Pi_{jt}^{(n),\theta}(\mathbf{x}_t^{(n),\theta}(\Omega_t); \Omega_t)$ with respect to $\overline{T}_{jt}$ via the envelope theorem eliminates $\partial_{x_j} \Pi_{jt}^{(n),\theta} \partial_{\overline{T}_j} x_{jt}^{(n),\theta} = 0$ a.e., leaving $\partial_{\overline{T}_j} R_{jt} + \beta \partial_{\overline{T}_j} H_{jt}^{(n-1)}$ plus rival feedback $\sum_{h \neq j} \partial_{x_h} \Pi_{jt}^{(n),\theta} \partial_{\overline{T}_j} x_{ht}^{(n),\theta}$. Because carrier and route-loss savings exhibit diminishing returns $\partial^2_{\overline{T}_j \overline{T}_j} R_{jt} \leq -c_R \varepsilon / \overline{T}_{jt}^{(1+2\alpha_T)/\alpha_T} < 0$ and expectations smooth rival policy kinks against the bounded-score truncated-Gaussian kernel of $\mathbf{z}_t$, (75) restricts the rival feedback derivative to $O((\varepsilon/\mu)\varepsilon^2 / (\overline{T}_{it} \overline{T}_{jt})^{(1+\alpha_T)/\alpha_T})$, preserving own-stock concavity and small cross-stock derivatives of $H_{j,t-1}^{(n)}$ for $n = 1, \ldots, H$.

Step 3 (Balanced-growth corridor and infinite-horizon convergence). Rescaling county stocks and increments by $g^t = 1.02^t$ via $\widetilde{T}_{jt} = g^{-t} \overline{T}_{jt}$ and $\bar{d}_j = g^{-t} d_{jt} = |\mathcal{C}_j|^{-1} \int_{\mathcal{C}_j} \iota_{j,1881}(\ell)\, d\ell$ gives $\widetilde{T}_{j,t+1} = g^{-1}(\widetilde{T}_{jt} + \bar{d}_j x_{jt})$. Discounted marginal carrier savings balance $m_I(1 + \chi_I x_{jt}) D_{jt}^I$ in the interior branch of (62) on a compact corridor $\mathcal{K} = \prod_{j=1}^J [L_j, U_j]$ at intensities $x_{jt}^{(n),\theta} \in [\underline{a}_j, \overline{a}_j] \subset (0, 1)$ satisfying $(g-1)L_j/\bar{d}_j \leq \underline{a}_j \leq \overline{a}_j \leq (g-1)U_j/\bar{d}_j$, which maps $\mathcal{K}$ into itself. Along $\mathcal{K}$, the state-weight drift is $\kappa = g^{1-\alpha_T}$ and $BL_a = O(\varepsilon/\mu)$, so $\beta\kappa + BL_a = \beta g^{1-\alpha_T} + O(\varepsilon/\mu) < 1$, and $(V_{jt}^{(H),\theta}, \mathbf{x}_t^{(H),\theta})$ contract geometrically as $H \to \infty$ to a unique infinite-horizon Markov perfect equilibrium. [← Back to Proposition 7] □

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Online Appendix

to "Corrective Overoptimism"

Brandon S.C. Chan

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S.A Independent Route Entry and Contestability

When transport carriage is imperfectly competitive, an independently controlled bypass route lowers carrier markups and generates shipper surplus that the entrant does not appropriate. This two-period extension illustrates how diagnostic expectations interact with imperfect route contestability and Mankiw and Whinston (1986) business stealing.

Operating equilibrium.— At date t, a chartered entrant may sink construction cost $F_t^I > 0$ to open an independent bypass at date $t+1$. Once the date-$t+1$ demand intercept v_{t+1} realizes, $n \in \{1, 2\}$ symmetric Cournot transport firms with constant marginal operating cost $c \geq 0$ choose service quantities against linear inverse demand:

$$\tau_{t+1}(D) = v_{t+1} - bD, \qquad b > 0, \quad \text{MC} = c \geq 0. \tag{S.A.1}$$

Writing $a_{t+1} \equiv (v_{t+1} - c)_+$, equilibrium aggregate service D_n, toll τ_n, and per-firm operating profit π_n with n independent routes are

$$D_n = \frac{n}{n+1}\frac{a_{t+1}}{b}, \qquad \tau_n = c + \frac{a_{t+1}}{n+1}, \qquad \pi_n = \frac{a_{t+1}^2}{b(n+1)^2}, \tag{S.A.2}$$

with ownership-adjusted concentration $H_n = 1/n$. Opening an independent bypass ($n = 1 \to 2$) lowers H_n from 1 to 1/2, compresses the markup from $a_{t+1}/2$ to $a_{t+1}/3$, and expands service from

D_1 to D_2 (Figure S.A.1, Panel b), whereas a route extension under incumbent control leaves $n = 1$ and $H_1 = 1$.

Diagnostic beliefs and private vs. social entry.— Suppose $v_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(\mu_{v,t}, \sigma_v^2)$ with $\sigma_v > 0$ and news $\Delta\mu_{v,t} \equiv \mu_{v,t} - \mathbb{E}_{t-1}[v_{t+1}]$. Under diagnosticity $\theta \geq 0$, the perceived mean is $\mu_{v,t}^{\theta} = \mu_{v,t} + \theta\Delta\mu_{v,t}$, and with standardized gap $z_t^{\theta} \equiv (\mu_{v,t}^{\theta} - c)/\sigma_v$, the perceived second moment of positive net demand is

$$M_t^{\theta} \equiv \mathbb{E}_t^{\theta}\left[(v_{t+1} - c)_+^2\right] \tag{S.A.3}$$

$$= \sigma_v^2\Big[\{1 + (z_t^{\theta})^2\}\Phi_N(z_t^{\theta}) + z_t^{\theta}\phi_N(z_t^{\theta})\Big], \tag{S.A.4}$$

which is strictly increasing in z_t^{θ}. For discount factor $\beta \in (0, 1)$, the duopoly entrant's perceived net present value is

$$V_t^{P,\theta} = \frac{\beta M_t^{\theta}}{9b} - F_t^I \geq 0. \tag{S.A.5}$$

Aggregate transport-market surplus with n independent carriers is

$$S_n(v) = \int_0^{D_n} (v - c - bq)\, dq = \frac{n(n+2)}{2(n+1)^2}\frac{(v-c)_+^2}{b}, \tag{S.A.6}$$

so moving from monopoly to duopoly raises direct transport surplus by $5(v-c)_+^2/(72b)$. Letting Γ_t^{ext} denote the discounted external spatial-reallocation and knowledge-diffusion gains not captured by route operators, objective social net present value is

$$V_t^S = \beta\left\{\frac{5M_t^0}{72b} + \Gamma_t^{\text{ext}}\right\} - F_t^I. \tag{S.A.7}$$

Since expected entrant profit $8M_t^0/(72b)$ exceeds the direct transport-surplus gain $5M_t^0/(72b)$ by the Mankiw and Whinston (1986) business-stealing transfer $3M_t^0/(72b) = M_t^0/(24b)$ from the incumbent, rational entry is socially insufficient whenever external spillovers satisfy $\Gamma_t^{\text{ext}} > M_t^0/(24b)$. Diagnostic optimism ($\Delta\mu_{v,t} > 0$) corrects rational under-entry when

$$\frac{\beta M_t^0}{9b} < F_t^I < \beta\left\{\frac{5M_t^0}{72b} + \Gamma_t^{\text{ext}}\right\} \quad \text{and} \quad F_t^I \leq \frac{\beta M_t^{\theta}}{9b}, \tag{S.A.8}$$

and induces socially excessive duplication when

$$\beta\left\{\frac{5M_t^0}{72b} + \Gamma_t^{\text{ext}}\right\} < F_t^I \leq \frac{\beta M_t^{\theta}}{9b}. \tag{S.A.9}$$

Figure S.A.1 illustrates the corrective threshold (Panel a) and the post-entry expansion in service and Marshallian surplus (Panel b), consistent with empirical evidence on carrier market power and network entry (Brancaccio et al., 2020; Preonas, 2024; Hausman, 2025; Caoui and Steck, 2025).

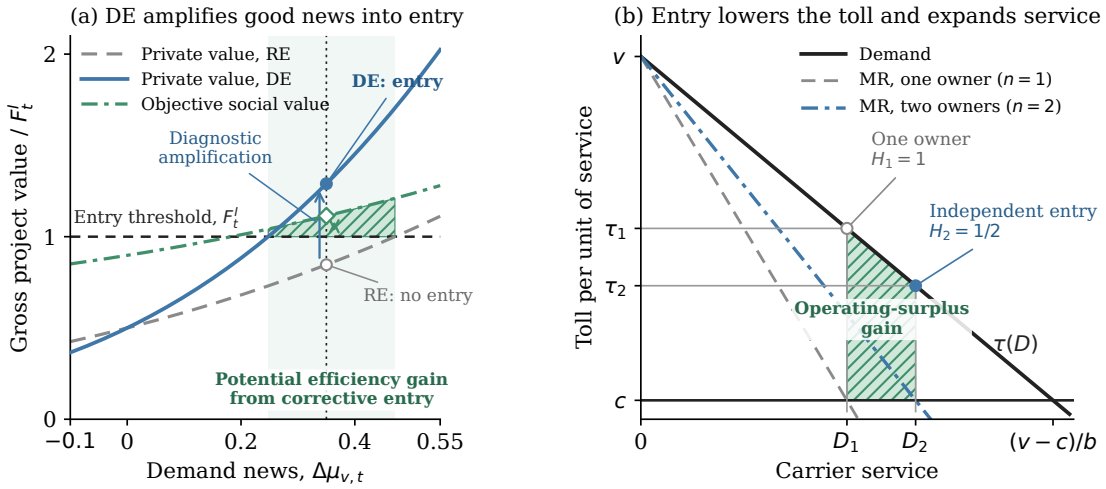


Figure S.A.1: Diagnostic Beliefs and Route Entry

Notes: Panel (a) sets $\beta = 0.9$, $c = 0$, $b = 1$, $\sigma_v = 1$, $\mathbb{E}_{t-1}[v_{t+1}] = 0$, $\theta = 0.9$, $\Gamma_t^{\text{ext}} = 0.065$, and $F_t^I = 0.1$. Panel (b) sets $v - c = b = 1$.

S.B Historical Data and Spatial Aggregation

National and regional occupational series use a common Primary–Secondary–Tertiary (PST) classification. National male (age-15-and-over) occupational shares for England and Wales come from Shaw-Taylor et al. (2019, Table 11) at the benchmark years 1710, 1761, 1817, 1851, 1861, 1871, and 1881, combined with the pre-census trajectory of Keibek (2017, Table 18) interpolated in additive log-ratio differences between adjacent CAMPOP anchors to form the twelve-date panel

$$\mathcal{Y}_O = \{1710, 1721, 1741, 1761, 1781, 1801, 1817, 1841, 1851, 1861, 1871, 1881\}.$$

Primary employment excludes mining; secondary industry includes manufacturing, mining, and construction; and tertiary employment comprises dealers, sellers, transport, and communications (with the sum of these three categories forming the denominator). Because Keibek (2017) classifies mining within primary employment, mining counts are transferred from primary to secondary before normalizing county shares or aligning to the CAMPOP national totals.

Regional occupational shares come from the county reconstructions in Keibek (2017) on fixed 1831 ancient-county boundaries (observed at 1721, 1741, 1761, 1781, 1801, 1817, 1841, and 1851, with 1710 interpolated in additive log ratios between 1701 and 1721) and the dedicated London benchmarks of Field and Shaw-Taylor (2019) (c. 1711, c. 1751, c. 1817, 1851, and 1881, mapped to the fixed 1851 London registration-county footprint). Log-interpolated county populations from Wrigley (2011, Table A2.6) and London population anchors supply the fixed exposure weights N_{by} that convert county and metropolitan occupational shares into relative employment mass. Figures S.B.1 and S.B.2 display the resulting primary and secondary employment shares across ancient counties.

To project county and London shares onto the twenty-five equal-latitude bands $b \in \{1, \ldots, 25\}$ on $\ell \in [0, 1]$, let B_{cb} denote the share of band b's land area lying in county c, $p_{sby} \in [0, 1]$ the share of sector $s \in \{A, M, T\}$ in band b and year y, and N_{by} the reconstructed band employment mass (with normalized weights $\omega_{by} \equiv N_{by} / \sum_{b'=1}^{25} N_{b'y}$ and county weights $W_{cy} \equiv \sum_{b=1}^{25} B_{cb} N_{by}$). The

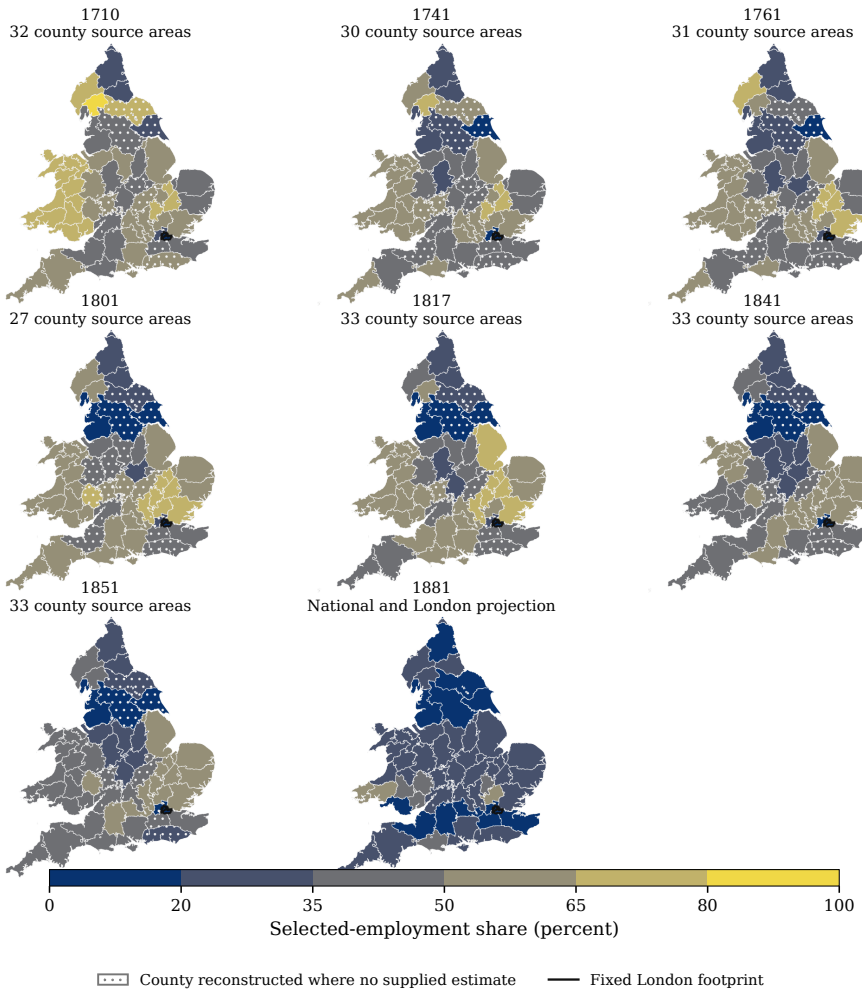


Figure S.B.1: Primary Employment Shares, 1710–1881

Notes: County primary shares through 1851 combine Keibek (2017) and Field and Shaw-Taylor (2019) (London, black outline), and 1881 projects the reconstructed latitude-band surface. Dotted counties indicate reconstructed values.

implied county share is

$$\widehat{p}_{csy} = \frac{\sum_{b=1}^{25} B_{cb} N_{by} p_{sby}}{\sum_{b=1}^{25} B_{cb} N_{by}}.$$

At each pre-1881 date $y \in \mathcal{Y}_O \setminus \{1881\}$, the twenty-five band shares $\{p_{sby}\}$ solve the constrained quadratic program

$$\min_{\{p_{sby} \in [0,1]\}} \sum_{s \in \{A,M,T\}} \left\{ \sum_{c} W_{cy} \left(\widehat{p}_{csy} - p_{csy}^{\text{obs}}\right)^2 + \lambda_{\text{occ}} \sum_{b=2}^{24} \left(p_{s,b+1,y} - 2p_{sby} + p_{s,b-1,y}\right)^2 \right\}$$

subject to $\sum_{s \in \{A,M,T\}} p_{sby} = 1$ for each band b and exact national occupational accounting $\sum_{b=1}^{25} \omega_{by} p_{sby} = P_{sy}^{\text{nat}}$, with spatial roughness weight $\lambda_{\text{occ}} = 0.05$. At the post-1851 census dates $y \in \{1861, 1871, 1881\}$, where ancient-county occupational tables are unavailable outside London, the band surface $\{p_{sby}\}$ updates the 1851 band profile in additive log ratios to match the observed London share $p_{\text{Lon},s,y}^{\text{obs}}$ on the metropolitan bands and the CAMPOP national totals P_{sy}^{nat}, and

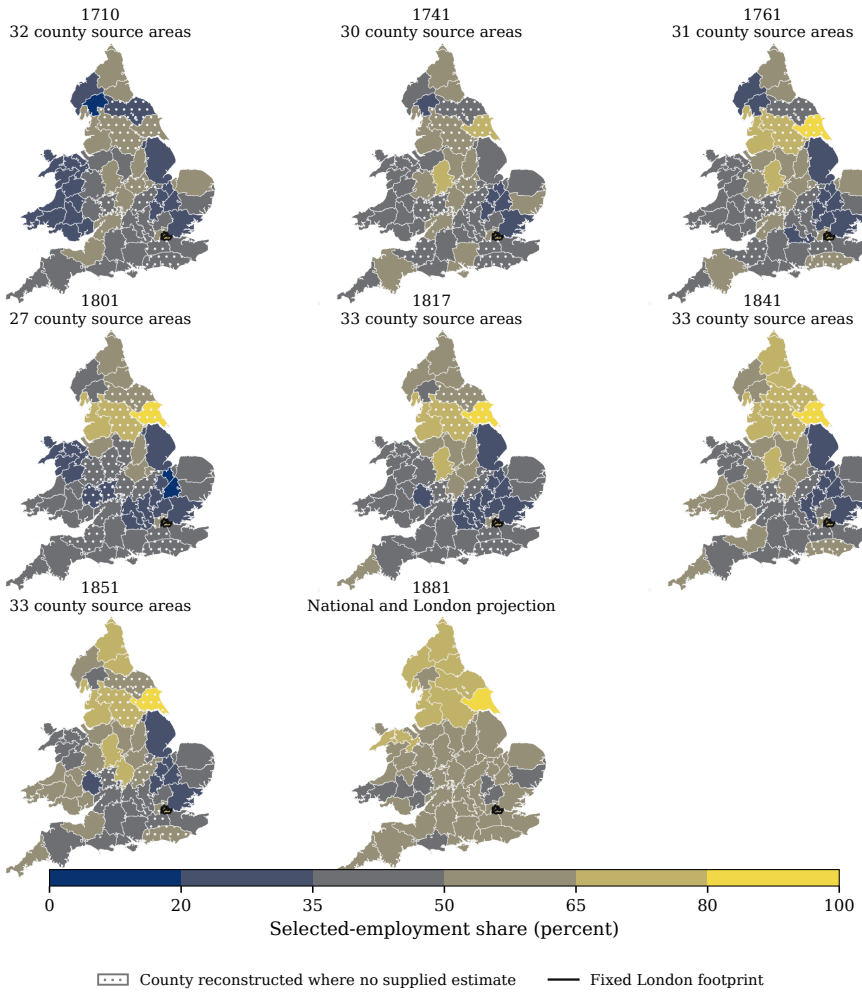


Figure S.B.2: Secondary Employment Shares, 1710–1881

Notes: County secondary shares through 1851 combine Keibek (2017) and Field and Shaw-Taylor (2019) (London, black outline), and 1881 projects the reconstructed latitude-band surface. Dotted counties indicate reconstructed values.

the 1881 county maps in Figures 3, S.B.1, and S.B.2 evaluate the area- and population-weighted projection $\widehat{p}_{c,s,1881} = \sum_{b=1}^{25} B_{cb} N_{b,1881} p_{s,b,1881} / \sum_{b=1}^{25} B_{cb} N_{b,1881}$ (holding London at $p^{\text{obs}}_{\text{Lon},s,1881}$). Each 25-band profile is prolonged piecewise constantly to the 496 interior cells $\{3, \ldots, 498\}$ of the $N = 500$ model grid supported by county centroids (with the four extreme coastal cells $\{1, 2, 499, 500\}$ outside the county-centroid latitude range assigned by nearest-neighbor boundary continuation; Figures S.B.3–S.B.4). The same fixed historical weights N_{by} are applied to data and simulated allocations. Figure S.B.5 reports the complementary 1851 and 1881 registration-subdistrict resident population profiles across the twenty-five latitude bands.

Robustness of regional local projections.— Table S.B.1 reports eight specification checks on the regional local projections in Panel A of Table 1. Row 1 reports two-way standard errors clustered by latitude band and year on the baseline TWFE specification, confirming that cross-corridor spatial correlation does not drive inference (the same one-year reversal holds in the national series, $\rho_1 = -0.325$, SE $= 0.074$). Row 2 replaces leave-one-out shift-share instruments with leave-corridor-out shifts excluding the two adjacent bands on each side ($|k - b| \geq 3$) so trunk projects

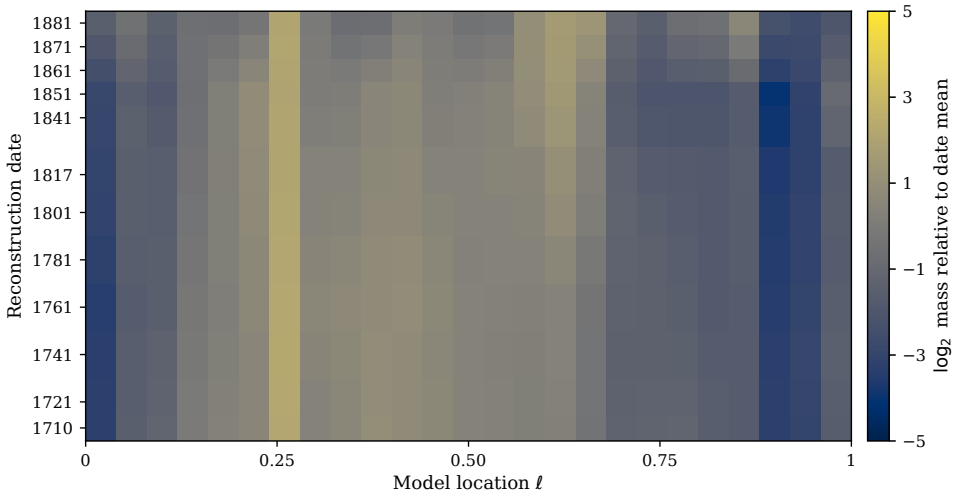


Figure S.B.3: Employment across the Model Interval

Notes: Within-date normalized adult-male employment mass across the twelve benchmark dates using county population weights from Wrigley (2011).

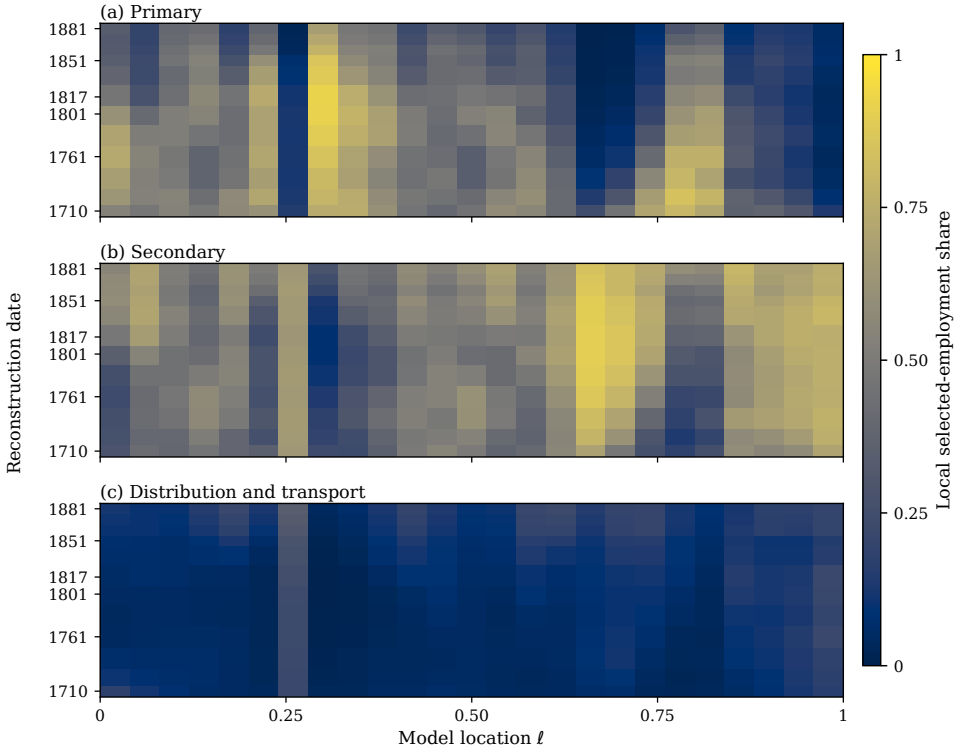


Figure S.B.4: Occupational Composition across the Model Interval

Notes: Primary, secondary, and distribution/transport employment shares across the twenty-five latitude bands at the twelve benchmark dates.

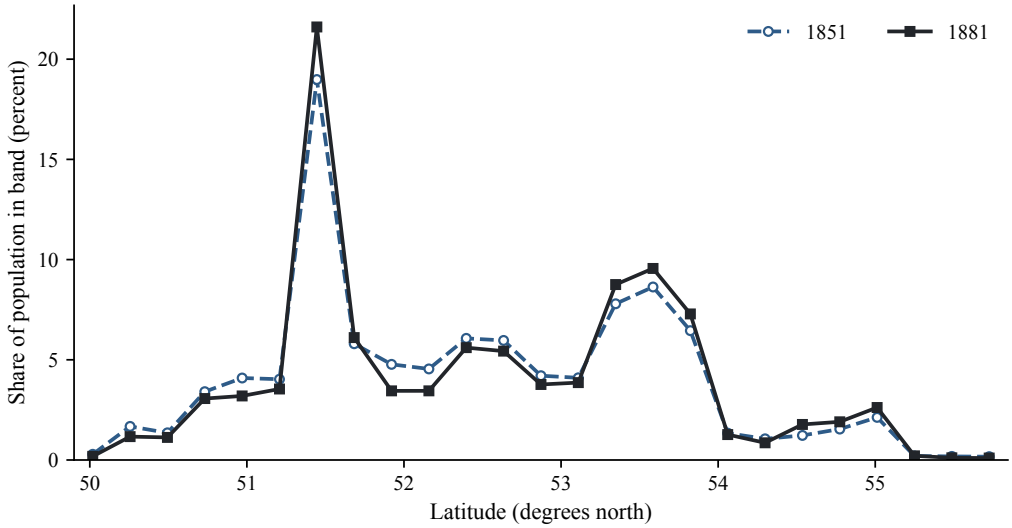


Figure S.B.5: Population across the Model Interval, 1851 and 1881

Notes: CAMPOP registration-subdistrict census population allocated by polygon area to the twenty-five model latitude bands (49.8998–55.8442°N).

spanning neighboring bands cannot correlate the instrument with local residuals. Row 3 detrends log addition rates via a one-sided backward-looking five-year moving average, $c_{b,t}^{1s} = \ln(10^{-3} + r_{b,t}^{I}) - \frac{1}{5}\sum_{j=1}^{5} \ln(10^{-3} + r_{b,t-j}^{I})$, which uses no future realizations after date t and shows positive one-year project-completion momentum ($\rho_1^{1s} = +0.159$) followed by significant reversals at $h = 3, 4, 5$. Row 4 estimates cumulative changes $\Delta_h \ln(10^{-3} + r_{b,t}^{I})$ on unfiltered log addition rates with band and year fixed effects ($\beta_h^{\text{diff}} \in [-0.834, -0.632]$), capturing rapid partial mean reversion ($1 + \beta_h^{\text{diff}} \in [0.166, 0.368]$), whereas level projections on detrended cycles capture drops below local trend. Rows 5 and 6 replace $\log(10^{-3}+r_{b,t}^{I})$ with the level addition rate $r_{b,t}^{I}$ and $\text{arsinh}(100\, r_{b,t}^{I})$, and Rows 7 and 8 split the sample at 1830 into the pre-railway (1710–1829) and railway (1830–1880) eras.

Table S.B.1: Alternative Local-Projection Specifications for Regional Infrastructure Additions

Specification	(1) $h=1$	(2) $h=2$	(3) $h=3$	(4) $h=4$	(5) $h=5$
1. Baseline TWFE (two-way band & year clustered SE)	−0.221	−0.215	−0.121	−0.028	−0.011
	(0.022)	(0.024)	(0.023)	(0.023)	(0.022)
2. Leave-corridor-out 2SLS ($\lvert k-b\rvert \geq 3$)	−0.240	−0.220	−0.096	−0.108	0.003
	(0.050)	(0.036)	(0.043)	(0.048)	(0.036)
3. Strictly one-sided 5-year moving-average filter $c^{1s}_{b,t}$	0.159	−0.017	−0.115	−0.169	−0.196
	(0.020)	(0.016)	(0.019)	(0.018)	(0.014)
4. Unfiltered cumulative change $\Delta_h \ln(10^{-3} + r^I_{b,t})$	−0.632	−0.735	−0.785	−0.806	−0.834
	(0.035)	(0.033)	(0.036)	(0.036)	(0.033)
5. Untransformed level addition rate $\mathrm{HP}_{6.25}[r^I_{b,t}]$	−0.138	−0.284	−0.136	−0.056	0.052
	(0.017)	(0.026)	(0.015)	(0.016)	(0.046)
6. Inverse hyperbolic sine $\mathrm{HP}_{6.25}[\mathrm{arsinh}(100\, r^I_{b,t})]$	−0.184	−0.226	−0.142	−0.050	−0.001
	(0.020)	(0.023)	(0.020)	(0.017)	(0.019)
7. Pre-railway turnpike and canal era (1710–1829)	−0.255	−0.201	−0.101	−0.027	−0.004
	(0.020)	(0.022)	(0.021)	(0.023)	(0.023)
8. Railway era (1830–1880)	0.067	−0.342	−0.274	−0.034	−0.082
	(0.050)	(0.045)	(0.028)	(0.035)	(0.036)

Notes: All specifications include latitude-band and year fixed effects on the 25-band regional transport panel. Standard errors in parentheses are clustered by latitude band (except Row 1, which clusters two-way by band and year).

S.C Stochastic Primitives

This appendix derives the properties of the two stochastic primitives of Section III that the main text uses: the bounded law of aggregate sectoral productivity gaps, with the diagnostic one-step law it implies, and the spatially correlated discovery field, which keeps the knowledge frontier finite.

S.C.1 Bounded sectoral shocks and the diagnostic law

Let $\boldsymbol{\rho} = \mathrm{diag}(\rho_A, \rho_M, \rho_T)$ with $|\rho_s| < 1$, and let $\varphi_{\boldsymbol{\Sigma}}$ be the centered Gaussian density with innovation covariance $\boldsymbol{\Sigma} = \mathbf{D}_z \boldsymbol{\Omega}_z \mathbf{D}_z$, where $\mathbf{D}_z = \mathrm{diag}(\sigma_A, \sigma_M, \sigma_T)$ and $\boldsymbol{\Omega}_z$ is the correlation matrix with off-diagonal entries $(\varrho_{AM}, \varrho_{AT}, \varrho_{MT})$, parameterized via the partial correlation $\varrho_{MT|A} \in (-1, 1)$ as $\varrho_{MT} = \varrho_{AM}\varrho_{AT} + \varrho_{MT|A}\sqrt{(1-\varrho^2_{AM})(1-\varrho^2_{AT})}$ so that $\boldsymbol{\Omega}_z$ is positive definite and the diagonal entries of $\boldsymbol{\Sigma}$ are $\Sigma_{ss} = \sigma_s^2$. Because $\boldsymbol{\rho}$ is diagonal, the untruncated stationary standard deviation of z_s is $\nu_s = \sigma_s/\sqrt{1-\rho_s^2}$ whatever the cross-sector correlations. The compact symmetric support box, box probability, and truncated conditional mean are

$$\begin{aligned} \mathcal{B} &= \prod_{s\in\{A,M,T\}} [-c\nu_s, c\nu_s], \quad \Phi_{\mathcal{B}}(\mathbf{m}) = \int_{\mathcal{B}} \varphi_{\boldsymbol{\Sigma}}(\mathbf{u}-\mathbf{m})\, d\mathbf{u}, \\ M(\mathbf{m}) &= \frac{\int_{\mathcal{B}} \mathbf{u}\, \varphi_{\boldsymbol{\Sigma}}(\mathbf{u}-\mathbf{m})\, d\mathbf{u}}{\Phi_{\mathcal{B}}(\mathbf{m})}, \end{aligned} \tag{S.C.1}$$

with $c = 3$ in the benchmark and $c = 4$ reserved for robustness, and the physical kernel in (10) is $f(\mathbf{z}' \mid \mathbf{z}) = \varphi_{\boldsymbol{\Sigma}}(\mathbf{z}' - \boldsymbol{\rho}\mathbf{z})/\Phi_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z})$ for $\mathbf{z}' \in \mathcal{B}$. Since f is continuous and strictly positive on the compact set $\mathcal{B}\times\mathcal{B}$, it is bounded below by some $\underline{f} > 0$, so $\Pr(\mathbf{z}_{t+1} \in E \mid \mathbf{z}_t) \geq \underline{f}\,|E|$ for every Borel set $E \subseteq \mathcal{B}$ with Lebesgue measure $|E|$. By this Doeblin minorization, f has a unique stationary distribution π on $\mathcal{B}$, approached geometrically from every initial state, and the normalizer in (19),

$$\mathbb{E}_\pi[e^{z_s}] = \int_{\mathcal{B}} e^{z_s}\, \pi(d\mathbf{z}), \qquad s \in \{A, M, T\}, \tag{S.C.2}$$

gives $e^{z_{st}}/\mathbb{E}_\pi[e^{z_s}]$ unit stationary physical mean. For any truncated Gaussian law $\mathbf{z} \sim \mathrm{TN}_{\mathcal{B}}(\mathbf{m}, \boldsymbol{\Sigma})$, completing the square in $\mathbf{u}'\mathbf{e}_i - \frac{1}{2}(\mathbf{u}-\mathbf{m})'\boldsymbol{\Sigma}^{-1}(\mathbf{u}-\mathbf{m})$ gives the conditional mean of effective productivity, with $\mathbf{m} = \boldsymbol{\rho}\mathbf{z}_t$ under physical beliefs and $\mathbf{m} = \mathbf{m}_t^\theta$ under diagnostic beliefs:

$$\mathbb{E}\left[\frac{A_i e^{z_i}}{\mathbb{E}_\pi[e^{z_i}]} \,\middle|\, \mathbf{m}\right] = \frac{A_i}{\mathbb{E}_\pi[e^{z_i}]}\, e^{m_i + \frac{1}{2}\sigma_i^2}\, \frac{\Phi_{\mathcal{B}}(\mathbf{m} + \boldsymbol{\Sigma}\mathbf{e}_i)}{\Phi_{\mathcal{B}}(\mathbf{m})}. \tag{S.C.3}$$

For the diagnostic law, the box normalizers $\Phi_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t)$ and $\Phi_{\mathcal{B}}(\boldsymbol{\rho}\bar{\mathbf{z}}_t)$ in the representativeness ratio Λ_t do not depend on the future state and cancel from (11). The remaining kernel $\varphi_{\boldsymbol{\Sigma}}(\mathbf{u} - \boldsymbol{\rho}\mathbf{z}_t)^{1+\theta}\varphi_{\boldsymbol{\Sigma}}(\mathbf{u} - \boldsymbol{\rho}\bar{\mathbf{z}}_t)^{-\theta}$ has log-quadratic term $-\frac{1}{2}\mathbf{u}'\boldsymbol{\Sigma}^{-1}\mathbf{u}$, because the exponents $1+\theta$ and $-\theta$ sum to one, and linear term $\mathbf{u}'\boldsymbol{\Sigma}^{-1}[(1+\theta)\boldsymbol{\rho}\mathbf{z}_t - \theta\boldsymbol{\rho}\bar{\mathbf{z}}_t]$. Restricted to $\mathcal{B}$, it is therefore a truncated Gaussian with covariance $\boldsymbol{\Sigma}$ and location $\boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}(\mathbf{z}_t - \bar{\mathbf{z}}_t)$, which is (12).

S.C.2 Spatially correlated discoveries and frontier moments

If discoveries were drawn independently across sites, any interval of active research would contain infinitely many independent Pareto draws, which are unbounded almost surely, and the spatial supremum (39) would be infinite. The discovery field (24) keeps the Bernoulli–Pareto law at every site but draws discoveries from Gaussian fields with continuous sample paths, which bounds the frontier and its moments.

LEMMA S.C.1: *Fix sector $i \in \{A, M\}$, bounded positive inherited knowledge A_{it}, pre-discovery plans (e_{it}, ϕ_{it}), and discoveries $G_{it}(\ell) = G_i(\phi_{it}(\ell); \ell, \omega)$ generated by (24) with correlation length $d_0 > 0$. Then the frontier $F_{it}(\ell)$ in (39) is finite almost surely, and for every $0 < b < a_i$ and $F_{it}^0(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell - r|} A_{it}(r)$,*

$$\mathbb{E}_t[F_{it}(\ell)^b] < \infty, \tag{S.C.4}$$

$$\mathbb{E}_t[F_{it}(\ell)^b] = [F_{it}^0(\ell)]^b + b\int_{F_{it}^0(\ell)}^{\infty} v^{b-1} \Pr_t(F_{it}(\ell) > v)\, dv. \tag{S.C.5}$$

For $a_i > 1$, next-period expected knowledge under (40) satisfies

$$\mathbb{E}_t[A_{i,t+1}(\ell)] = e^{-\lambda_t(\ell)} A_{it}(\ell)\left[1 + \frac{e_{it}(\ell)\phi_{it}(\ell)}{a_i - 1}\right] + (1 - e^{-\lambda_t(\ell)})\mathbb{E}_t[F_{it}(\ell)]. \tag{S.C.6}$$

Proof. Because X_{it}^g has continuous sample paths on the compact set $[0,1]$, $\overline{X} = \sup_{r\in[0,1]}[-X_{it}^g(r)]$ is finite almost surely, and the Borell–TIS inequality (Adler et al., 2015, Theorem 2.6.1) with unit variance gives $\bar{\mu} = \mathbb{E}[\overline{X}] < \infty$ and $\Pr(\overline{X} \geq \bar{\mu} + x) \leq e^{-x^2/2}$ for $x > 0$. Pathwise, $1 \leq G_{it}(r) \leq \Phi(X_{it}^g(r))^{-1/a_i} \leq \Phi(-\overline{X})^{-1/a_i}$, and the Gaussian tail bound $\Phi(-x) \geq c_\Phi(1+x)^{-1}e^{-x^2/2}$ ($x \geq 0$, $c_\Phi > 0$) gives $\sup_{r\in[0,1]} G_{it}(r)^b \leq C(1 + \overline{X}_+)^{b/a_i} \exp\{(b/(2a_i))\overline{X}_+^2\}$, which is integrable whenever $b < a_i$: with $Y = (\overline{X} - \bar{\mu})_+$ and $c = b/(2a_i) < \frac{1}{2}$, the Borell–TIS tail gives $\mathbb{E}[e^{c'Y^2}] = 1 + \int_0^\infty 2c'xe^{c'x^2}\Pr(Y > x)\,dx \leq 1 + 2c'/(1-2c') < \infty$ for every $c' < \frac{1}{2}$, and the polynomial factor and the shift by $\bar{\mu}$ are absorbed by taking c' slightly above c. Since $F_{it}(\ell) \leq \sup_r A_{it}(r)\, \sup_r G_{it}(r)$, the frontier is finite almost surely and (S.C.4) holds. Because $G_{it} \geq 1$ implies $F_{it} \geq F_{it}^0$, the layer-cake identity gives (S.C.5). Finally, $\mathbb{E}_t[G_{it}(\ell) - 1] = \phi_{it}(\ell)/(a_i - 1)$ under (22)–(23), and taking expectations in $A_{i,t+1} = e^{-\lambda_t}A_{it}^+ + (1 - e^{-\lambda_t})F_{it}$ with (38) and $\mathbb{E}_t[F_{it}] < \infty$ ($b = 1 < a_i$) yields (S.C.6). □

S.D Estimation Design, Moment Construction, and Covariance

This section details the parameterization, empirical observation maps, moment formulas, and weighting matrix used in the Simulated Method of Moments (SMM) estimation of Section IV.C.

S.D.1 Estimated and fixed parameters

Under diagnostic expectations, the SMM criterion (78) jointly estimates $p = 26$ free structural parameters: $\theta \geq 0$, the 16 preference, innovation, spatial diffusion, transport, and construction-cost parameters (η, γ, a_A, a_M, f_A, f_M, χ_A, χ_M, ω_E, δ, λ_0, α_T, κ_0, r_A, m_I, χ_I), and the 9 aggregate sectoral shock parameters

$$\boldsymbol{\psi}^z = (\rho_A, \rho_M, \rho_T, \sigma_A, \sigma_M, \sigma_T, \varrho_{AM}, \varrho_{AT}, \varrho_{MT|A})', \tag{S.D.1}$$

while the manufacturing wedge scale $\bar{\psi}_M = 0.251$ (SE $= 0.032$) is calibrated from the gross-operating-surplus identity (77) (Table 5). Under rational expectations, the model is re-estimated subject to $\theta = 0$ ($p = 25$ free parameters). Table 4 summarizes the identifying margins and Table 3 lists the fixed parameters.

The initial 1710 log-productivity profiles $\ln A_{i,1710}(\ell)$ ($i \in \{A, M\}$) use a five-knot natural cubic spline basis on $\{0, 1/4, 1/2, 3/4, 1\}$ with four shape parameters per sector and zero second derivatives at $\ell \in \{0, 1\}$. Goods units normalize $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$ for $i \in \{A, M\}$, and at each candidate parameter vector the eight spline parameters are concentrated out on the 1710 occupational cross-section. Nominal units fix $p_A(1, t) = 1$, household mass starts at $\bar{L}_{1710} = 100$ and evolves via (9), and the 1710 multimode transport stock has cross-band mean one. Every candidate solves the conditional dated equilibrium and the Markov perfect county investment equilibrium on the $N = 500$ grid across $J = 10$ counties, matching the $M = 30$ moments in Table 6 ($M - p = 4$ overidentifying degrees of freedom under DE and 5 under RE).

S.D.2 Observation maps and empirical roles

At each calendar date y, sectoral employment aggregates over grid cells of width $\Delta\ell_q$ (Replication Appendix R.A.3.1) and national occupational shares are formed from predetermined production and carrier hiring:

$$\begin{aligned} \mathcal{L}_A(y) &= \sum_{q=1}^{N} \Delta\ell_q e_{Aq}(y) L_{Aq}(y), \quad \mathcal{L}_M(y) = \sum_{q=1}^{N} \Delta\ell_q e_{Mq}(y) L_{Mq}(y), \\ \mathcal{L}_T(y) &= \sum_{q=1}^{N} \Delta\ell_q L_{Tq}(y), \qquad\qquad \pi^{\text{occ}}_{o,y} = \frac{\mathcal{L}_o(y)}{\sum_{o' \in \{A,M,T\}} \mathcal{L}_{o'}(y)}, \quad o \in \{A, M, T\}, \end{aligned} \tag{S.D.2}$$

satisfying $\sum_{o \in \{A,M,T\}} \mathcal{L}_o(y) = \sum_{q=1}^{N} \Delta\ell_q n_q(y) = \bar{L}_y$ since innovation is goods-financed (26) and hires no separate research headcount. National occupational shares aggregate the model domain, whereas dated spatial covariances aggregate pure-site employment onto the 496 supported cells of the twenty-five latitude bands. Table S.D.1 summarizes the empirical series and their quantitative roles.

Observed 1710 transport capacity initializes $T_0(\ell)$, whereas post-1710 stocks enter only as empirical moment targets. Because the source-bootstrap covariance matrix conditions on the historical spatial and demographic reconstructions, the asymptotic standard errors and Hansen (1982) J-statistics are interpreted conditionally on those reconstructed surfaces.

S.D.3 Complete historical profiles

To capture both the timing and steepness of the Industrial Revolution, the targeted moments summarize the complete temporal trajectories of the national-account, occupational, and transport-stock series plotted in Figure S.D.1. For an n-date series $z = (z_1, \ldots, z_n)'$, let $H_n \in \mathbb{R}^{(n-1)\times n}$ have

Table S.D.1: Empirical Objects and Their Quantitative Roles

Object	Transformation	Role in estimation and validation
Factor payments	Decade-specific nominal wage–output ratios and agricultural rent–wage ratios	Set production labor elasticities (α_A, α_M) externally (Table 3) and discipline manufacturing wedge scale $\bar{\psi}_M$
Annual national accounts	Linear trends in log real GDP, $\ln(Y_M/Y_A)$, and $\ln(p_M/p_A)$ over complete profiles; aggregate real wage $w/\mathcal{P}$	Discipline aggregate growth, structural transformation, and relative prices (Table 6, Panel A), with the real wage path $w/\mathcal{P}$ reserved for untargeted validation (Figure 7, Panel b)
Sectoral activity cycles	Ravn–Uhlig ($\lambda_{\text{HP}} = 6.25$) HP-filtered variances, autocovariances, and cross-covariances	Discipline aggregate shock persistence $\boldsymbol{\rho}$ and innovation covariance $\boldsymbol{\Sigma}$ (Panel D)
Occupation surfaces	Linear trends in π_A^{occ} and π_T^{occ} and mean dated spatial covariances $(C_{l,y}, C_{e,y}, C_{N,y}, V_{T,y}^{\text{occ}})$	Discipline initial productivity $A_i(\ell, 1710)$, Stone–Geary (η, γ), and spatial sorting (Panels A, C)
Multimode transport stock	Linear and quadratic trends, 1830–1850 growth, post-1850 slope change, spatial log-variance $V_{T,y}$, and investment dynamics	Discipline construction cost scale m_I, adjustment convexity χ_I, and diagnosticity θ (Panels B, E)
Manufacturing & energy records	Normalized fire-insurance wedge exposure $\omega_M(\ell)$ and coalfield cost profile $\xi(\ell)$	Supply predetermined spatial wedge and energy-access schedules
Patents, railways & carrier accounts	Geocoded patent–railway panels, working accounts, and route concentration	Reserved for external checks of transport-mediated diffusion and carriage outside the estimation criterion

orthonormal rows orthogonal to $\mathbf{1}_n$:

$$H_n \mathbf{1}_n = 0, \qquad H_n H_n' = I_{n-1}, \qquad H_n' H_n = I_n - \frac{\mathbf{1}_n \mathbf{1}_n'}{n}. \tag{S.D.3}$$

Because $H_n' H_n z = z - \bar{z}\mathbf{1}_n$, retaining $H_n z \in \mathbb{R}^{n-1}$ preserves the complete demeaned trajectory while remaining invariant to the choice of orthonormal basis under covariance rotation. The three national-account profiles are log aggregate real GDP (1710–1881, $n = 172$), log industrial output relative to agriculture $\ln(Y_M/Y_A)$ (1710–1881, $n = 172$; Broadberry et al. 2022 linked at 1870 to Feinstein 1972, Table 54), and log relative industrial price $\ln(p_M/p_A)$ (1710–1870, $n = 161$), whose OLS linear trends on calendar time $w_{\text{lin}} = (y - \bar{y})/\sum_k (y_k - \bar{y})^2$ form the first three moments in Panel A of Table 6.

S.D.4 Occupations and spatial reallocation

Evaluating (S.D.2) across the twelve reconstruction dates $y \in \mathcal{Y}_O$ supplies the national occupational profiles $(\pi_{A,y}^{\text{occ}}, \pi_{T,y}^{\text{occ}})$ (spanning the three-sector simplex since $\sum_{o \in \{A,M,T\}} \pi_{o,y}^{\text{occ}} = 1$), whose OLS linear trends over 1710–1881 complete Panel A of Table 6 (5 moments in total). On the 496 supported cells d carrying the twenty-five band occupational shares $\pi_{o,d,y}$, with equal-cell mean $\langle \cdot \rangle$, standardized latitude l_d, predetermined coalfield energy exposure e_d, and adjacent-cell averaging matrix A, the four dated spatial moments at each $y \in \mathcal{Y}_O$ are

$$\begin{aligned} C_{l,y} &= \big\langle l_d \big(\pi_{M,d,y} - \langle \pi_{M,\cdot,y} \rangle\big) \big\rangle, \quad & C_{e,y} &= \big\langle e_d \big(\pi_{M,d,y} - \langle \pi_{M,\cdot,y} \rangle\big) \big\rangle, \\ C_{N,y} &= \big\langle \widetilde{\pi}_{M,d,y} (A \widetilde{\pi}_{M,\cdot,y})_d \big\rangle, & V_{T,y}^{\text{occ}} &= \big\langle \big(\pi_{T,d,y} - \langle \pi_{T,\cdot,y} \rangle\big)^2 \big\rangle, \end{aligned} \tag{S.D.4}$$

where $\widetilde{\pi}_{M,d,y} = \pi_{M,d,y} - \langle \pi_{M,\cdot,y} \rangle$, and their sample means across $y \in \mathcal{Y}_O$ form the 4 moments in Panel C of Table 6.

S.D.5 Infrastructure accumulation and dispersion

Aggregating installed route capacity $T_t(\ell)$ onto the twenty-five latitude bands $h \in \{1, \dots, 25\}$ at each annual date $y \in \{1710, \dots, 1881\}$ yields the national mean stock $\overline{T}_y$ and cross-band log-stock variance $V_{T,y}$:

$$\overline{T}_y = \frac{1}{25}\sum_{h=1}^{25} T_{h,y}, \qquad V_{T,y} = \frac{1}{25}\sum_{h=1}^{25}\left(\log T_{h,y} - \frac{1}{25}\sum_{k=1}^{25}\log T_{k,y}\right)^2. \tag{S.D.5}$$

In Panel B of Table 6 (5 moments), the trajectory of $\ln \overline{T}_y$ is summarized by four linear functionals of $H_{172}\ln \overline{T}$—the full-sample OLS linear trend, the orthogonal quadratic trend $2\,(q'\ln \overline{T})/(q'q)$ on century-rescaled calendar time $\tilde{t}_y = (y - \bar{y})/100$ (with $q_y = \tilde{t}_y^2 - \overline{\tilde{t}^2}$ orthogonalized to $\tilde{t}$), the 1830–1850 OLS growth rate, and the post-1850 slope change (the difference between the 1850–1881 and 1830–1850 OLS growth rates)—together with the full-sample mean of $V_{T,y}$.

S.D.6 Investment dynamics and local returns

From the annual national commitment rate r_y^I and its HP-filtered log cycle c_y^I,

$$r_y^I = \frac{\overline{T}_{y+1} - \overline{T}_y}{\overline{T}_y}, \qquad c_y^I = \mathrm{HP}_{6.25}\left[\log(10^{-3} + r_y^I)\right], \tag{S.D.6}$$

the 7 investment-dynamics moments in Panel E of Table 6 are the standard deviation $\sigma(c_t^I) = \mathrm{RMS}(c^I)$, the contemporaneous and lag-1 output-cycle slopes (b_0, b_1) from $c_y^I = \alpha + b_0\bar{c}_{Yy} + b_1\bar{c}_{Y,y-1} + u_y$ where $\bar{c}_{Yy} = \frac{1}{3}(c_{Ay} + c_{My} + c_{Ty})$, the first-order autocorrelation ρ_1 ($c_{y+1}^I = \alpha + \rho_1 c_y^I + u_{y+1}$, equivalent to the cumulative change slope $\beta_1^{\mathrm{diff}} = \rho_1 - 1$), the one-year lead output slope b_{lead} from $\bar{c}_{Y,y+1} = \alpha + b_{\mathrm{lead}} c_y^I + u_{y+1}$, and the two-way fixed-effects decadal county capacity-addition slopes (g_1, g_2) on the first and second decadal lags of county population growth across the 41 ancient counties over 1791–1851. Matched agricultural land rents and production-weighted agricultural wages at year y over grid cells of width $\Delta\ell_q$ (Replication Appendix R.A.3.1) satisfy

$$r_y^A = \frac{\sum_{q=1}^N \Delta\ell_q e_{Aq} R_q}{\sum_{q=1}^N \Delta\ell_q e_{Aq}}, \qquad w_y^A = \frac{\sum_{q=1}^N \Delta\ell_q e_{Aq} L_{Aq} w_q}{\sum_{q=1}^N \Delta\ell_q e_{Aq} L_{Aq}}, \tag{S.D.7}$$

deflated by the agricultural producer price index (Clark, 2002).

S.D.7 Sectoral fluctuations

Applying the annual Ravn and Uhlig (2002) Hodrick–Prescott filter ($\lambda_{\mathrm{HP}} = 6.25$, second-difference matrix D_2) identically to historical and simulated annual log sectoral outputs ($n = 172$ years),

$$c_s = \left[I_n - (I_n + 6.25 D_2' D_2)^{-1}\right]\log Y_s, \tag{S.D.8}$$

yields the 9 sectoral cycle moments in Panel D of Table 6—the variances v_s, first-order autocovariances a_s, and cross-sector covariances $k_{ss'}$ for $s, s' \in \{A, M, T\}$:

$$\begin{aligned} v_s &= \frac{1}{n}\sum_{y=1}^{n}(c_{s,y} - \bar{c}_s)^2, \qquad a_s = \frac{1}{n-1}\sum_{y=2}^{n}(c_{s,y-1} - \bar{c}_{s,-})(c_{s,y} - \bar{c}_{s,+}), \\ k_{ss'} &= \frac{1}{n}\sum_{y=1}^{n}(c_{s,y} - \bar{c}_s)(c_{s',y} - \bar{c}_{s'}). \end{aligned} \tag{S.D.9}$$

S.D.8 External benchmarks and covariance

Geocoded British patent and railway opening panels (Coll Ardanuy et al., 2021; Berkes et al., 2026), railway working accounts, and ownership-adjusted carrier concentration indices are reserved as external validation checks outside the SMM objective.

Stacking Panels A–E of Table 6 forms the $M = 30$ empirical moment vector $\mathbf{m}^{\text{data}} \in \mathbb{R}^{30}$ (5 aggregate trends in Panel A, 5 transport-network moments in Panel B, 4 spatial employment covariances in Panel C, 9 HP-filtered sectoral moments in Panel D, and 7 investment-dynamics moments in Panel E). For each moment $k \in \{1, \dots, 23\}$ in Panels A–D with projection weight c_k on the complete profile, $\widehat{\text{SE}}_k = \sqrt{c_k' \widehat{\Sigma}_{\text{prof}} c_k}$, where $\widehat{\Sigma}_{\text{prof}}$ is the Ledoit and Wolf (2004) shrinkage-regularized source-bootstrap covariance matrix from paired time-series resamples of the Broadberry et al. (2022) panel and spatial block-bootstrap resamples of the twenty-five latitude-band histories; for $k \in \{24, \dots, 30\}$ in Panel E, $\widehat{\text{SE}}_k$ comes from the Newey–West HAC covariance of the national investment regressions and the county block-bootstrap covariance.

With inverse-variance weighting $W = \text{diag}(1/\widehat{\text{SE}}_1^2, \dots, 1/\widehat{\text{SE}}_{30}^2)$ and simulated mean $\overline{\mathbf{m}}(\boldsymbol{\psi}) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{m}(\boldsymbol{\psi}; u_b)$, the SMM criterion is

$$Q(\boldsymbol{\psi}) = \left[\overline{\mathbf{m}}(\boldsymbol{\psi}) - \mathbf{m}^{\text{data}}\right]' W \left[\overline{\mathbf{m}}(\boldsymbol{\psi}) - \mathbf{m}^{\text{data}}\right] = \sum_{k=1}^{30} \left(\frac{\overline{m}_k(\boldsymbol{\psi}) - m_k^{\text{data}}}{\widehat{\text{SE}}_k} \right)^2. \tag{S.D.10}$$

Table S.D.2 reports all $M = 30$ targeted moments, their symbols in (S.D.3)–(S.D.9), empirical estimates $\widehat{m}_k$, standard errors $\widehat{\text{SE}}_k$, weights $W_{kk} = 1/\widehat{\text{SE}}_k^2$, and squared studentized residuals $q_k(\widehat{\Theta}) = W_{kk}(\overline{m}_k(\widehat{\Theta}) - \widehat{m}_k)^2$ under DE and RE. With $J(\widehat{\boldsymbol{\psi}}) = D_{\boldsymbol{\psi}} \overline{\mathbf{m}}(\widehat{\boldsymbol{\psi}})$ and empirical moment covariance $\widehat{\Sigma}_m$, misspecification-robust standard errors follow from the sandwich estimator

$$\widehat{V}_{\boldsymbol{\psi}} = (J'WJ)^{-1} J'W \widehat{\Sigma}_m W J (J'WJ)^{-1}. \tag{S.D.11}$$

S.D.9 Industrial output and the construction account

County construction purchases $I_{jy} = \int_{\mathcal{C}_j} I_y(\ell)\, d\ell$ at cost $C^I_{jy} = \int_{\mathcal{C}_j} p_M(\ell, y) I_y(\ell)\, d\ell$ are a final use of manufactured output deducted from county land rent; manufacturing output $Y_{My} = \int_0^1 Y_M(\ell, y)\, d\ell$ and producer receipts $V_{iy} = \int_0^1 P_i(\ell, y) Y_i(\ell, y)\, d\ell$ ($P_i = p_i - r_i \tau_{j(\ell)}$) already include these investment goods. Following Broadberry et al. (2022), the chain-linked agriculture–manufacturing volume index Y_t^{AM} ($Y_{1710}^{AM} = 1$) weights sectoral quantities within each segment by benchmark value shares V_{ib_s} (with benchmark years $b_s \in \{1710, 1759, 1801, 1841\}$ linked at 1740/41, 1780/81, and 1820/21):

$$\frac{Y_t^{AM}}{Y_{t_s}^{AM}} = \frac{\sum_{i \in \{A,M\}} V_{ib_s}\, Y_{it} / Y_{ib_s}}{\sum_{i \in \{A,M\}} V_{ib_s}\, Y_{it_s} / Y_{ib_s}}. \tag{S.D.12}$$

Figure S.D.2 compares the baseline width-adjusted multimode transport stock (76) with the unadjusted route-mile series.

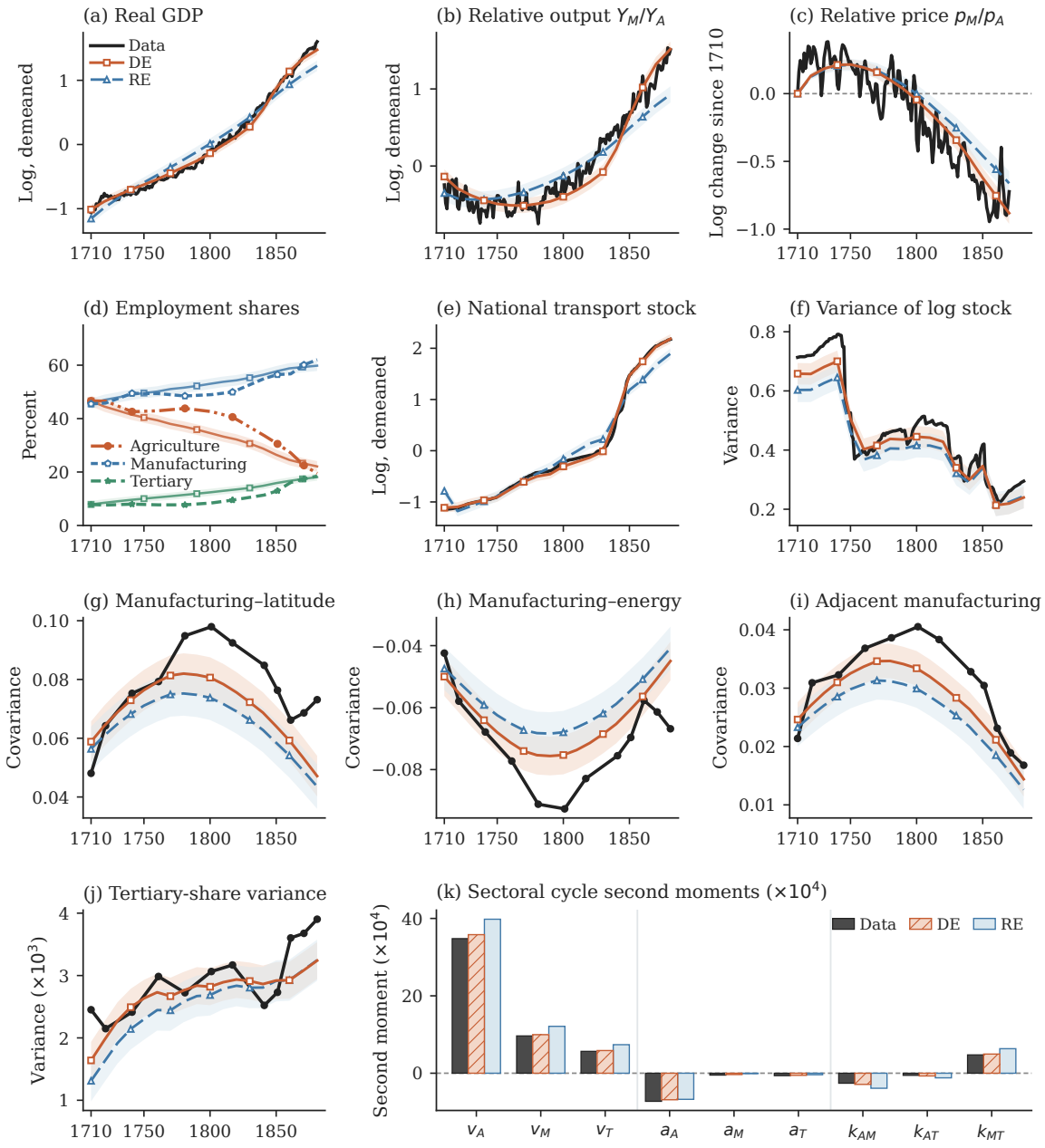


Figure S.D.1: Complete-Series Empirical Moment Profiles, 1710–1881

Notes: Historical profiles and simulated trajectories (means and 5th–95th percentile bands across 1,000 draws) under diagnostic (DE, $\widehat{\theta} = 0.739$) and rational (RE, $\theta = 0$) expectations underlying Table 6. Panels (a)–(e) plot demeaned log real GDP and Y_M/Y_A, log change in p_M/p_A since 1710, occupational shares at the twelve reconstruction dates, and demeaned log transport stock (76); Panels (f)–(k) plot cross-band log-stock variance $V_{T,y}$ (S.D.5), manufacturing spatial covariances $(C_{l,y}, C_{e,y}, C_{N,y})$ (S.D.4), tertiary spatial variance $V_{T,y}^{\text{occ}}$ $(\times 10^3)$, and HP-filtered sectoral cycle variances, autocovariances, and cross-covariances $(v_s, a_s, k_{ss'})$ $(\times 10^4)$ (S.D.9).

Table S.D.2: Complete List of Targeted SMM Moments, Standard Errors, and Weights ($M = 30$)

k	Moment description	Symbol	$\widehat{m}_k$	$\widehat{\mathrm{SE}}_k$	$W_{kk} = 1/\widehat{\mathrm{SE}}_k^2$	$q_k(\widehat{\Theta}^{\mathrm{DE}})$	$q_k(\widehat{\Theta}^{\mathrm{RE}})$
	Panel A. Aggregate growth and structural transformation						
1	Log real GDP trend (%/yr)	β_{GDP}	1.473	0.210	22.7	0.052	0.27
2	Log relative output $\ln(Y_M/Y_A)$ trend (%/yr)	$\beta_{M/A}$	1.085	0.572	3.05	0.040	0.25
3	Log relative price $\ln(p_M/p_A)$ trend (%/yr)	β_{p_M/p_A}	−0.757	0.616	2.63	0.041	0.18
4	Agricultural employment share trend (pp/yr)	β_{π_A}	−0.146	0.023	1,833.3	0.077	1.61
5	Tertiary employment share trend (pp/yr)	β_{π_T}	0.060	0.007	18,939.4	0.092	0.54
	Panel B. Transport network growth and spatial dispersion						
6	Log transport stock: linear trend (%/yr)	$\beta_{\overline{T},1}$	1.948	0.103	94.3	0.002	3.39
7	Log transport stock: quadratic trend (×100)	$\beta_{\overline{T},2}$	2.923	0.217	21.3	0.018	12.90
8	Log transport stock: 1830–1850 growth (%/yr)	$g_{\overline{T},1830\text{–}50}$	7.470	0.759	1.73	0.006	12.36
9	Log transport stock: post-1850 slope change (%/yr)	$\Delta g_{\overline{T},\text{post-1850}}$	−5.101	0.881	1.29	0.002	8.98
10	Cross-band variance of log transport stock	$\overline{V}_T$	0.469	0.148	45.9	0.047	0.16
	Panel C. Cross-sectional employment covariances						
11	Manufacturing share–latitude covariance	$\overline{C}_l$	0.077	0.026	1,471.2	0.062	0.21
12	Manufacturing share–energy covariance	$\overline{C}_e$	−0.070	0.024	1,804.4	0.061	0.24
13	Spatial autocovariance of manufacturing share	$\overline{C}_N$	0.030	0.006	24,758.3	0.080	0.51
14	Cross-band variance of tertiary share	$\overline{V}_T^{\mathrm{occ}}$	0.0029	0.0011	831,010.2	0.052	0.15
	Panel D. HP-filtered log sectoral output moments ($\times 10^4$)						
15	Variance: Agriculture (A)	v_A	34.79	4.60	0.05	0.050	1.19
16	Variance: Manufacturing (M)	v_M	9.64	1.40	0.51	0.052	3.12
17	Variance: Transport (T)	v_T	5.66	0.93	1.17	0.049	3.44
18	First-order autocovariance: Agriculture (A)	a_A	−7.27	2.41	0.17	0.029	0.05
19	First-order autocovariance: Manufacturing (M)	a_M	−0.49	0.68	2.13	0.043	0.36
20	First-order autocovariance: Transport (T)	a_T	−0.63	0.39	6.52	0.043	0.37
21	Cross-sector covariance: (A, M)	k_{AM}	−2.59	1.43	0.49	0.043	0.82
22	Cross-sector covariance: (A, T)	k_{AT}	−0.55	0.83	1.46	0.036	0.61
23	Cross-sector covariance: (M, T)	k_{MT}	4.74	0.78	1.65	0.072	4.21
	Panel E. Infrastructure investment dynamics						
24	Standard deviation of cyclical investment, $\sigma(c_t^I)$	$\sigma(c_t^I)$	0.431	0.043	537.8	0.008	4.01
25	Investment on current output cycle (b_0)	b_0	2.954	1.431	0.49	0.009	0.39
26	Investment on lagged output cycle (b_1)	b_1	1.585	1.257	0.63	0.004	1.53
27	First-order autocorrelation of investment (ρ_1)	ρ_1	−0.325	0.074	181.4	0.065	3.68
28	Lead output cycle on investment (b_{lead})	b_{lead}	−0.0051	0.0039	65,197.6	0.003	0.16
29	County investment on population growth, lag 1 (g_1)	g_1	−0.162	0.050	400.0	0.324	2.54
30	County investment on population growth, lag 2 (g_2)	g_2	−0.496	0.050	400.0	0.063	49.65
	Subtotal: Panels A–C ($k = 1, \ldots, 14$)					0.63	41.74
	Subtotal: Panel D ($k = 15, \ldots, 23$)					0.42	14.16
	Subtotal: Panel E ($k = 24, \ldots, 30$)					0.48	61.96
	Total weighted SMM criterion $Q(\widehat{\Theta}) = \sum_{k=1}^{30} q_k(\widehat{\Theta})$					1.53	117.86

Notes: Targeted SMM moments ($k = 1, \ldots, 30$), symbols from (S.D.3)–(S.D.9), empirical estimates $\widehat{m}_k$, standard errors $\widehat{\mathrm{SE}}_k$, inverse-variance weights $W_{kk} = 1/\widehat{\mathrm{SE}}_k^2$, and squared studentized residuals $q_k(\widehat{\Theta}) = \left[(\overline{m}_k(\widehat{\Theta}) - \widehat{m}_k)/\widehat{\mathrm{SE}}_k\right]^2$ under DE ($\widehat{\theta} = 0.739$) and RE ($\theta = 0$).

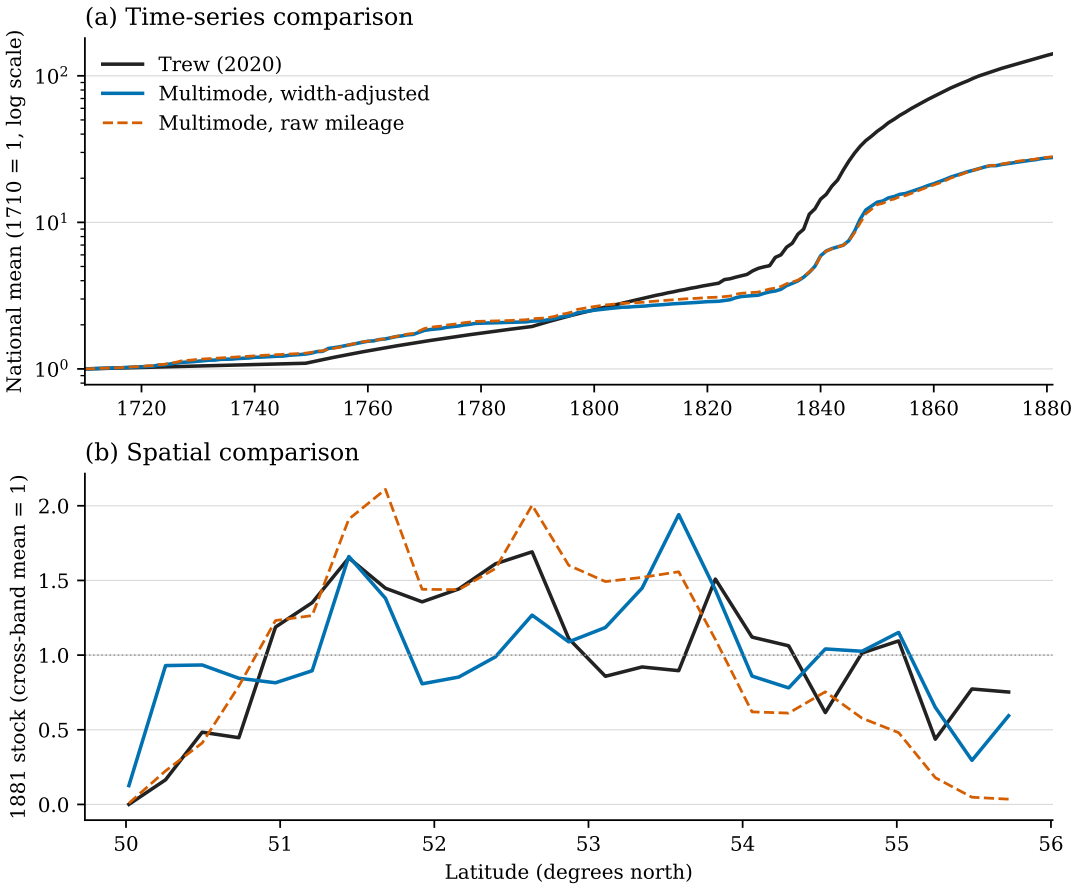


Figure S.D.2: Alternative Constructions of the Multimode Transport Stock

Notes: Panel (a) reports national mean transport stocks normalized to one in 1710. Panel (b) reports 1881 band-level stocks normalized by each series' cross-band mean, with and without the east–west land-width adjustment.

S.E Identification of the Structural Moment Map

Identification proceeds in four steps: (i) single-valuedness of the population moment map from Propositions 6 and 7, (ii) the factor-payment restriction linking fixed production elasticities to payroll shares and the manufacturing wedge, (iii) local point identification of $\boldsymbol{\psi}$ from the Jacobian rank and Hessian conditions of Rothenberg (1971), Iskrev (2010), Komunjer and Ng (2011), and Andrews et al. (2017), and (iv) analytical separation of θ from (m_I, χ_I) and $\boldsymbol{\rho}$.

S.E.1 Normalizations and observational equivalence

Fix the unit normalizations $p_A(1,t) = 1$, $\bar{L}_{1710} = 100$, and $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$, and let $\mathcal{T} \subset \mathbb{R}^p$ denote the open parameter domain containing the full parameter vector $\boldsymbol{\psi}$ (either unrestricted with $\theta \in (0, \bar{\theta})$ or restricted to $\theta = 0$). For a complete primitive shock history $u \sim \mathsf{Q}$, let $\mathscr{Y}(\boldsymbol{\psi}, u)$ denote the unique equilibrium trajectory of observables and $\Psi(\mathscr{Y}(\boldsymbol{\psi}, u)) \in \mathbb{R}^{d_m}$ the retained moment vector. The population moment map and identified set at $m_0 \in \mathbb{R}^{d_m}$ are

$$m(\boldsymbol{\psi}) = \int \Psi(\mathscr{Y}(\boldsymbol{\psi}, u))\, d\mathsf{Q}(u), \tag{S.E.1}$$

$$\mathcal{I}(m_0) = \{\boldsymbol{\psi} \in \mathcal{T} : m(\boldsymbol{\psi}) = m_0\}. \tag{S.E.2}$$

A parameter vector $\boldsymbol{\psi}_0 \in \mathcal{T}$ is locally point identified if there exists an open neighborhood $\mathcal{N} \ni \boldsymbol{\psi}_0$ such that $\mathcal{I}(m(\boldsymbol{\psi}_0)) \cap \mathcal{N} = \{\boldsymbol{\psi}_0\}$.

PROPOSITION S.E.1: *Suppose the conditions of Propositions 6 and 7 (and Corollary A.1) hold at every state reached from the initial condition for each $\boldsymbol{\psi} \in \mathcal{T}$. If $\Psi(\mathscr{Y}(\boldsymbol{\psi}, u))$ is Q-integrable, then the population moment map (S.E.1) is single valued on $\mathcal{T}$.*

Proof. By Propositions 6 and 7 and Corollary A.1, each primitive shock history u generates a unique measurable outcome path $\mathscr{Y}(\boldsymbol{\psi}, u)$, whose Q-integral defines a unique moment vector $m(\boldsymbol{\psi})$. □

S.E.2 Direct restrictions and local rank

PROPOSITION S.E.2: *For every $\boldsymbol{\psi} \in \mathcal{T}$, the conditional hiring rule (30) implies that expected operating payroll relative to expected net producer revenue at active sites satisfies*

$$\frac{w(\ell,t)L_A(\ell,t)}{\mathbb{E}_t[P_A(\ell,t)_+ Y_A(\ell,t)]} = \alpha_A, \qquad \frac{w(\ell,t)L_M(\ell,t)}{\mathbb{E}_t[P_M(\ell,t)_+ Y_M(\ell,t)]} = \frac{\alpha_M}{1 + \bar{\psi}_M \omega_M(\ell)}, \tag{S.E.3}$$

for all $\phi_{it}^*(\ell) \in [0,1)$ (and holds statewise whenever $P_i(\ell,t,\omega) > 0$, $\phi_{it}^*(\ell) = 0$, and spot prices and tolls are deterministic).

Proof. Since $Y_i(\ell,\omega) = Z_i(\ell)G_i(\ell,\omega)L_i^*(\ell)^{\alpha_i}$ whenever $P_i(\ell,\omega) > 0$, (30) gives

$$\mathbb{E}_t\big[P_i(\ell,\omega)_+ Y_i(\ell,\omega)\big] = Z_i(\ell) m_i(\phi_i^*(\ell);\ell) L_i^*(\ell)^{\alpha_i} = \frac{1 + \psi_i(\ell)}{\alpha_i} w(\ell) L_i^*(\ell),$$

yielding (S.E.3). □

THEOREM S.E.1: *Fix an interior $\boldsymbol{\psi}_0 \in \mathcal{T}$. Suppose Proposition S.E.1 holds and m is C^1 on a neighborhood of $\boldsymbol{\psi}_0$. If the population Jacobian satisfies*

$$D_{\boldsymbol{\psi}} m(\boldsymbol{\psi}_0) \in \mathbb{R}^{d_m \times p} \quad \textit{with} \quad \operatorname{rank} D_{\boldsymbol{\psi}} m(\boldsymbol{\psi}_0) = p, \tag{S.E.4}$$

then $\boldsymbol{\psi}_0$ is locally point identified by the retained moments, and the rank is invariant to nonsingular whitening and smooth interior reparameterizations.

Proof. By (S.E.4), $D_{\psi} m(\psi_0)$ contains a nonsingular $p \times p$ minor, and applying the inverse function theorem to the corresponding p-moment subvector yields local injectivity around ψ_0. □

S.E.3 Pseudo-true values and branch regularity

Let $g(\psi) = m^{\text{data}} - m(\psi)$ and $Q(\psi) = g(\psi)' W g(\psi)$ for positive-definite W.

THEOREM S.E.2: *Fix an interior point ψ^* where g is C^2 and satisfies first-order stationarity*

$$2Dg(\psi^*)' W g(\psi^*) = 0. \tag{S.E.5}$$

If the exact criterion Hessian

$$H(\psi^*) = 2Dg(\psi^*)' W Dg(\psi^*) + 2 \sum_{k=1}^{d_m} \left[W g(\psi^*) \right]_k \nabla^2 g_k(\psi^*) \tag{S.E.6}$$

is positive definite, then ψ^* is a strict local minimizer of Q in the global pseudo-true set

$$\mathcal{P} = \arg\min_{\psi \in \mathcal{T}} g(\psi)' W g(\psi), \tag{S.E.7}$$

and at exact fit ($g(\psi^*) = 0$) the rank condition (S.E.4) suffices for $H(\psi^*) \succ 0$.

Proof. Equations (S.E.5)–(S.E.6) are the gradient and Hessian of $Q(\psi)$, so $H(\psi^*) \succ 0$ yields strict local convexity around ψ^*; at $g(\psi^*) = 0$, $H(\psi^*) = 2Dg(\psi^*)' W Dg(\psi^*) \succ 0$ if and only if $\operatorname{rank} Dg(\psi^*) = p$. □

S.E.4 Joint identification of diagnosticity θ

Separate identification of θ from the construction cost parameters (m_I, χ_I) and the aggregate shock persistence vector $\boldsymbol{\rho}$ follows from the block-triangular structure of the moment map across Panels B, D, and E of Table 6 and the contrasting signs of their partial derivatives on investment–output comovement b_0 and first-order investment autocorrelation ρ_1. Linearizing the interior branch of the clipped investment rule (62), $x_{jt}^{\theta} = \chi_I^{-1}[\tilde{q}_{jt}^{\theta}/(m_I D_{jt}^I) - 1]$, around the deterministic transition path under the one-period diagnostic reference lag ($J_{\text{DE}} = 1$, $\bar{\mathbf{z}}_t = \mathbb{E}_{t-1}[\mathbf{z}_t]$) shows that a date-$t$ aggregate innovation $\boldsymbol{\varepsilon}_t$ shifts perceived date-$(t+1)$ productivity by $(1+\theta)\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$, whereas at date $t+1$ (under $\mathbb{E}_t[\boldsymbol{\varepsilon}_{t+1}] = \mathbf{0}$) the reference forecast updates to $\boldsymbol{\rho}\mathbf{z}_t$ and the date-t diagnostic belief wedge $+\theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$ vanishes while the elevated stock $\overline{T}_{t+1}$ lowers marginal continuation value. By contrast, under $\theta = 0$, lowering adjustment convexity χ_I (raising $-\chi_I$) or raising shock persistence ρ_M increases both b_0 and ρ_1:

$$\begin{aligned} &\frac{\partial b_0}{\partial \theta} > 0, \qquad \frac{\partial \rho_1}{\partial \theta} < 0, \qquad \frac{\partial b_{\text{lead}}}{\partial \theta} < 0, \\ &\frac{\partial b_0}{\partial(-\chi_I)}\bigg|_{\theta=0} > 0, \qquad \frac{\partial \rho_1}{\partial(-\chi_I)}\bigg|_{\theta=0} > 0, \qquad \frac{\partial b_0}{\partial \rho_M}\bigg|_{\theta=0} > 0, \qquad \frac{\partial \rho_1}{\partial \rho_M}\bigg|_{\theta=0} > 0. \end{aligned} \tag{S.E.8}$$

Because the sectoral output variances and first-order autocovariances (v_s, a_s) in Panel D pin down $(\boldsymbol{\rho}, \boldsymbol{\Sigma})$ and the long-run linear trend of log transport stock in Panel B pins down the construction cost scale m_I, no movement in $-\chi_I$ can simultaneously raise b_0 and lower ρ_1, separating θ from $(m_I, \chi_I, \boldsymbol{\rho})$. Full column rank of $D_{\psi} m(\psi_0)$ in (S.E.4) is verified numerically at the estimates in Table 5. Table S.E.1 verifies that the fit advantage of diagnostic expectations is not driven by an extra free parameter, moment-block weighting, or literal-specification assumptions: Panel A re-estimates

all $p = 25$ remaining structural parameters at fixed values $\theta \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$ and compares each restricted specification against $\theta = 0$ via the misspecification-robust Rivers and Vuong (2002) equal-parameter z-statistic; Panel B reports the Andrews et al. (2017) local identifying information shares across moment blocks—showing that the investment-dynamics and transport-network blocks (Panels E and B) supply 85.7 percent (49.8 and 35.9 percent) of the identifying information for $\widehat{\theta}$—and confirms that $\widehat{\theta}$ remains stable across leave-one-block-out estimations ($\widehat{\theta} \in [0.666, 0.747]$, holding the nuisance block identified primarily by the omitted moment panel at its baseline estimate while θ, m_I, χ_I, and the remaining parameters are re-estimated so that $d_m > p_{\text{sub}}$ in every sub-sample) as well as under equal-block studentized weighting ($\widehat{\theta}_{\text{eq}} = 0.748$, SE $= 0.104$).

Table S.E.1: Restricted SMM Profile Across Fixed θ and Leave-One-Block-Out Estimates

Specification / statistic	$\theta = 0.0$ (RE)	$\theta = 0.3$	$\theta = 0.6$	$\theta = 0.9$	$\theta = 1.2$
Panel A. Restricted SMM estimation across fixed θ ($p = 25$ free parameters)					
Estimated free parameters p	25	25	25	25	25
Sub-criterion: Panels A–C ($Q_{\text{A–C}}$, 14 moments)	41.74	15.37	2.28	2.16	14.70
Sub-criterion: Panel D (Q_{D}, 9 moments)	14.16	4.88	0.48	0.61	4.91
Sub-criterion: Panel E (Q_{E}, 7 moments)	61.96	20.81	2.43	2.77	18.60
SMM criterion, $Q(\widehat{\Theta}(\theta))$ (30 moments)	117.86	41.06	5.19	5.54	38.21
Rivers–Vuong (2002) z-statistic vs. $\theta = 0$	—	8.56	6.42	4.35	2.36
Rivers–Vuong (2002) p-value vs. $\theta = 0$	—	[< 0.0001]	[< 0.0001]	[< 0.0001]	[0.0184]
Adjustment cost convexity $\widehat{\chi}_I(\theta)$	1.589	1.728	1.867	2.006	2.145
Construction cost scale $\widehat{m}_I(\theta)$	0.020	0.021	0.022	0.023	0.024
Manufacturing AR(1) persistence $\widehat{\rho}_M(\theta)$	0.561	0.550	0.538	0.526	0.515
Manufacturing innovation s.d. $\widehat{\sigma}_M(\theta)$	0.042	0.041	0.039	0.038	0.037
Technological catch-up rate $\widehat{\lambda}_0(\theta)$	0.175	0.169	0.163	0.157	0.150
Iceberg transport cost scale $\widehat{\kappa}_0(\theta)$	0.073	0.076	0.079	0.082	0.085
Panel B. Moment-block information shares and leave-one-block-out estimates of $\widehat{\theta}$					
Moment block	Panel A	Panel B	Panel C	Panel D	Panel E lags
Andrews–Gentzkow–Shapiro (2017) info. share (%)	1.3%	35.9%	0.3%	12.7%	49.8%
Leave-one-block-out $\widehat{\theta}$ (SE)	0.739 (0.090)	0.723 (0.095)	0.739 (0.090)	0.747 (0.091)	0.666 (0.166)
Sub-sample criterion $Q_{\text{sub}}(\widehat{\Theta})$	1.22	1.44	1.27	1.05	1.06

Notes: Panel A reports restricted SMM estimates fixing $\theta \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$ ($p = 25$ free parameters) with misspecification-robust Rivers and Vuong (2002) equal-fit tests against $\theta = 0$ (p-values in brackets). Panel B reports Andrews et al. (2017) local identifying information shares for $\widehat{\theta}$ alongside leave-one-block-out estimates (standard errors in parentheses; equal-block studentized weighting yields $\widehat{\theta}_{\text{eq}} = 0.748$, SE $= 0.104$).

Computational and Replication Appendix

to "Corrective Overoptimism"

Brandon S.C. Chan

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R.A Computational Methods and Verification

R.A.1 County investment algorithm

At each date t and inherited state $\Omega_t = (A_{At}, A_{Mt}, T_t, \mathbf{z}_t, \overline{\mathbf{z}}_t, t)$, the county infrastructure game is solved on the statutory box $\mathcal{X}_t = \prod_{j=1}^{J}[0, \bar{x}_{jt}] \subseteq \prod_{j=1}^{J}[0, o_{jt}]$. A candidate portfolio $\mathbf{x} \in \mathcal{X}_t$ is admissible if its conditional dated equilibrium (Definition 1) yields positive wages and prices, satisfies Stone–Geary subsistence $w(\ell) + b_t(\omega) \geq \gamma p_A(\ell, \omega)$ across all sites and states, and clears goods, labor, county service, and dividend ledgers with construction demand $I_t(\ell; \mathbf{x})$ absorbed from manufacturing output.

For any trial portfolio $\mathbf{x}$, the inner solver computes the pre-discovery plan—residence $n(\ell)$, sector $e_i(\ell) \in \{0, 1\}$, research probability $\phi_i(\ell) \in [0, 1)$, monetary innovation bill $\Psi_i(\phi_i(\ell); w(\ell), \ell)$, and hired labor $(L_i(\ell), L_T(\ell))$—and then solves the statewise post-discovery allocation $(Y_i, p_i, H_i, \tau_j, b_t)$ across $N_\omega = 32$ stratified spatial discovery draws $\omega \in \{1, \ldots, N_\omega\}$ from the Ornstein–Uhlenbeck Gaussian copula ($d_0 = 0.10$) and Pareto step-size law. Current county return $R_{jt}(\mathbf{x}; \Omega_t)$ equals expected county land rent minus the expected spot cost of committed construction goods. Continuation values $\mathbb{E}_t^\theta[V_{j,t+1}^\theta(\Omega_{t+1})]$ in (57) are represented on the reduced county-level state $(\overline{\mathbf{T}}_{t+1}, \overline{\ln \mathbf{A}}_{A,t+1}, \overline{\ln \mathbf{A}}_{M,t+1}, \mathbf{z}_{t+1}, \overline{\mathbf{z}}_t)$ via a second-order Chebyshev–Smolyak polynomial basis $H_{jt}^{(n-1)}$ iterated backward from the post-1881 balanced-growth continuation value (Appendix A.7.2), where $\overline{T}_{j,t+1} = \overline{T}_{jt} + d_{jt}x_{jt}$ updates county j's average route stock (54) and $\overline{\ln A}_{ij,t+1}$ tracks county-mean log knowledge under (40). Conditional on each draw ω, expectations over the diagnostic law (12) of next-period productivity gaps, $\mathbf{z}_{t+1} \sim \mathrm{TN}_{\mathcal{B}}(\mathbf{m}_t^\theta, \boldsymbol{\Sigma})$ on $\mathcal{B} = \prod_{s\in\{A,M,T\}}[-c\nu_s, c\nu_s]$ with $\mathbf{m}_t^\theta = \boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}[\mathbf{z}_t - M(\boldsymbol{\rho}\mathbf{z}_{t-1})]$, are evaluated by a $5 \times 5 \times 5$ Gauss–Hermite quadrature rule (125 nodes) mapped to $\mathcal{B}$ via the Cholesky factor of $\boldsymbol{\Sigma}$ and renormalized on the truncated support.

Fixed points of the county best-response map solve the projected pseudogradient system

$$\mathbf{r}_{t,\alpha}(\mathbf{x}) = \mathbf{x} - \mathrm{proj}_{\mathcal{X}_t}\left[\mathbf{x} + \alpha\widehat{\mathbf{g}}_t^\theta(\mathbf{x})\right] = 0, \qquad \alpha > 0, \tag{R.A.1}$$

where $\widehat{\mathbf{g}}_{jt}^\theta(\mathbf{x}) = \partial_{x_j}\widehat{\Pi}_{jt}^\theta(x_j, \mathbf{x}_{-j})$ is county j's unilateral payoff derivative along the implicit conditional equilibrium and differentiated Chebyshev–Smolyak continuation surface. Zeros of $\mathbf{r}_{t,\alpha}(\mathbf{x})$ satisfy the KKT conditions (A.1), equivalently (62), and are unique under Proposition 7.

Numerically, the solver initializes from inherited reference portfolios and alternates damped Anderson-accelerated pseudogradient steps with coordinate-wise unilateral sweeps, evaluating every candidate by recomputing the full conditional dated equilibrium and continuation value.

At a converged candidate $\widehat{\mathbf{x}}$, each county's maximum unilateral deviation gain is checked on $\{0, 1/4, 1/2, 3/4, 1\}\bar{x}_{jt}$ plus local refinement around $\widehat{x}_j$:

$$\epsilon_j(\widehat{\mathbf{x}}) = \max_{u \in [0, \bar{x}_{jt}]} \{\widehat{\Pi}^{\theta}_{jt}(u, \widehat{\mathbf{x}}_{-j}) - \widehat{\Pi}^{\theta}_{jt}(\widehat{\mathbf{x}})\} \leq 10^{-10},$$

before advancing the state to $t+1$ via (54).

R.A.2 Joint-equilibrium differentiation

To differentiate simulated moment paths with respect to the transformed parameter vector z, we differentiate the conditional private equilibrium and the continuous county investment KKT system jointly along each simulated history $b \in \{1, \ldots, B\}$ and date t. Let s_{bt} denote the inherited state, u_{bt} the conditional private-equilibrium unknowns, and $\mathbf{x}_{bt}$ the county investment vector, and let $\mathcal{R}_{bt}$ stack the pre-discovery plan conditions and the finite-volume spot-market relations of Section R.A.3.1:

$$\mathcal{R}_{bt}(u_{bt}, \mathbf{x}_{bt}, s_{bt}, z) = 0, \qquad \overline{\mathbf{g}}^{\theta}_{bt}(\mathbf{x}_{bt}; u_{bt}, s_{bt}, z) \text{ satisfies (A.1).} \tag{R.A.2}$$

Partition the county coordinates into lower-bound ($\mathcal{L}_{bt}$), interior ($\mathcal{I}_{bt}$), and upper-bound ($\mathcal{U}_{bt}$) active sets. Substituting $x_{j,bt} = 0$ on $\mathcal{L}_{bt}$ and $x_{j,bt} = \bar{x}_{j,bt}$ on $\mathcal{U}_{bt}$, the interior equilibrium system is

$$\mathcal{H}_{bt}(q_{bt}, s_{bt}, z) = \begin{bmatrix} \mathcal{R}_{bt} \\ \overline{\mathbf{g}}^{\theta}_{bt,\mathcal{I}_{bt}} \end{bmatrix} = 0, \qquad q_{bt} = (u'_{bt}, \mathbf{x}^{I\prime}_{bt,\mathcal{I}_{bt}})'. \tag{R.A.3}$$

PROPOSITION R.A.1: *Fix a finite set of shock histories and a parameter z_0. At every reached history and date, suppose: (i) $\mathcal{H}_{bt}$ is continuously differentiable with nonsingular Jacobian $\mathcal{H}_{q,bt}$; (ii) bound coordinates satisfy strict complementarity; (iii) the own-concavity and rival-sensitivity bounds in (63) and (64) hold with positive slack; and (iv) all active private-equilibrium faces and state transitions $\mathcal{G}_{bt}$ are locally regular. Then a neighborhood of z_0 preserves the KKT active sets and unique equilibrium branch, and $(u_{bt}, \mathbf{x}_{bt})$ is continuously differentiable on that neighborhood.*

Proof. By the implicit-function theorem applied to (R.A.3), q_{bt} is locally C^1 with fixed active sets under strict complementarity and contraction slack. Induction over the finite horizon and intersection across the B histories yield a common neighborhood of z_0. □

If $c_q(z)$ enumerates the signed complementarity, curvature, and branch slacks with $m_q = c_q(z_0) > 0$ and Lipschitz constant $L_q = \sup_{\|z - z_0\| \leq \bar{r}} \|Dc_q(z)\|_* < \infty$, the local branch-preservation radius is

$$r_{\text{pol}} = \min\left\{\bar{r}, \min_{q: L_q > 0} \frac{m_q}{2L_q}\right\}. \tag{R.A.4}$$

PROPOSITION R.A.2: *Under the hypotheses of Proposition R.A.1, let $S_{bt} = D_z s_{bt}$ and $Q_{bt} = D_z q_{bt}$. The sequence-space path sensitivities satisfy the forward lower-triangular recursion*

$$\begin{aligned} Q_{bt} &= -\mathcal{H}^{-1}_{q,bt}\left(\mathcal{H}_{s,bt} S_{bt} + \mathcal{H}_{z,bt}\right), \\ S_{b,t+1} &= \mathcal{G}_{s,bt} S_{bt} + \mathcal{G}_{q,bt} Q_{bt} + \mathcal{G}_{z,bt}, \end{aligned} \tag{R.A.5}$$

and the simulated moment Jacobian of $\widehat{m}_B(z) = B^{-1} \sum_{b=1}^{B} \Psi(\mathscr{Y}_b(z), z)$ is

$$D_z \widehat{m}_B(z) = \frac{1}{B} \sum_{b=1}^{B} \left[\Psi_{\mathscr{Y},b} D_z \mathscr{Y}_b(z) + \Psi_{z,b}\right]. \tag{R.A.6}$$

Proof. Implicit differentiation of (R.A.3) yields the first equation in (R.A.5), and the chain rule applied to $s_{b,t+1} = \mathcal{G}_{bt}(s_{bt}, q_{bt}, z)$ yields the second, following the sequence-space Jacobian principle of Auclert et al. (2021). Averaging across the B common-random-number histories gives (R.A.6). □

PROPOSITION R.A.3: *Let $\widehat{Q}_B(z) = \widehat{g}_B(z)'W\widehat{g}_B(z)$ denote the finite-sample SMM criterion evaluated by solving the full endogenous model. At a fully evaluated center z_k, any local or fixed-policy directional matrix $\widetilde{J}_k$ used solely to propose candidate steps p_k preserves the exact estimator and asymptotic inference whenever step acceptance ($\widehat{Q}_B(z_k + p_k) < \widehat{Q}_B(z_k)$), stationarity verification, and terminal Jacobians are evaluated using the fully endogenous model.*

Proof. Because $\widehat{Q}_B(z_k + p_k)$, convergence gates, and terminal covariance matrices recompute the 1710 initialization, dated private equilibria, continuation appraisals, and county KKT solutions from scratch, $\widetilde{J}_k$ enters only the step-proposal generator and leaves the scored objective, accepted iterates, and statistical inference invariant (Nocedal and Wright, 2006). □

R.A.3 Parameterization and initialization

As detailed in Online Appendix S.D.1, the parameter vector $\boldsymbol{\psi}$ is mapped to unconstrained optimization coordinates $z \in \mathbb{R}^p$ via logarithmic transforms for positive parameters and hyperbolic-tangent transforms for bounded correlations and shares. The initial 1710 log-productivity profiles $\ln A_{i,1710}(\ell)$ are represented on the five-knot natural cubic spline basis $\{0, 1/4, 1/2, 3/4, 1\}$ and normalized to unit mean $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$ for $i \in \{A, M\}$. The spatial grid comprises $N = 500$ cells partitioned into $J = 10$ counties, with moments projected onto the 496-cell historical support and twenty-five latitude bands.

R.A.3.1 Spatial grid and conservative finite-volume scheme

The unit interval is partitioned into ordered cells $q \in \{1, \dots, N\}$ with midpoints ℓ_q, widths $\Delta\ell_q > 0$ ($\sum_q \Delta\ell_q = 1$), and county assignment $j(q)$, each county $\mathcal{C}_j$ being a union of whole cells. Cell quantities are evaluated at ℓ_q, and face flows and prices carry half-integer indices. Writing $\kappa_q = \kappa_0 e^{-T_t(\ell_q)}$ and $S_{iq} = Y_{iq} - n_q c_{iq} - e_{iq}\Psi_{iq}/p_{iq} - I_{iq}$ ($I_{Aq} = 0$, $I_{Mq} = I_q$), the conservative discretization of goods conservation (69), spatial no-arbitrage (45), and county service clearing (51) is

$$H_{i,q+1/2} - H_{i,q-1/2} = \Delta\ell_q\big[S_{iq} - \kappa_q|H_{iq}|\big], \tag{R.A.7}$$

$$p_{i,q+1/2} - p_{i,q-1/2} \in \Delta\ell_q(\kappa_q p_{iq} + r_i\tau_{j(q)})\,\mathrm{Sign}(H_{iq}), \tag{R.A.8}$$

$$D_j = \sum_{q\in\mathcal{C}_j}\Delta\ell_q \sum_{i\in\{A,M\}} r_i(Y_{iq} + |H_{iq}|) \le K_j = \sum_{q\in\mathcal{C}_j}\Delta\ell_q Y_{Tq},$$

$$0 \le Y_{iq} \le e_{iq}Z_{iq}G_{iq}L_{iq}^{\alpha_i}, \qquad \tau_j \ge 0, \quad \tau_j(K_j - D_j) = 0, \tag{R.A.9}$$

with closed borders $H_{i,1/2} = H_{i,N+1/2} = 0$, numeraire $p_{A,N+1/2} = 1$, and $\mathrm{Sign}(0) = [-1, 1]$. The cell-centered flow solves $H_{iq} + \frac{1}{2}\Delta\ell_q\kappa_q|H_{iq}| = H_{i,q-1/2} + \frac{1}{2}\Delta\ell_q S_{iq}$, which is strictly increasing in H_{iq}, making the forward flow sweep single-valued whenever

$$\max_q \frac{\Delta\ell_q\kappa_q}{2} < 1, \tag{R.A.10}$$

a condition verified as part of the equilibrium certificate.

Remark R.A.1 (Local branch regularity and spot-market solver). On a fixed complementarity and flow-sign branch with positive prices, wages, capacities, and Stone–Geary slack, $\mathcal{R}_{bt}$ in (R.A.2) is C^1; if $D_u\mathcal{R}_{bt}$ is nonsingular, the implicit function theorem gives a unique local C^1 spot-market branch with sensitivity $\partial_\vartheta u_{bt} = -[D_u\mathcal{R}_{bt}]^{-1}\partial_\vartheta\mathcal{R}_{bt}$ (hypothesis (iv) of Proposition R.A.1; continuum global uniqueness is Proposition 6). Numerically, for each of the $N_\omega = 32$ stratified discovery draws, (R.A.7)–(R.A.9) is solved to relative tolerance 10^{-10} by an active-set Newton–Fischer–Burmeister continuation scheme that alternates forward-backward price-flow sweeps under (R.A.10) with damped Newton updates on the $J = 10$ county toll complementarities, dividend balance, and labor-clearing utility $\bar{u}$.

R.A.3.2 Closed-form quote-to-plan inversion and 1710 initialization

By Propositions 1–2, any statewise quote bank $\{p_{Aq}(\omega), p_{Mq}(\omega), \tau_{j(q)}(\omega), b(\omega)\}$ and candidate 1710 profile $A_{i,1710}(\ell)$ map in closed form into the pre-discovery plan at every cell $q \in \{1, \dots, N\}$:

1. Expected Stone–Geary mobility (18) uniquely determines $\bar{u}$ and local contract wage
$$w_q(\bar{u}) = \frac{\bar{u} + \mathbb{E}_0[(\gamma p_{Aq}(\omega) - b(\omega))/\mathcal{P}_q(\omega)]}{\mathbb{E}_0[1/\mathcal{P}_q(\omega)]}$$
such that total hired labor integrates to $\bar{L}_{1710}$.

2. Given $w_q(\bar{u})$ and expected net unit revenues $\mu_{iq} = \mathbb{E}_0[(p_{iq}(\omega) - r_i\tau_{j(q)}(\omega))_+]$, the bid rents (30)–(32) determine optimal research $\phi^*_{iq} \in [0, 1)$, innovation bill $\Psi_i(\phi^*_{iq}; w_q, \ell_q)$, production labor L^*_{iq}, and sector assignment $e_{iq} \in \{0, 1\}$.

3. Given expected tolls $\bar{\tau}_{j(q)} = \mathbb{E}_0[\tau_{j(q)}(\omega)]$, (50) determines carrier labor L^*_{Tq} and county capacity $K_j = \sum_{q\in\mathcal{C}_j} \Delta\ell_q Y_{Tq}$, while labor clearing fixes resident density $n_q = e_{Aq}L^*_{Aq} + e_{Mq}L^*_{Mq} + L^*_{Tq}$.

Substituting this pre-discovery plan into the spot-market conditions (R.A.11) defines the dated equilibrium operator $\mathcal{T}_{1710}(w, p, \tau) = (w, p, \tau)$ of Definition 1:
$$0 \le Y^0_{iq}(\omega) \le e_{iq}Z^0_{iq}G^0_{iq}(\omega)[L^0_{iq}]^{\alpha_i}, \quad D^0_j(\omega) \le K^0_j, \quad \tau^0_j(\omega) \ge 0, \quad \tau^0_j(\omega)[K^0_j - D^0_j(\omega)] = 0. \tag{R.A.11}$$
Concentrating out the initial productivity spline parameters on the 1710 occupational cross-section via (S.D.2) disciplines the 1710 spatial employment and urban profile. At each date t, the available frontier on the ordered grid is evaluated in $O(N)$ operations per sector via a two-pass dynamic program:
$$F_{iq,t} = \max_{r\in\{1,\dots,N\}} e^{-\delta|\ell_q - \ell_r|}A^+_{ir,t} = \max\{F^{\rightarrow}_{iq}, F^{\leftarrow}_{iq}\}, \tag{R.A.12}$$
where $F^{\rightarrow}_{iq} = \max\{A^+_{iq,t}, e^{-\delta(\ell_q - \ell_{q-1})}F^{\rightarrow}_{i,q-1}\}$ and $F^{\leftarrow}_{iq} = \max\{A^+_{iq,t}, e^{-\delta(\ell_{q+1} - \ell_q)}F^{\leftarrow}_{i,q+1}\}$.

R.A.4 Simulation and optimization sequence

For each specification, the estimator simulates B common-random-number histories over the 172 dates $t \in \{1710, \dots, 1881\}$ ($B_{\text{screen}} = 64$ during multistart screening and $B = 1{,}000$ for final optimization, evaluation, and inference), initializing from a low-discrepancy Halton sequence in the p-dimensional transformed box followed by Gauss–Newton dogleg trust-region refinement. Local stationarity is verified on the minimal normalized positive basis
$$\mathcal{D}_p = \left\{e_1, \dots, e_p, -\frac{1}{\sqrt{p}}\mathbf{1}_p\right\}. \tag{R.A.13}$$

LEMMA R.A.1: $\mathcal{D}_p$ *in* (R.A.13) *positively spans* $\mathbb{R}^p$ *with minimum cardinality* $p+1$*, so for any* $\nabla Q \neq 0$ *at least one* $d \in \mathcal{D}_p$ *satisfies* $\nabla Q' d < 0$*. Supplementing* $\mathcal{D}_p$ *with the* $2p$ *signed-coordinate poll* $\{\pm e_i\}_{i=1}^p$ *guards against finite-step curvature reversals.*

Proof. For any $v \in \mathbb{R}^p$, $t = \max\{0, -\min_i v_i\} \geq 0$ gives $v = \sum_{i=1}^p (v_i + t)e_i + \sqrt{p}\, t(-p^{-1/2}\mathbf{1}_p)$ with nonnegative weights, whereas any p vectors have a pointed nonnegative cone. □

Terminal derivatives use a scaled Helmert regular simplex $D \in \mathbb{R}^{p\times(p+1)}$ with unit columns satisfying

$$D\mathbf{1}_{p+1} = 0, \qquad DD' = \frac{p+1}{p} I_p. \tag{R.A.14}$$

Evaluating the $p+1$ vertices at unit-box radii $h_1 = 0.01$ and $h_2 = 0.005$ with residual matrix G_h and center residual g_0 yields

$$J_h = (G_h - g_0 \mathbf{1}'_{p+1}) D' \left[h \frac{p+1}{p} \right]^{-1}. \tag{R.A.15}$$

Convergence requires that the unit-box gradient satisfy

$$\|\nabla_u Q\|_\infty \leq 5 \times 10^{-4} \max\{1, Q\}, \tag{R.A.16}$$

that no simplex or $2p$ coordinate-poll vertex improves Q by more than $10^{-8}\max\{1, Q\}$, and that the whitened Jacobians $L_Q^{-1} J_{h_1}$ and $L_Q^{-1} J_{h_2}$ ($L_Q L'_Q = W^{-1}$) have full column rank p at tolerance 10^{-10}, singular-value ratio at least 10^{-3}, and relative Frobenius drift at most 0.10. The half-step Jacobian $J = J_{h_2}$ forms $H_{\mathrm{GN}} = 2J'WJ$ and the sandwich covariance (S.D.11).